



College of Health Sciences, School of Medicine, Department of
Biochemistry

Neurotransmitters

By
Dr. Abdisa Tufa

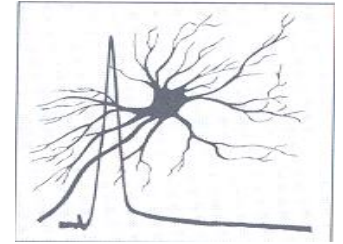
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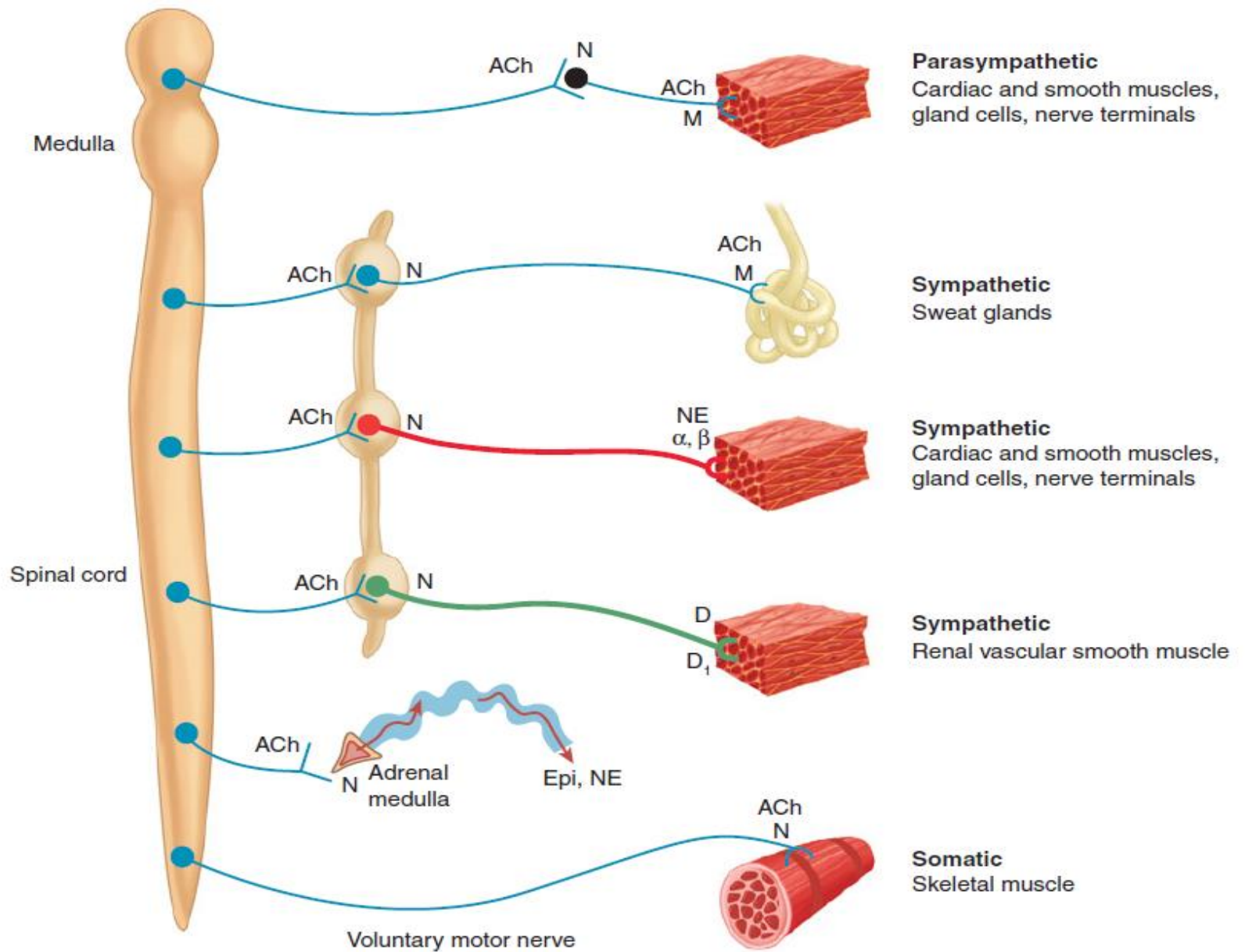
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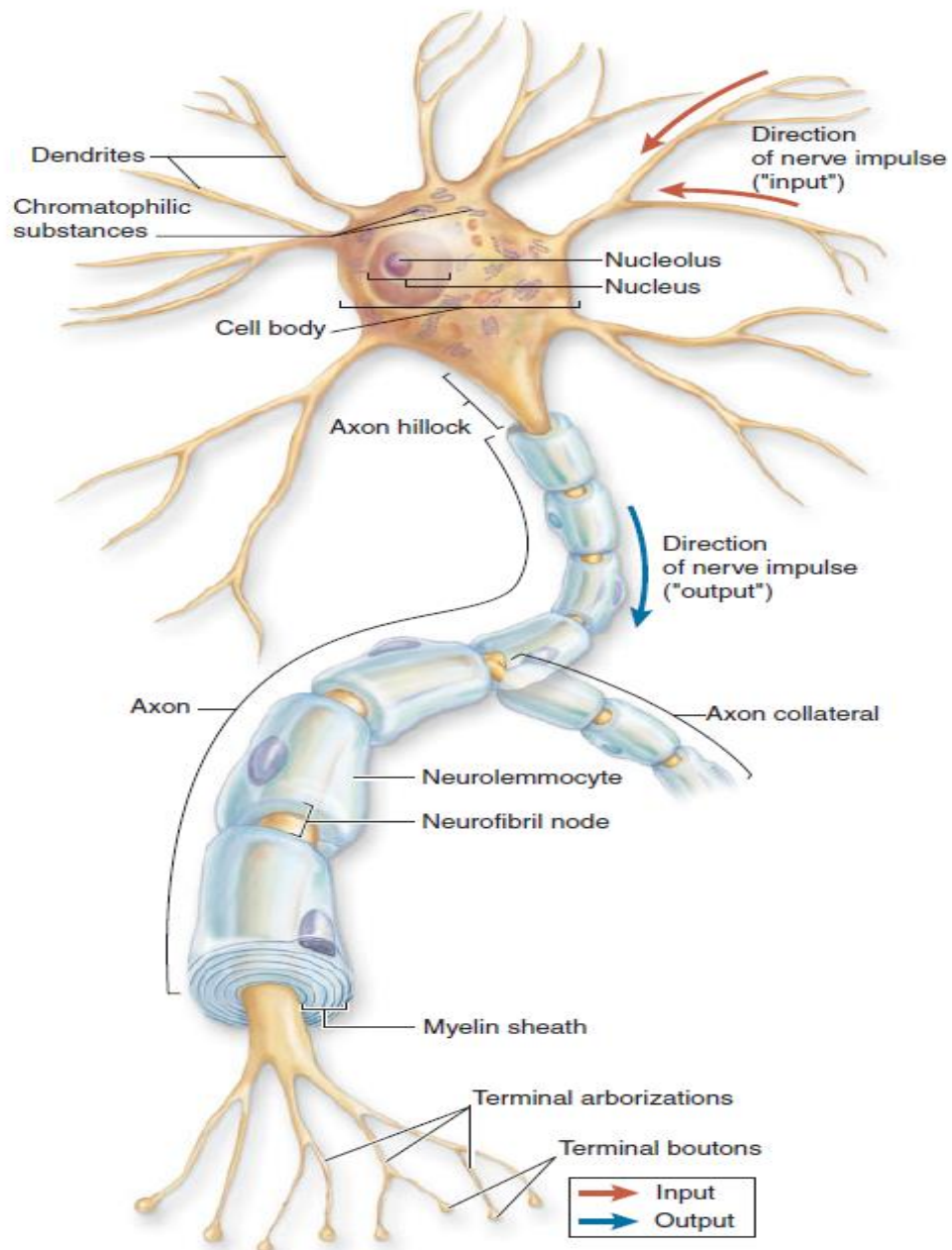
Biochemistry of Nerve Cells



December, 2023



Overview of the Nervous System.



General Structure of a Nerve Cell.

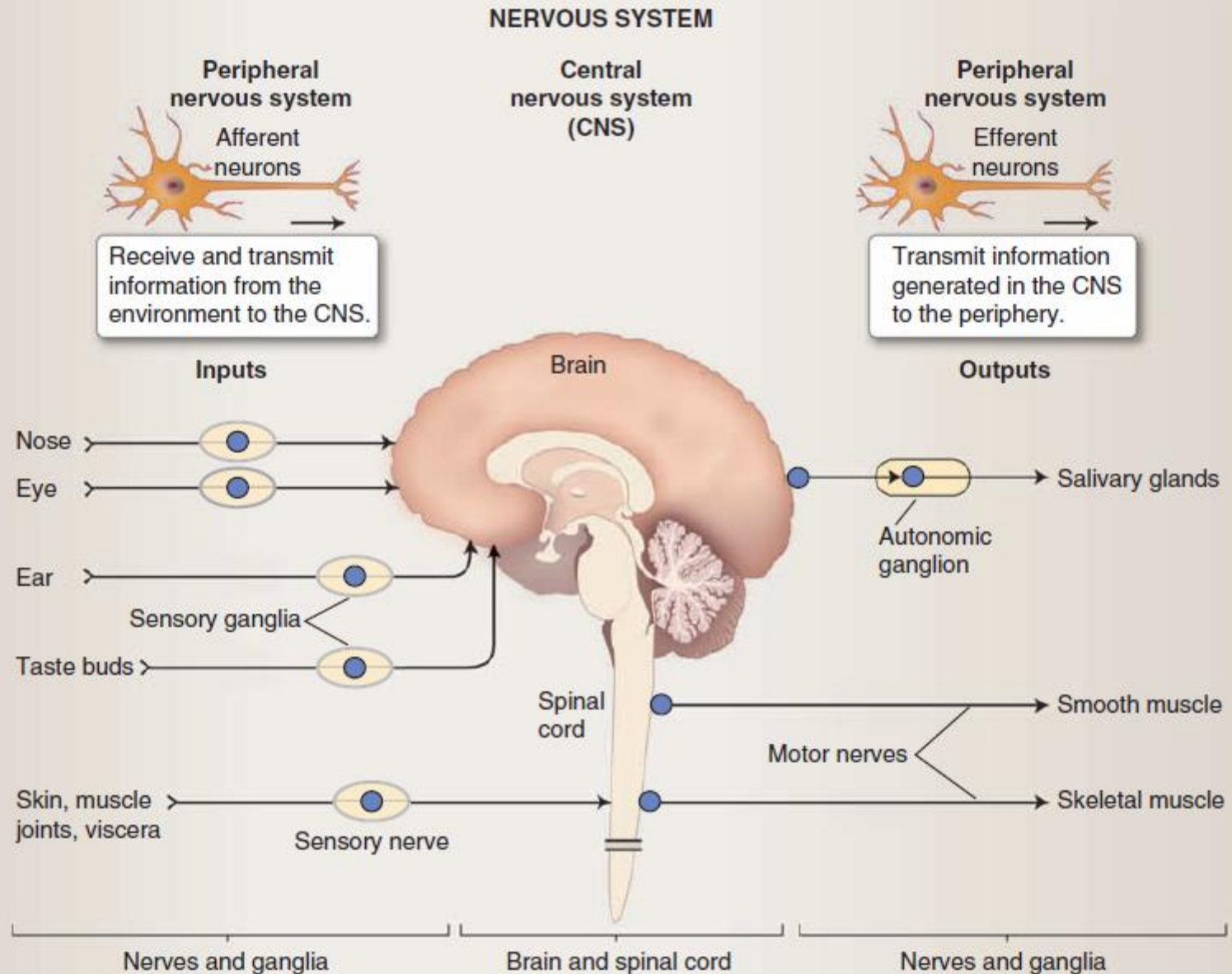
Objectives:

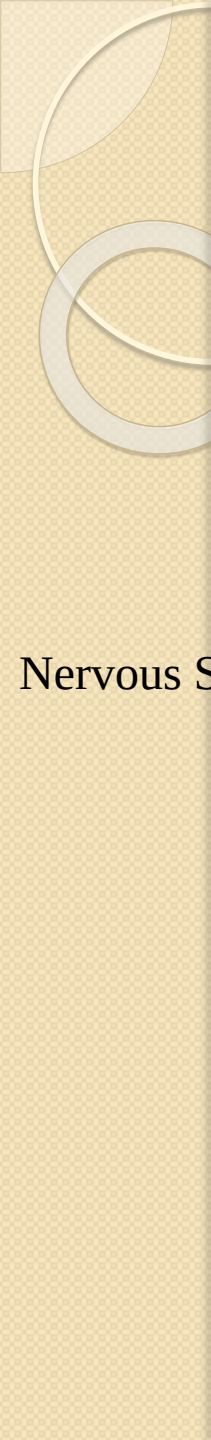
- ❖ *Apprehending* : Structural interaction and Biochemistry of Neurotransmitters
- ❖ Biochemical Aspects of CNS Disorder

Nervous System

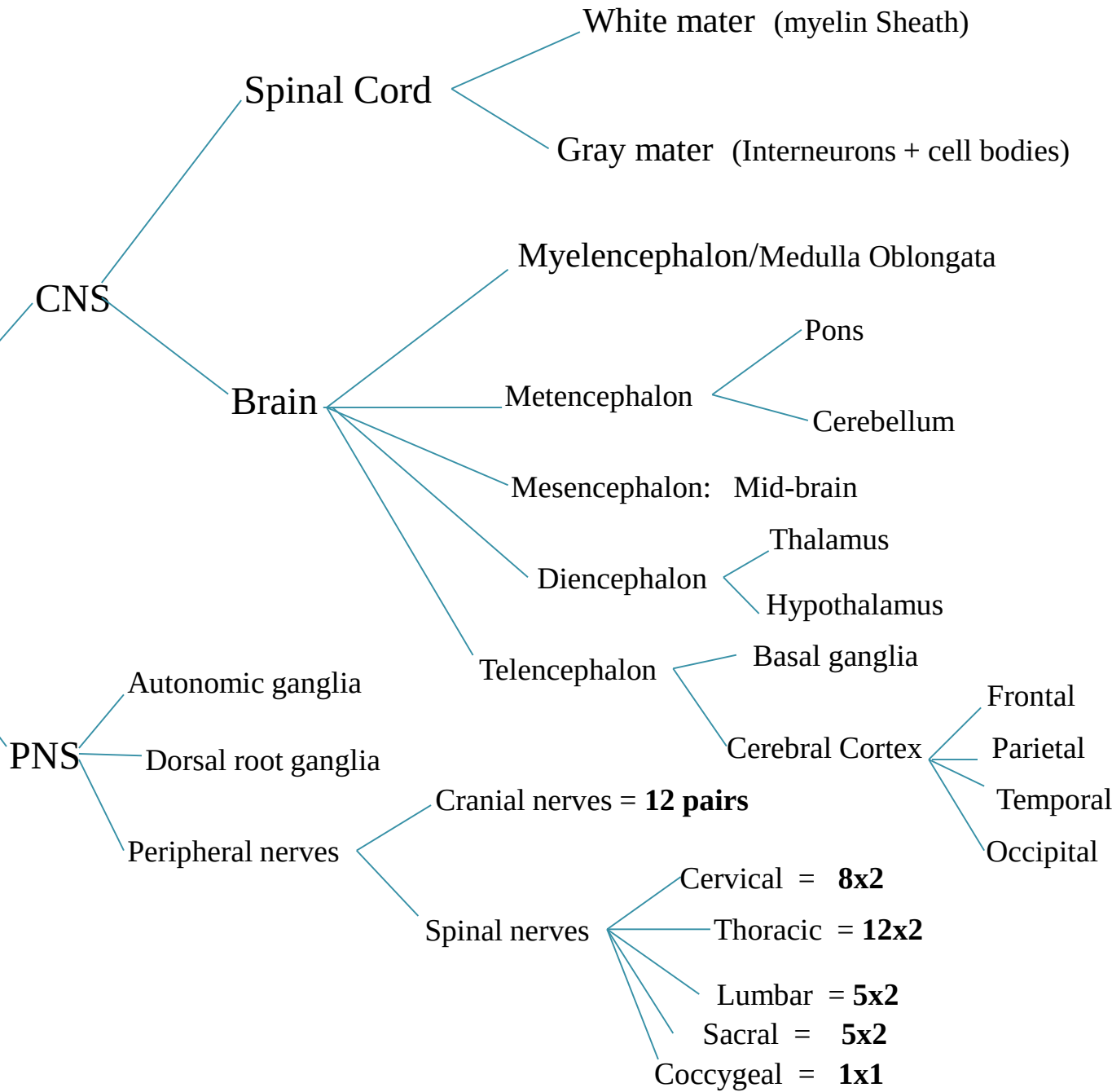
- ✚ Sensory detection (+ *sensory receptors*)
- ✚ Information processing (+ *Learning & memory*)
 - ✓ Transmission of information via neural networks.
 - ✓ Transformation of information... (neural integration)
 - ✓ Perception of sensory information...
 - ✓ Storage and retrieval of information (memory...)
 - ✓ Planning and implementation of motor commands
 - ✓ Thought processes and conscious awareness
 - ✓ Learning
 - ✓ Emotion and motivation...
- ✚ Expression of behavior...

The Biochemistry of Nerve Cell





Nervous System



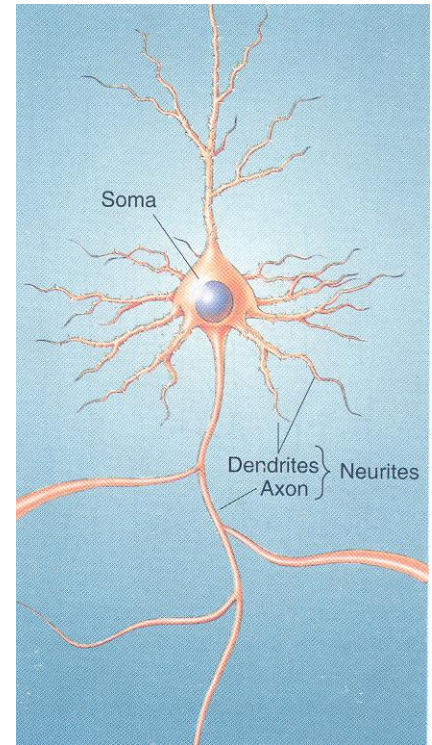
1.0. The Neuron

- a. Structural and functional unit of the nervous system.

Neuron Doctrine: Santiago Ramon y Cajal.

- b. Human brain: 10^{12} neurons.

10^{15} synapses.



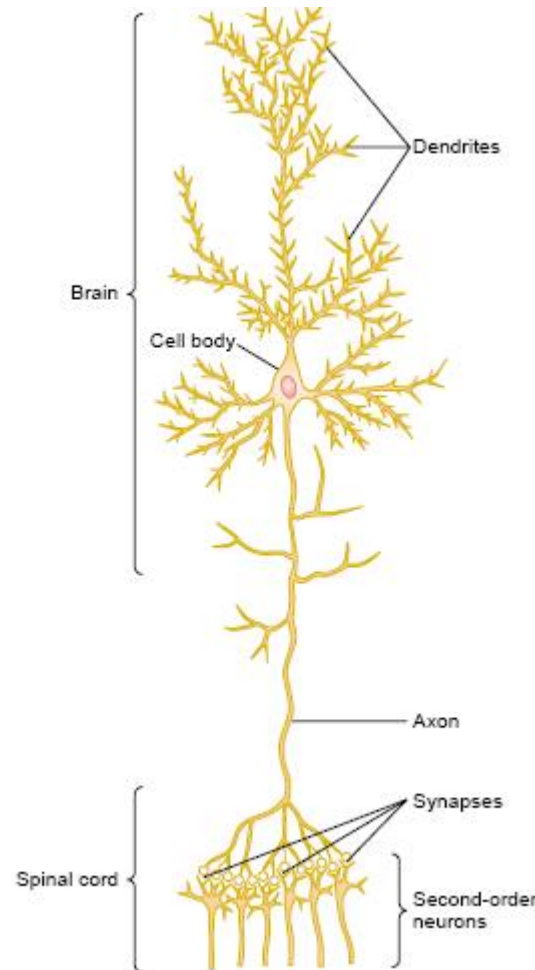
1.1. Cellular Structures and Function of a Neuron

a. Soma

b. Dendrites

c. Axon

d. Presynaptic Terminal



1. The nervous system has two classes of cells:
 - Neurons
 - Neuroglia.

2. The neuron is the functional unit of the nervous system.
 - Anatomical unit
 - Embryological unit
 - Metabolic unit
 - Information processing unit
 - *Law of Dynamic Polarization*

(‘in miniature the integrative capacity of the entire NS’)

3. Major functions of neurons:

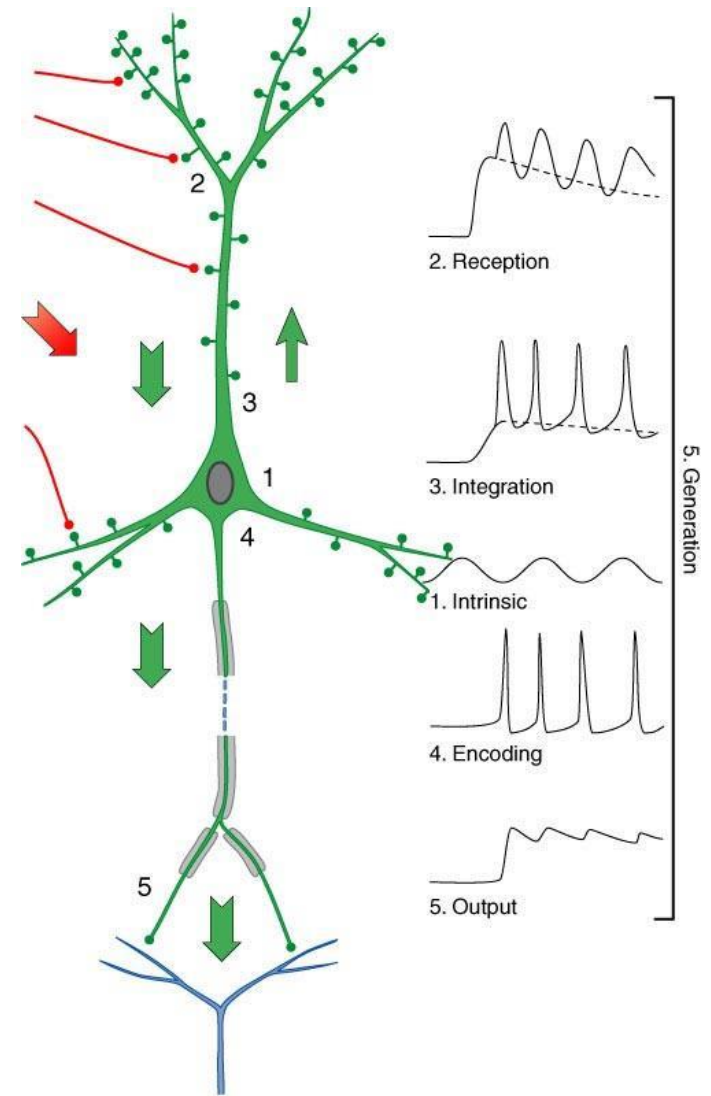
i. Impulse reception:

→ Internal environment
(*pH, Glucose, ABP, Temp...etc*)

→ External environment
(*Special senses*)

ii. Impulse conduction: APs (+ Graded)

iii. Impulse transmission: *Chemical* *Electrical*



4. Neuroglial cells

- a. Regulate the microenvironment of neurons
- b. Provide myelin sheath that speed conduction velocities.

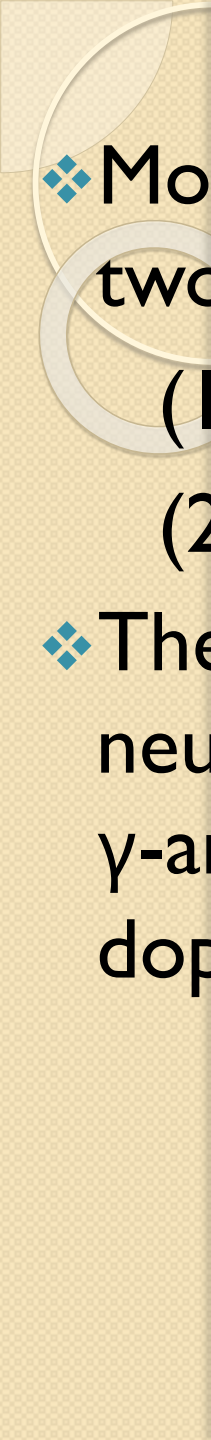
Chemical messengers

Three types of chemical messengers

- ❖ Hormones
- ❖ Neurotransmitters
- ❖ cytokines

Neurotransmitters

- ➡ Neurotransmitters are endogenous substances that act as chemical messengers by transmitting signals from a neuron to a target cell across a synapse.
- ➡ Chemical transmission is the major means by which nerves communicate with one another in the nervous system.
- ➡ The pre- and postsynaptic events are highly regulated and subject to use-dependent changes that are the basis for plasticity and learning in the CNS.
- ➡ Although direct electrical connections also occur, these account for transmission of information between nerves only in specialized cases.



❖ Molecules that serve as neurotransmitters fall into two basic structural categories:

- (1) small nitrogen-containing molecules, and
- (2) neuropeptides.

❖ The major small nitrogen-containing molecule neurotransmitters include serotonin, glutamate, γ -aminobutyric acid (GABA), glycine, acetylcholine, dopamine, norepinephrine, and histamine.

- ❖ Additional neurotransmitters that fall into this category include epinephrine, aspartate, CO and nitric oxide.
- ❖ In general, each neuron synthesizes only those neurotransmitters that it uses for transmission through a synapse or to another cell.
- ❖ The neuronal tracts are often identified by their neurotransmitter; for example, a dopaminergic tract synthesizes and releases the neurotransmitter **dopamine**.

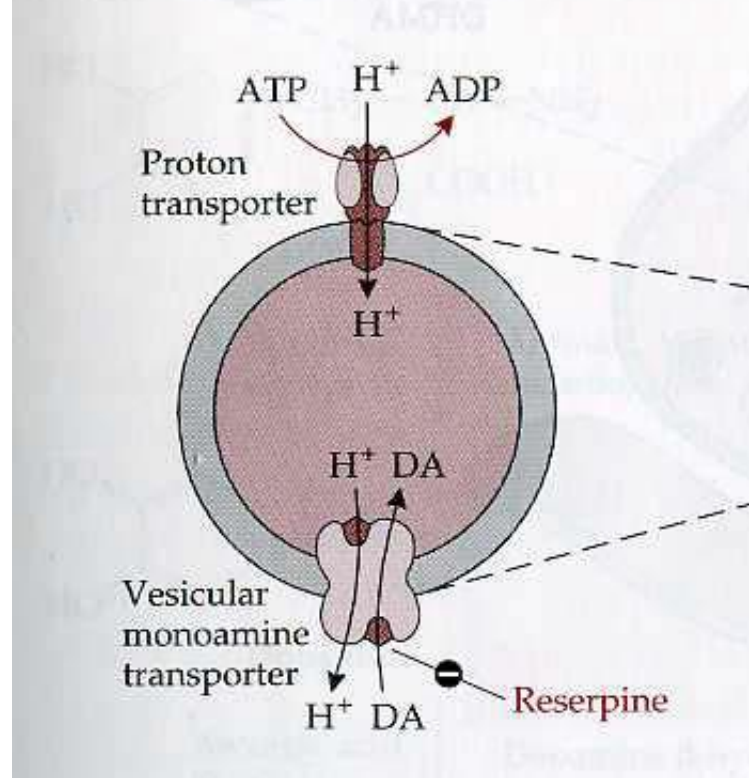
- ❖ Neuropeptides are usually small peptides, which are synthesized and processed in the CNS.
- ❖ Some of these peptides have targets within the CNS (such as endorphins, which bind to opioid receptors and block pain signals), whereas others are released into the circulation to bind to receptors on other organs (such as growth hormone and thyroid-stimulating hormone).

A. General Features of Neurotransmitter Synthesis

- ❖ A number of features are **common to the synthesis, secretion, and metabolism of most small nitrogen-containing neurotransmitters.**
- ❖ Most of these neurotransmitters are synthesized from **amino acids, intermediates of glycolysis and the TCA cycle in the cytoplasm of the presynaptic terminal.**

- ❖ The rate of synthesis is ***generally regulated to correspond to the rate of firing of the neuron.***
- ❖ Once synthesized, the neurotransmitters are transported into storage vesicles by an ***ATP-requiring pump linked with the proton gradient.***
- ❖ Release from the storage vesicle is triggered by the *nerve impulse that depolarizes the postsynaptic membrane and causes an influx of Ca^{2+} ions through **voltage-gated** calcium channels.*

Loading neurotransmitters into vesicles at the terminal



The proton pump hydrolyzes ATP and produces a proton gradient and membrane potential across the granule membrane. The amine transporter binds an uncharged amine and couples its transport INTO the vesicle with the H^+ exit OUT of it. The amine then becomes protonated at the acidic pH of the vesicle and is stuck inside the vesicle.

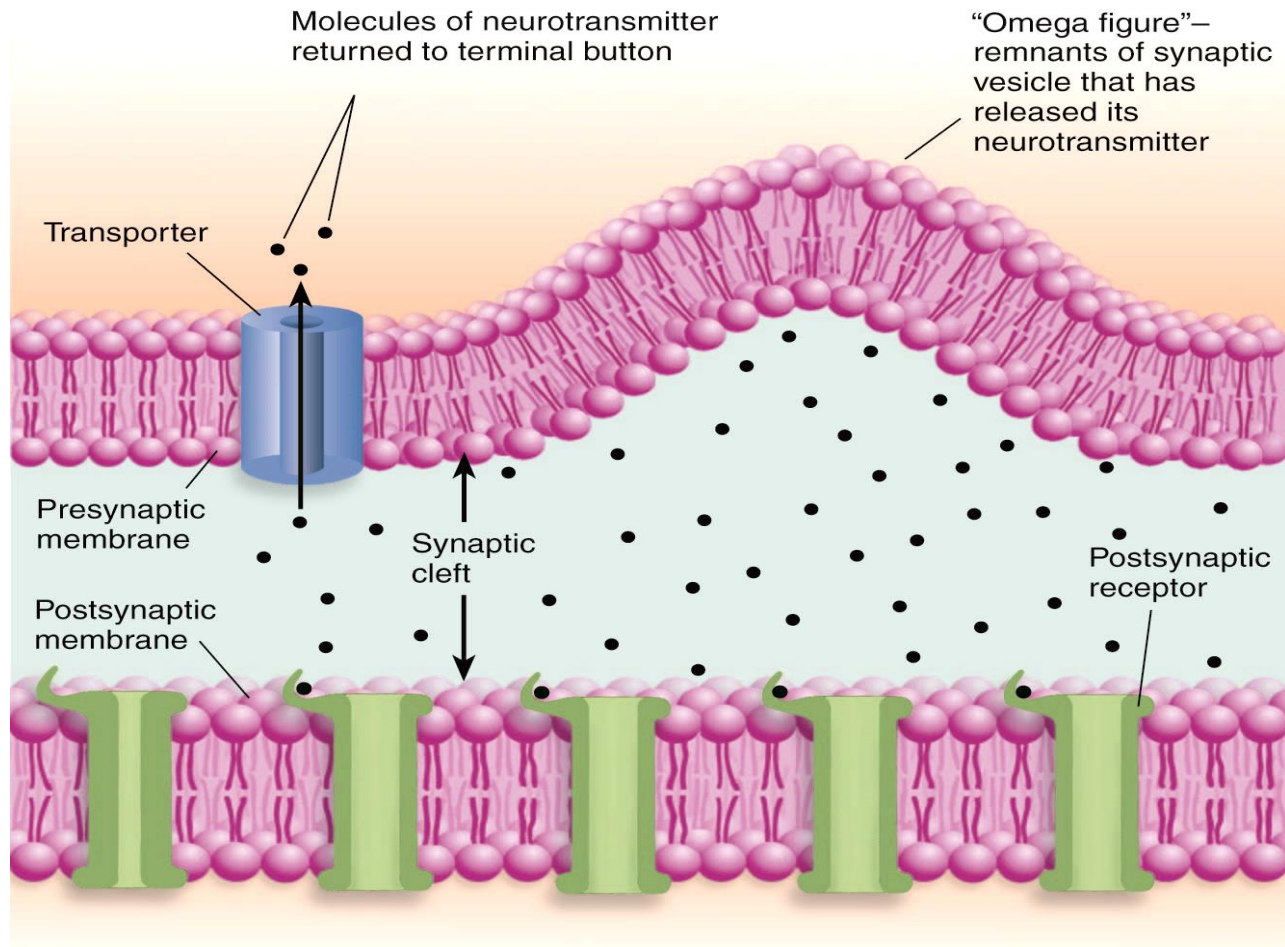
- ❖ The influx of Ca^{2+} promotes fusion of the vesicle with the synaptic membrane and release of the neurotransmitter into the synaptic cleft.
- ❖ The transmission across the synapse is completed by binding of the neurotransmitter to a receptor on the postsynaptic membrane
- ❖ The action of the neurotransmitter is terminated through;
 - ✓ **reuptake into the presynaptic terminal,**
 - ✓ **uptake into glial cells,**
 - ✓ **diffusion away from the synapse, or**
 - ✓ **enzymatic inactivation.**

Removal of neurotransmitter from the synapse

1.Reuptake

2.Enzymatic deactivation

3.Diffusion





❖ The enzymatic **inactivation** may occur in the **postsynaptic terminal**, the **presynaptic terminal** or an adjacent **astrocyte** microglia cell, or in **endothelial cells** in the brain capillaries.

❖ *Not all neurotransmitters* exhibit all of these features. Nitric oxide, because it is a gas, is an exception to most of these generalities.

❖ Some neurotransmitters are synthesized and secreted by both **neurons** and **other cells** (e.g., epinephrine, serotonin, histamine).



RECAPITULATIONS

Properties of Neurotransmitters

1. Synthesis in the presynaptic neurons.
2. Localized to vesicles in the presynaptic neurons
3. Released from the presynaptic neurons under physiological condition.
4. Rapidly removed from the synaptic cleft by uptake or degradation.
5. Presence of receptor on the post synaptic neurons.
6. Binding to the receptor elicits a biological response.

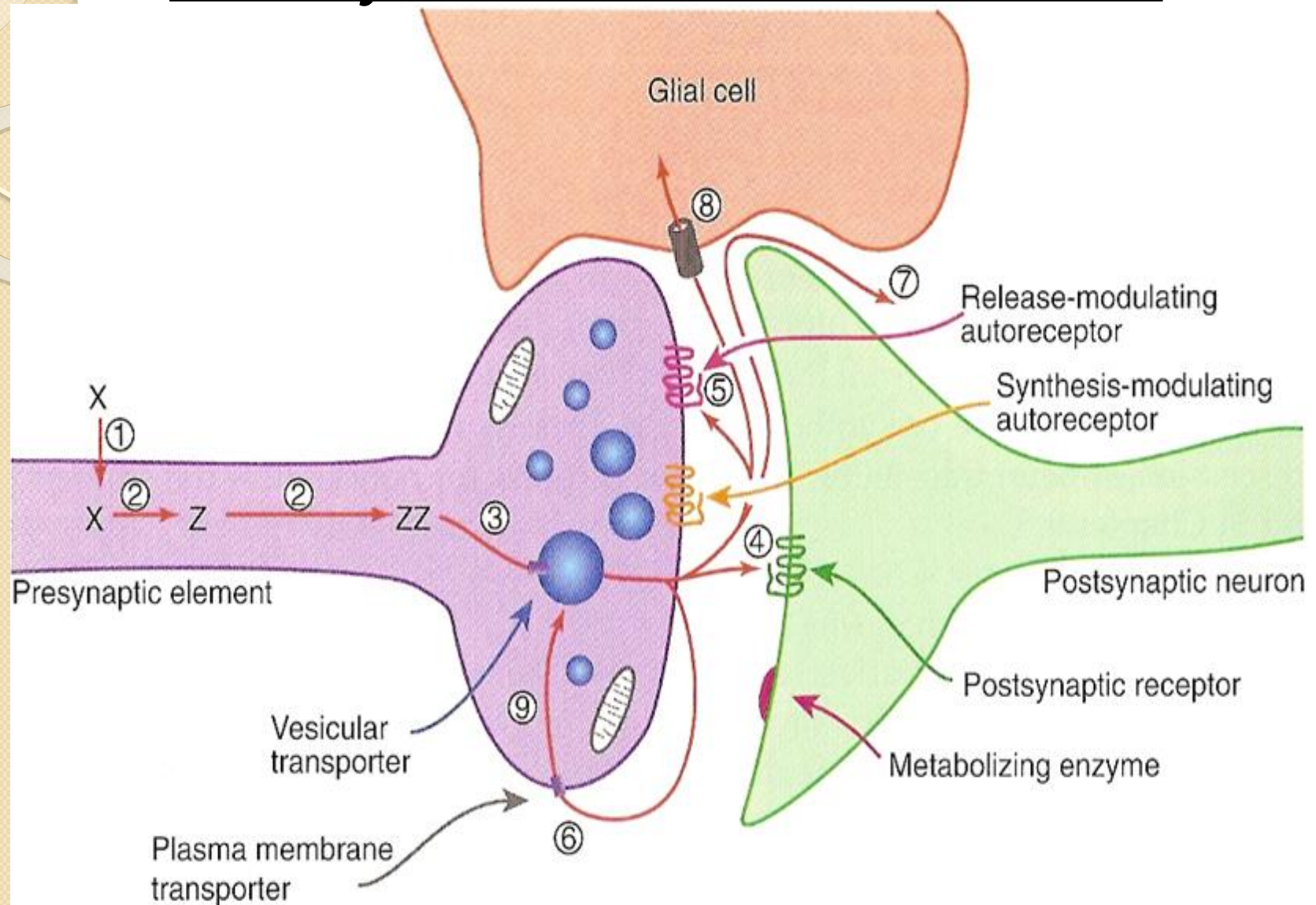
I. It must be synthesized in the presynaptic neuron -

Proof = Biochemical evidence (the biosynthetic enzymes are present in the presynaptic neuron)

Neurotransmitter Life Cycle

- ❑ Synthesis of NT's from precursor molecules
- ❑ Storage of NT's in vesicles
- ❑ Release of NT's into synaptic space
- ❑ Actions at postsynaptic receptors
- ❑ Reuptake by presynaptic neuron
- ❑ Metabolism/inactivation of NT's
- ❑ Removal of NT's via diffusion and glia
- ❑ Down-regulation of NT synthesis & release via presynaptic autoreceptors (feedback)

Life Cycle of a Neurotransmitter



Classification of NTs based on overall activity

EXCITATORY

Acetylcholine

Aspartate

Dopamine

Histamine

Norepinephrine

Epinephrine

Glutamate

Serotonin

INHIBITORY

GABA

Glycine

Classification NT based on size:

□ small-molecule neurotransmitters

- Glutamate
- Glycine
- GABA
- AcetylCholine
- Serotonin
- Histamin
- biogenic amines(EP,NEP,Dopamine)

□ Neuropeptides:

- melanocyte-stimulating hormone, adrenocorticotropin & β -endorphin

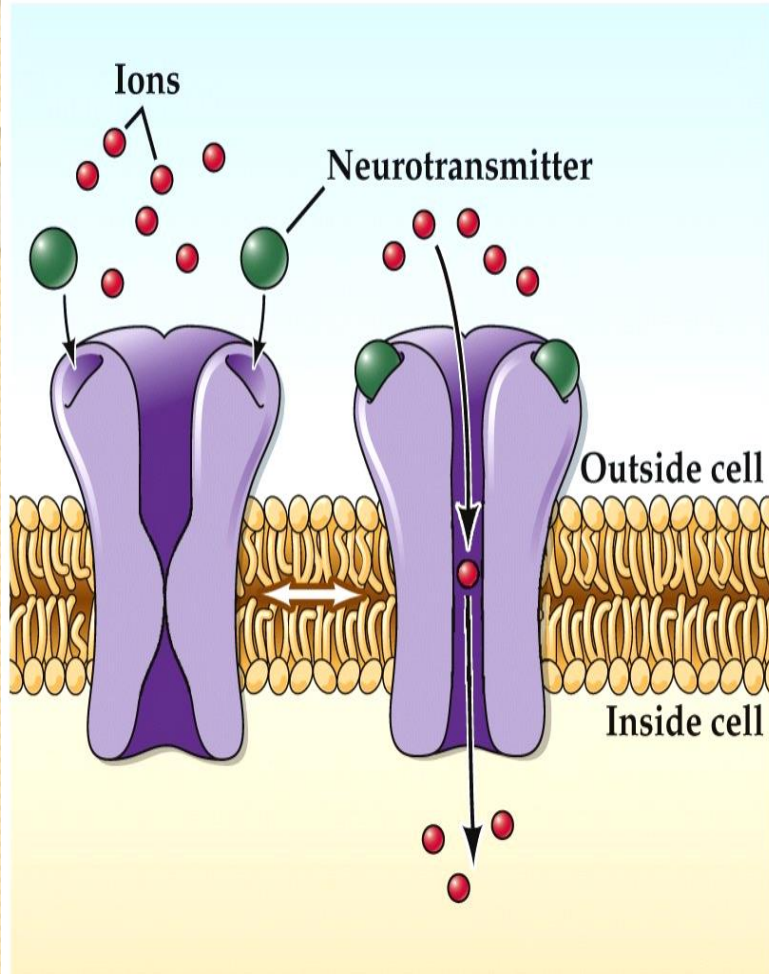
Source & Biosynthesis Sites of NTs

Transmitter Molecule	Source	Site of Synthesis
Acetylcholine	Choline	CNS, parasympathetic nerves
Serotonin 5-Hydroxytryptamine (5-HT)	Tryptophan	CNS, chromaffin cells of the gut, enteric cells
GABA	Glutamate	CNS
Glutamate		CNS
Aspartate		CNS
Glycine		spinal cord
Histamine	Histidine	hypothalamus
Epinephrine synthesis pathway	Tyrosine	adrenal medulla, some CNS cells
Norpinephrine synthesis pathway	Tyrosine	CNS, sympathetic nerves
Dopamine synthesis pathway	Tyrosine	CNS
Adenosine	ATP	CNS, periperal nerves
ATP		Sympathetic, sensory and enteric nerves
Nitric oxide, NO	Arginine	CNS, gastrointestinal tract

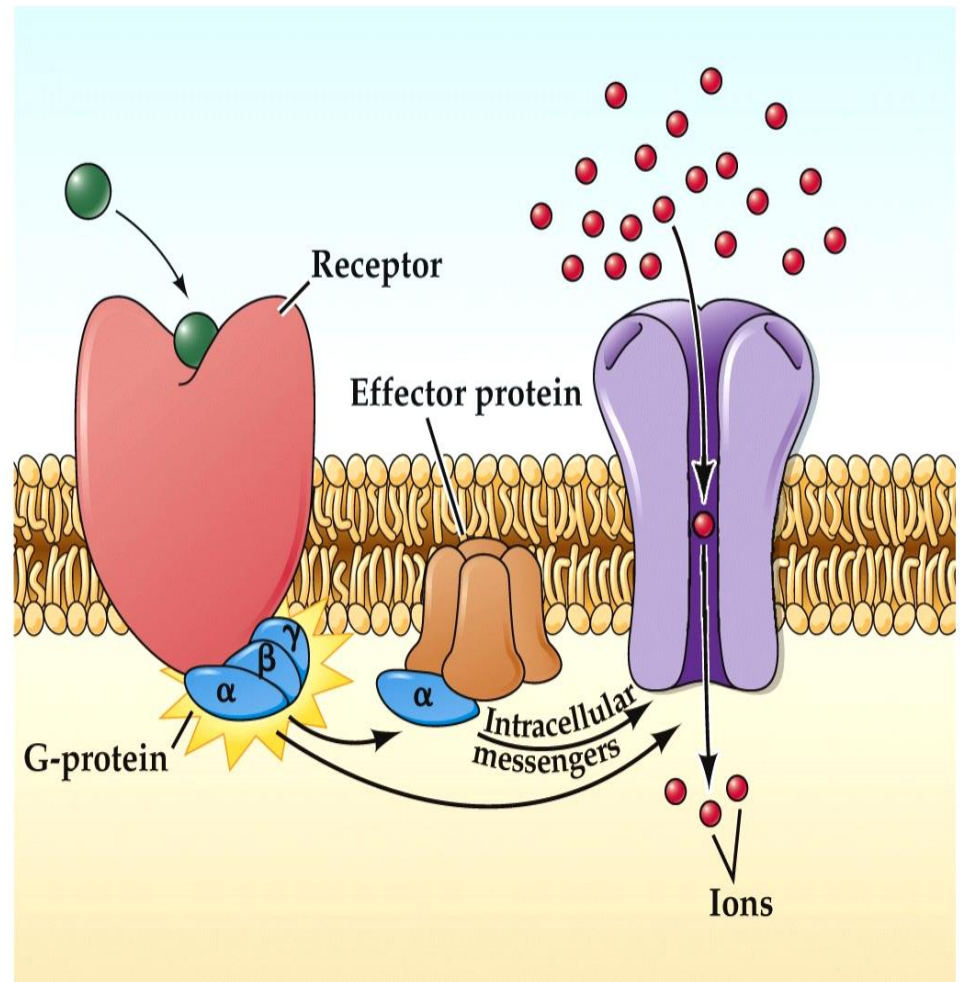


Receptors of Neurotransmitters

(A) Ligand-gated ion channels



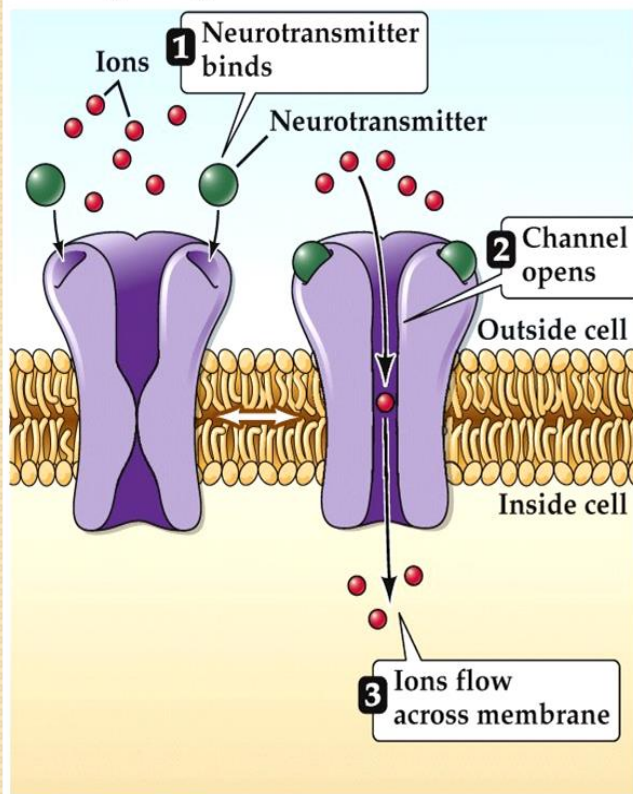
(B) G-protein-coupled receptors



Inotropic

- ❖ Fast effect (milliseconds)
- ❖ Only opens transmitter dependent channels

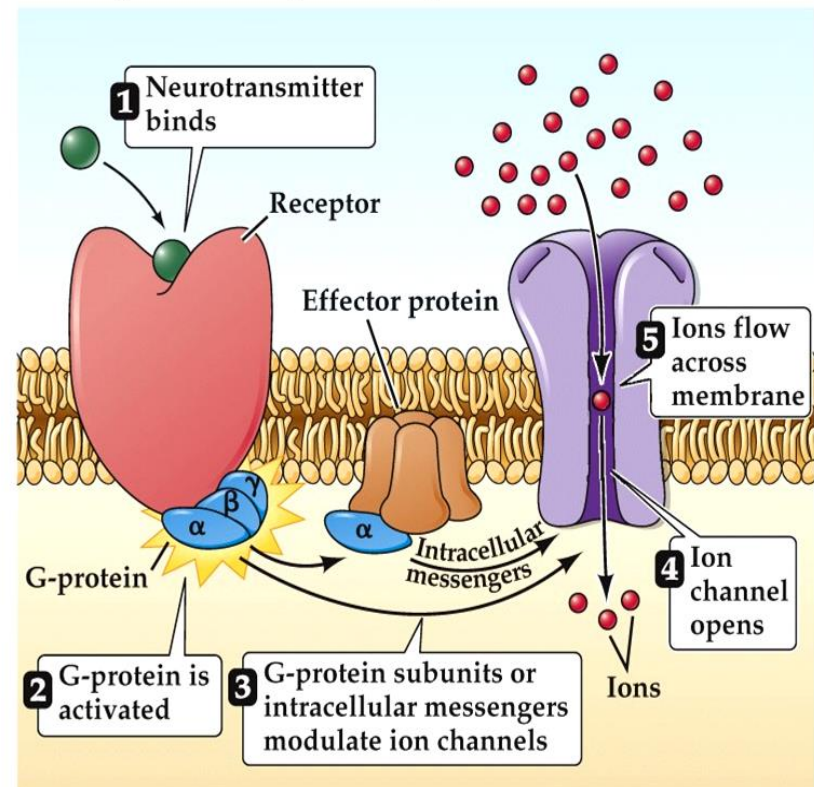
(A) Ligand-gated ion channels

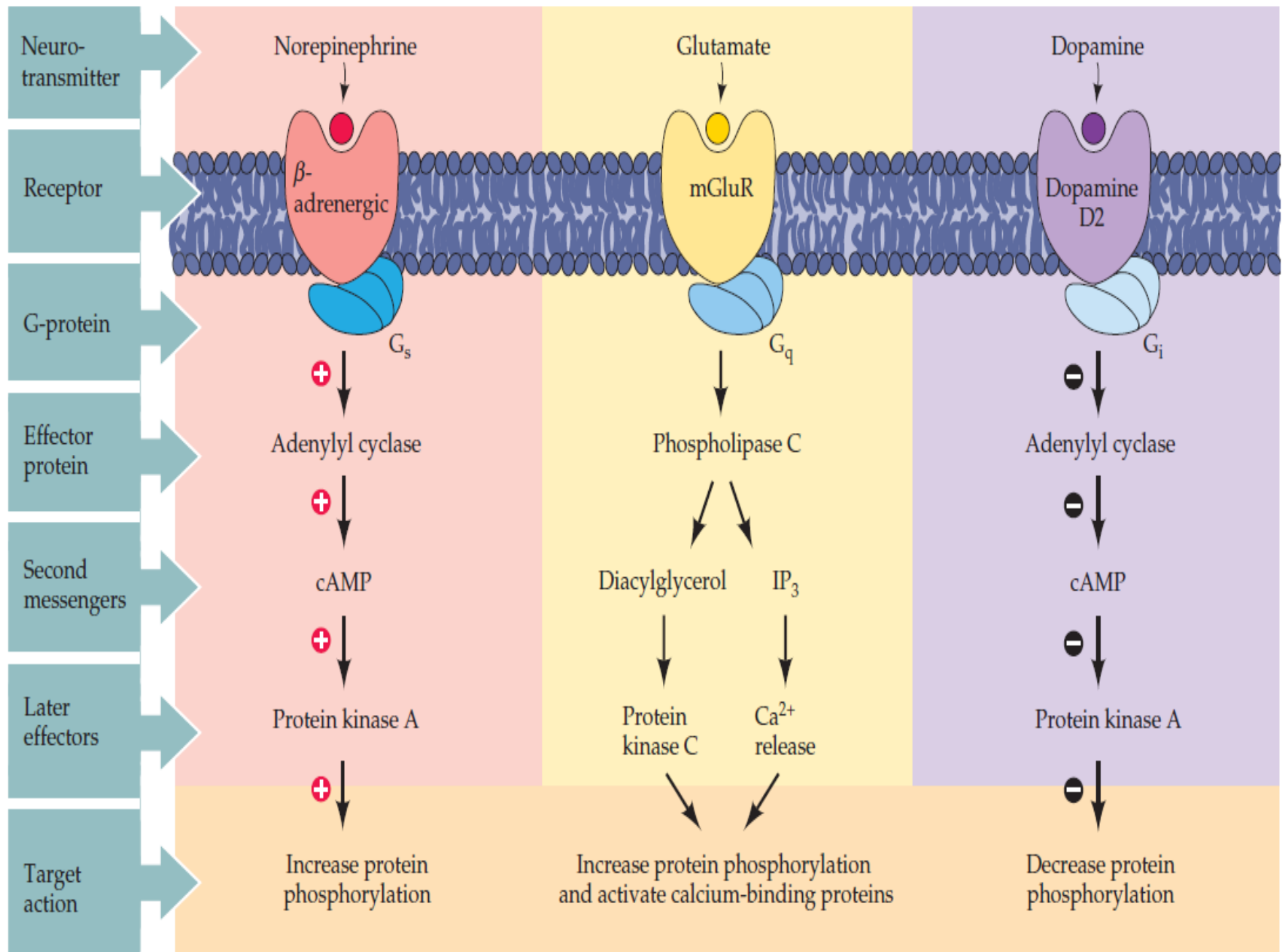


Metabotropic

- ❖ Slow effect (seconds/minutes)
- ❖ may open/close transmitter dependent channels

(B) G-protein-coupled receptors

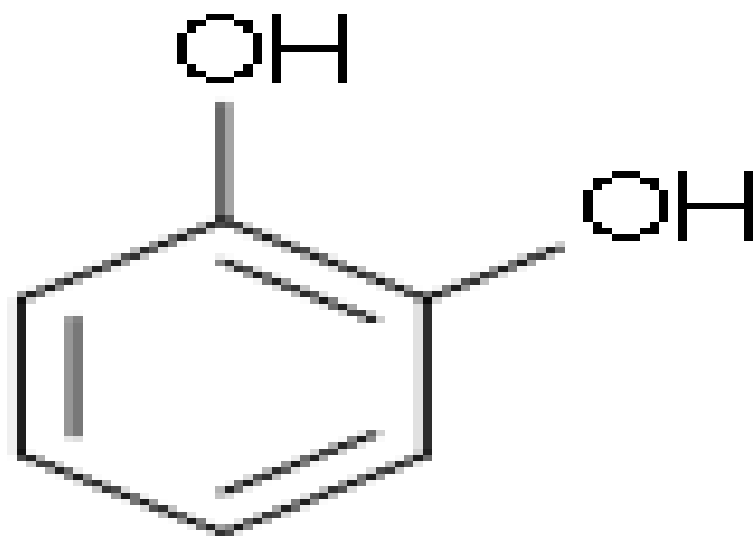




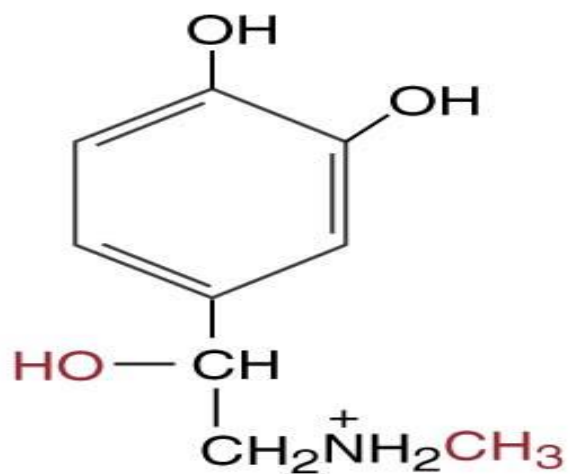
Dopamine, Norepinephrine, and Epinephrine

I. SYNTHESIS OF THE CATECHOLAMINE NEUROTRANSMITTERS

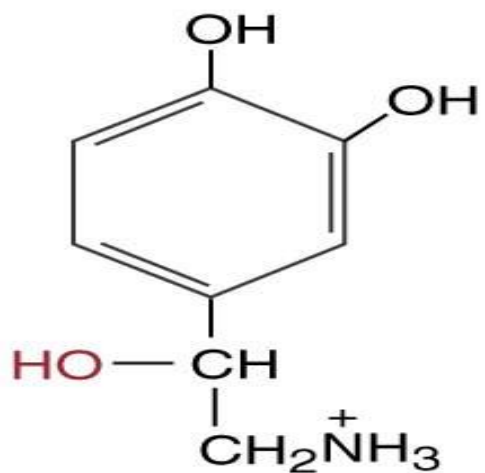
- ❖ These three neurotransmitters are synthesized in a common pathway from the amino acid ***L-tyrosine***.
- ❖ Tyrosine is *supplied in the diet or is synthesized in the liver from the essential amino acid* ***phenylalanine*** by ***phenylalanine hydroxylase***.



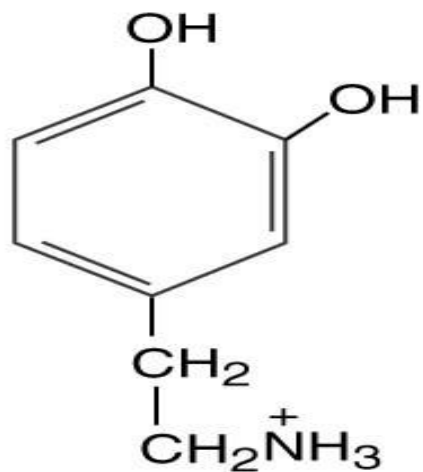
Catechol



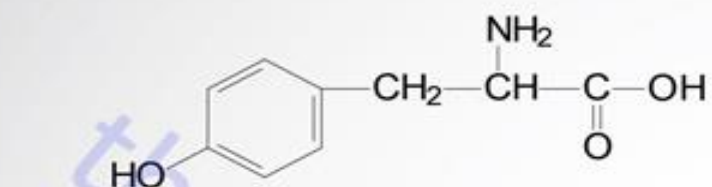
Epinephrine



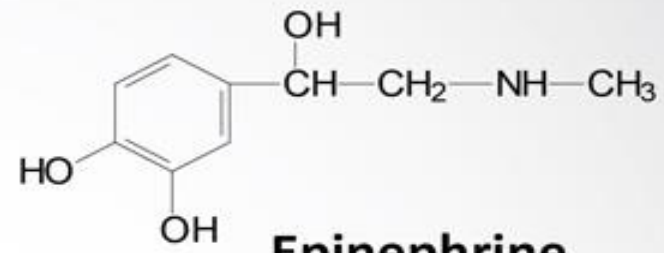
Norepinephrine



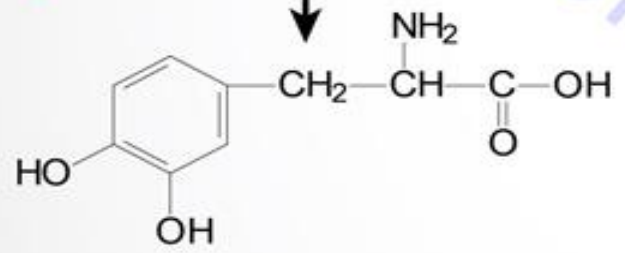
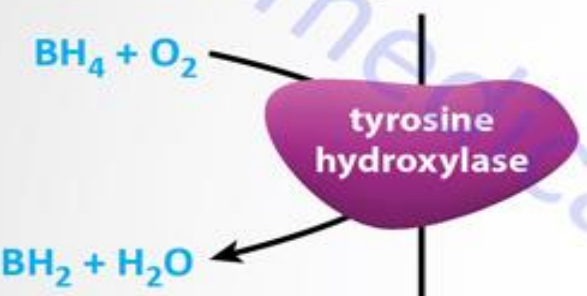
Dopamine



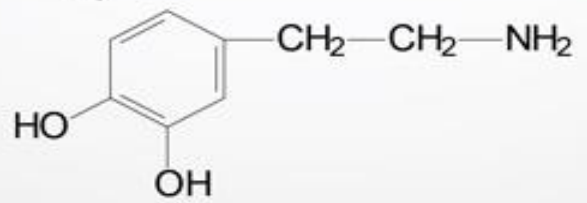
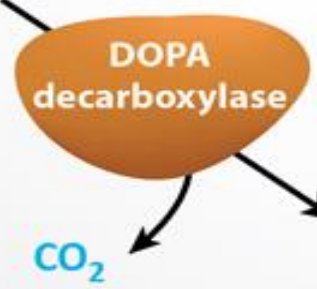
Tyrosine



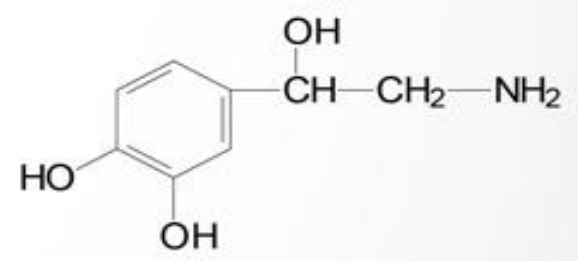
Epinephrine



DOPA

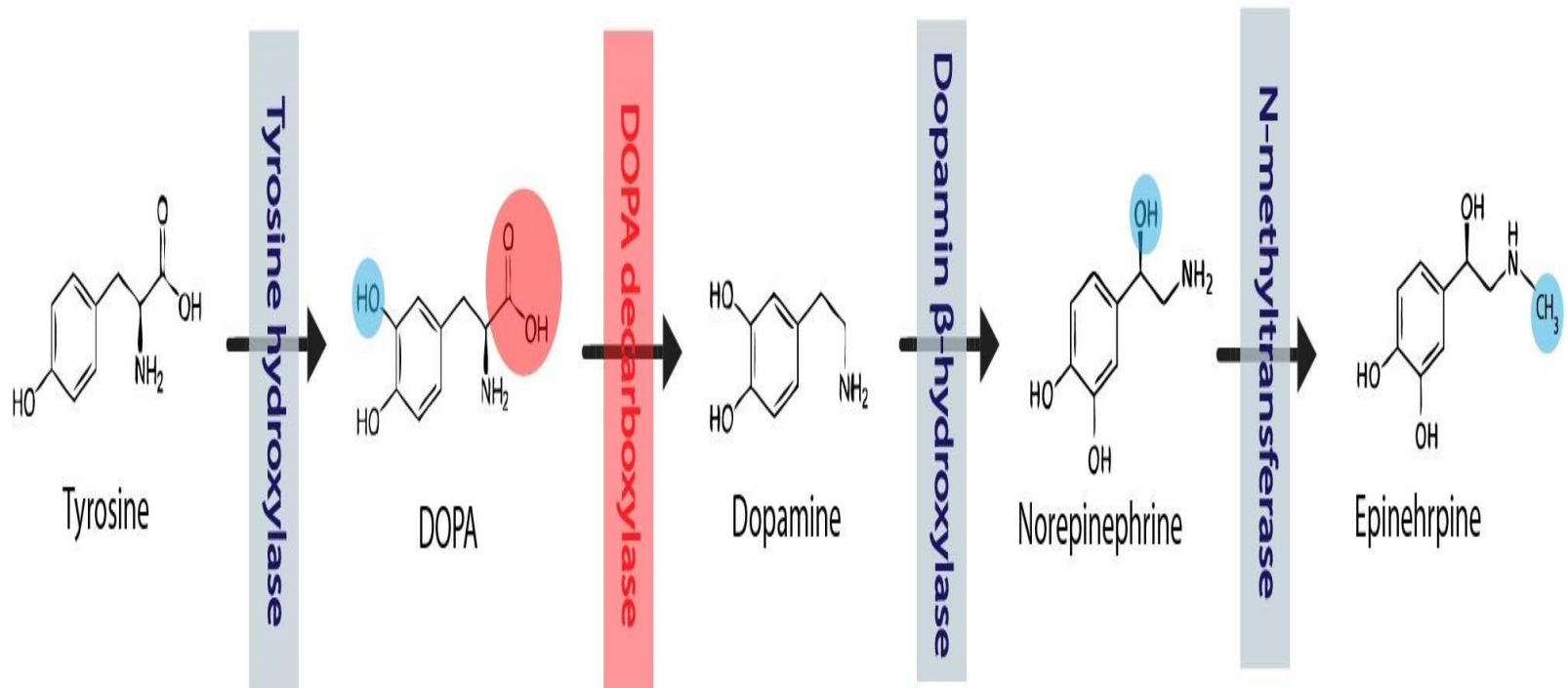


Dopamine



Norepinephrine



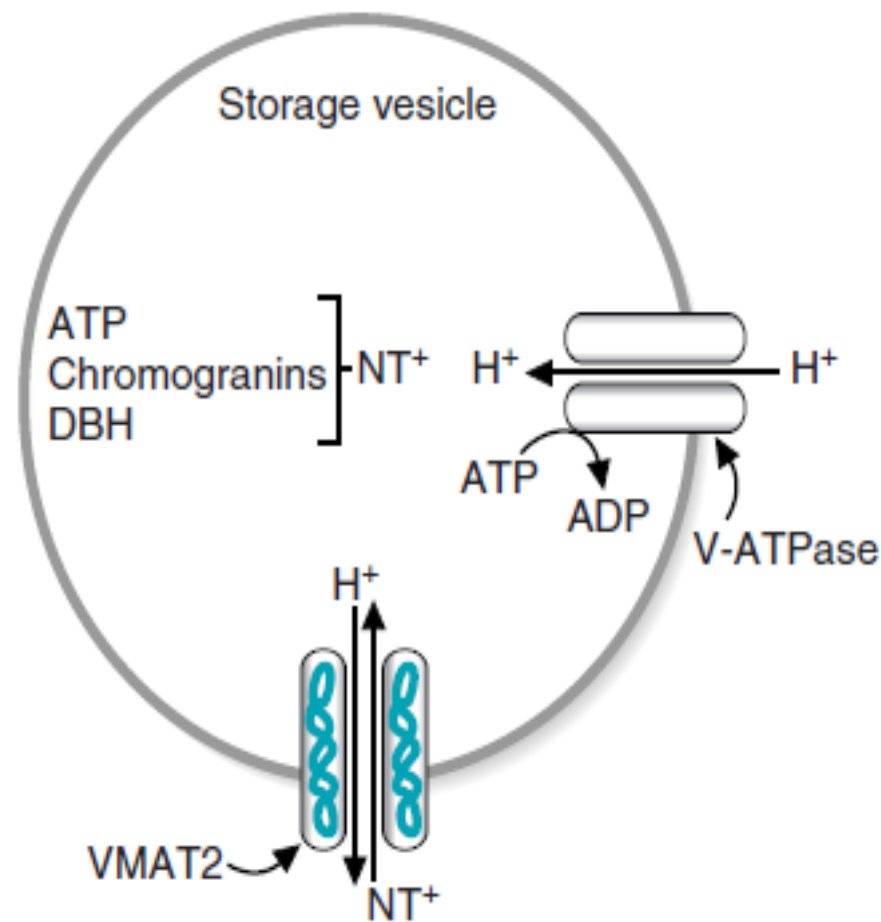


- 1) Ring hydroxylation
- 2) Decarboxylation – formation of dopamine
- 3) Side chain hydroxylation – formation of noradrenaline
- 4) N-methylation – formation of adrenaline

2. STORAGE AND RELEASE OF CATECHOLAMINES

- ❖ only low concentrations of catecholamines are free in the cytosol, whereas high concentrations are found within the storage vesicles.
- ❖ Conversion of tyrosine to L-DOPA and that of L-DOPA to dopamine occurs *in the cytosol*.
- ❖ Dopamine is then taken up into the storage vesicles.
- ❖ In norepinephrine-containing neurons, the final hydroxylation reaction occurs within the vesicles.
- ❖ The catecholamines are transported into vesicles by the protein **VMAT2** (**vesicle membrane associated transporter 2**).

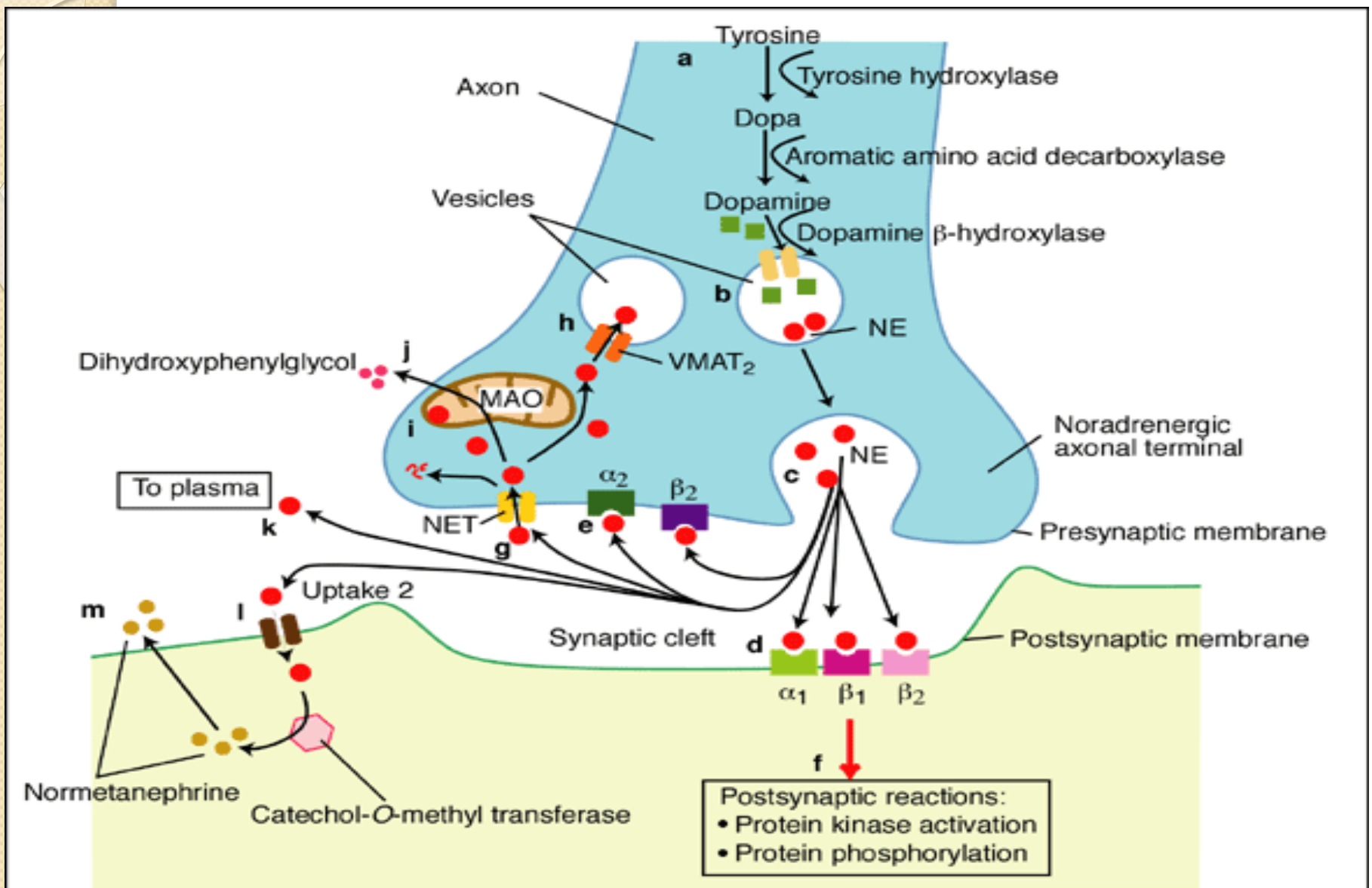
- ❖ The intravesicular concentration of catecholamines is approximately 0.5M, roughly 100 times the cytosolic concentration.
- ❖ In the vesicles, the catecholamines exist in a complex with **ATP** and **acidic proteins** known as **chromogranins**.
- ❖ Chromogranins are required for the biogenesis of the secretory vesicle. When released from the vesicle, chromogranins can be proteolytically clipped to form bioactive peptides.
- ❖ Elevated levels of **chromogranins** in the circulation may be found in patients harboring **neuroendocrine tumors**, such as a **pheochromocytoma**



- Fig. Transport of catecholamines into storage vesicles.** This is a secondary active transport based on the generation of a proton gradient across the vesicular membrane. NT^+ positively charged neurotransmitter (catecholamine); DBH dopamine -hydroxylase; VMAT2 vesicle membrane transporter 2; V-ATPase vesicular ATPase.

3. INACTIVATION AND DEGRADATION OF CATECHOLAMINE NEUROTRANSMITTERS

- ✚ The action of catecholamines is terminated through *reuptake into the presynaptic terminal and diffusion away from the synapse.*
- ✚ Degradative enzymes are present in the presynaptic terminal, and in adjacent cells, including glial cells and endothelial cells.

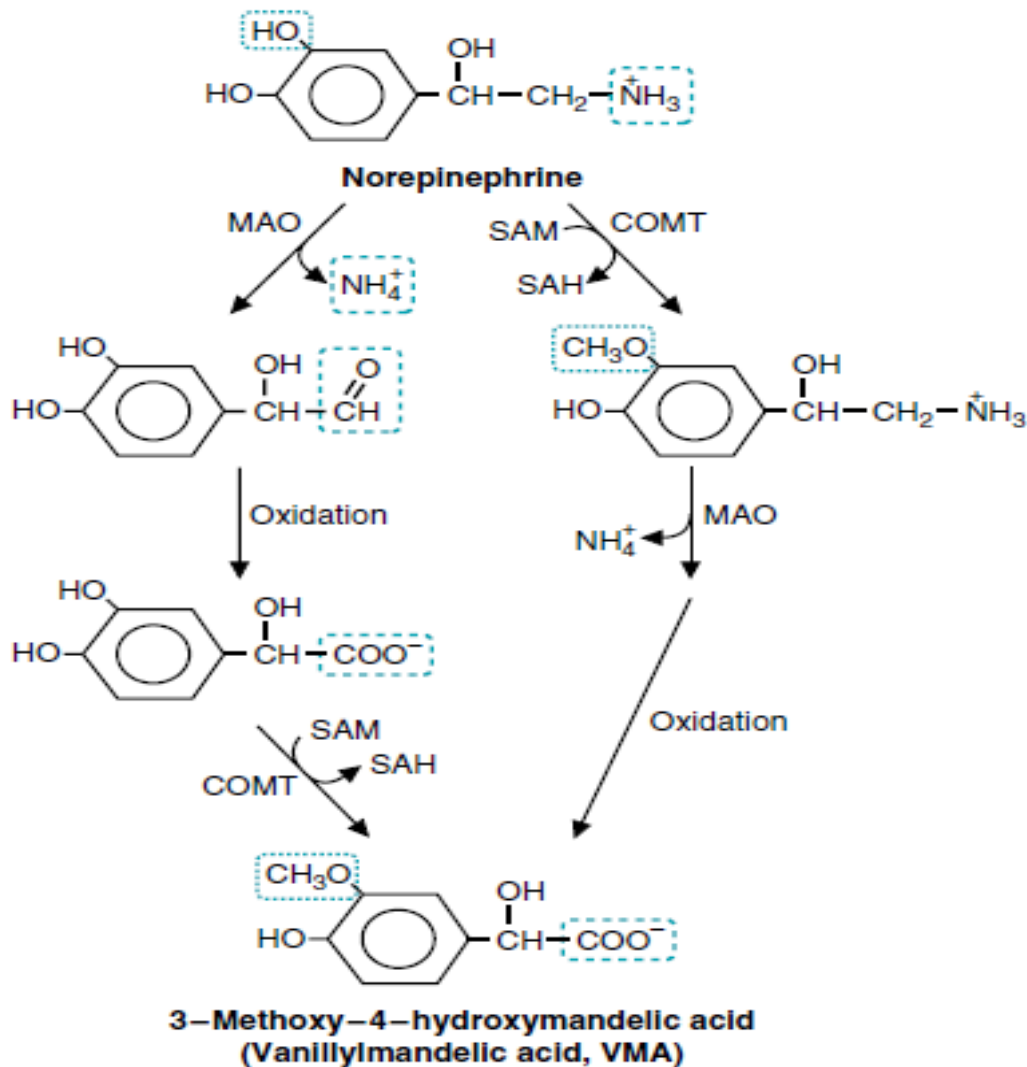


Release and re-uptake of norepinephrine

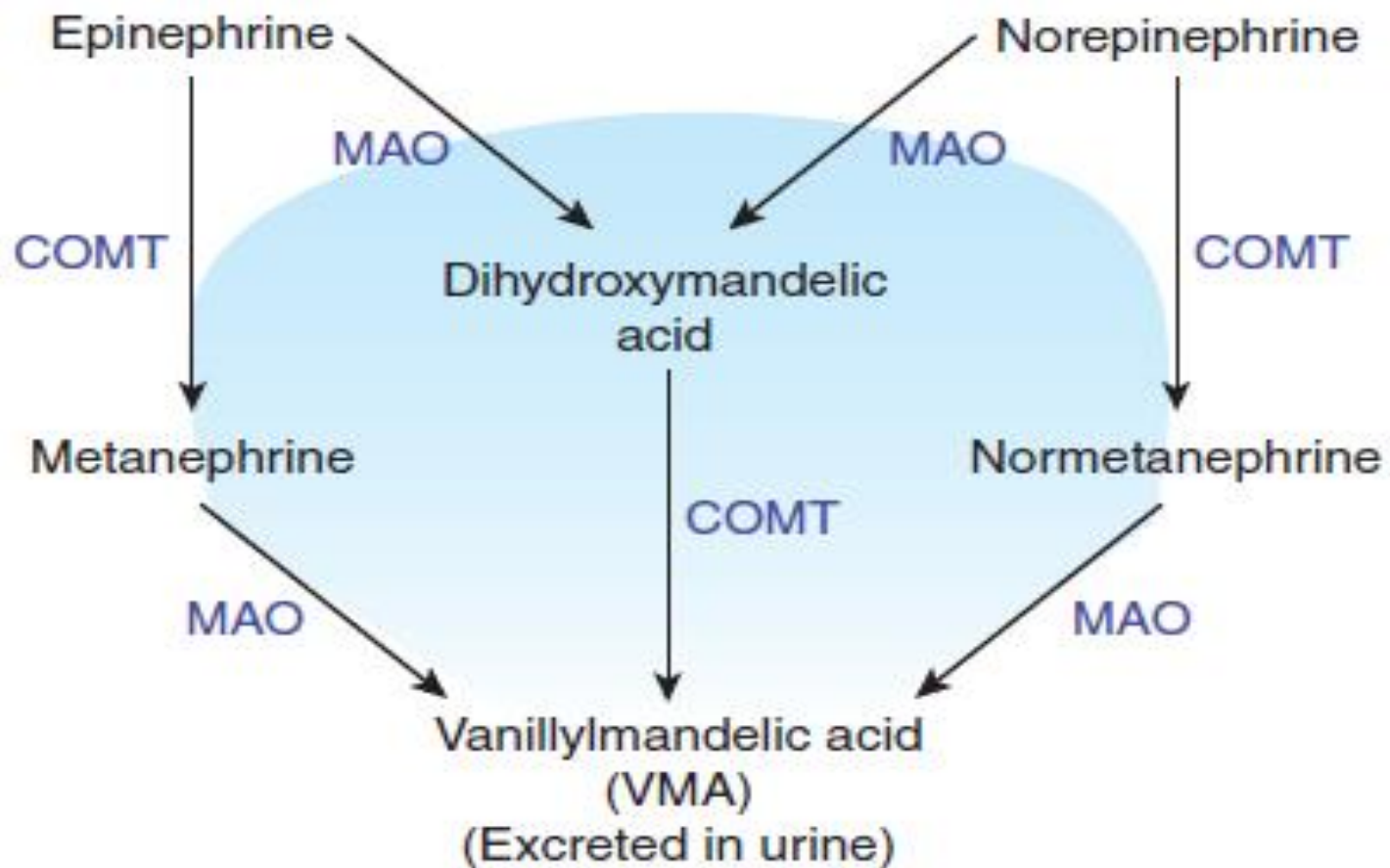
Two of the major reactions in the process of inactivation and degradation of catecholamines are **catalyzed by monamine oxidase (MAO) and catechol-O-methyltransferase (COMT).**

MAO is present on the **outer mitochondrial membrane** of many cells and oxidizes the carbon containing the amino group to an **aldehyde**, thereby releasing **ammonium ion**.

- ✚ In the presynaptic terminal, MAO inactivates catecholamines *that are not protected in storage vesicles*. (Thus, drugs that deplete storage vesicles indirectly increase catecholamine degradation.)
- ✚ There are two isoforms of MAO with different specificities of action:
 - MAO-A preferentially deaminates **norepinephrine** and **serotonin**, whereas
 - MAO-B acts on a wide spectrum of phenylethylamines (phenylethyl refers to a $-CH_2-$ group linked to a phenyl ring).
- ✚ MAO in the liver and other sites protects against the ingestion of dietary biogenic amines such as the tyramine found in many cheeses.



- Fig. Inactivation of catecholamines.**
Methylation and oxidation may occur in any order.
 Methylated and oxidized derivatives of norepinephrine and epinephrine are produced, and 3-methoxy-4-hydroxymandelic acid is the final product. These compounds are excreted in the urine. MAO monoamine oxidase; COMT catechol O-methyltransferase; SAM S-adenosylmethionine; SAH S-adenosylhomocysteine.



epinephrine and norepinephrine to vanillylmandelic acid (VMA).

Comment

- **Tyramine** is a degradation product of tyrosine that can lead to **headaches**, **palpitations**, **nausea** and **vomiting**, and elevated **ABP** if present in large quantities.
- **Tyramine** mimics norepinephrine and binds to norepinephrine receptors, stimulating them. Tyramine is inactivated by **MAO-A**, but if a person is taking a MAO inhibitor, foods containing tyramine should be avoided.

- ❖ **COMT** is also found in many cells, including the erythrocyte. *It works on a broad spectrum of extraneuronal catechols and those that have diffused away from the synapse.*
- ❖ **COMT** transfers a methyl group from SAM to a hydroxyl group on the catecholamine or its degradation product .
- ❖ Because the inactivation reaction requires **SAM**, it is indirectly dependent on **vitamins B12** and **folate**.

- ❖ The action of MAO and COMT can occur in almost any order, thereby resulting in a large number of degradation products and intermediates, many of which appear in the urine.
- ❖ Cerebrospinal homovanillylmandelic acid (**HVA**) is an indicator of dopamine degradation.

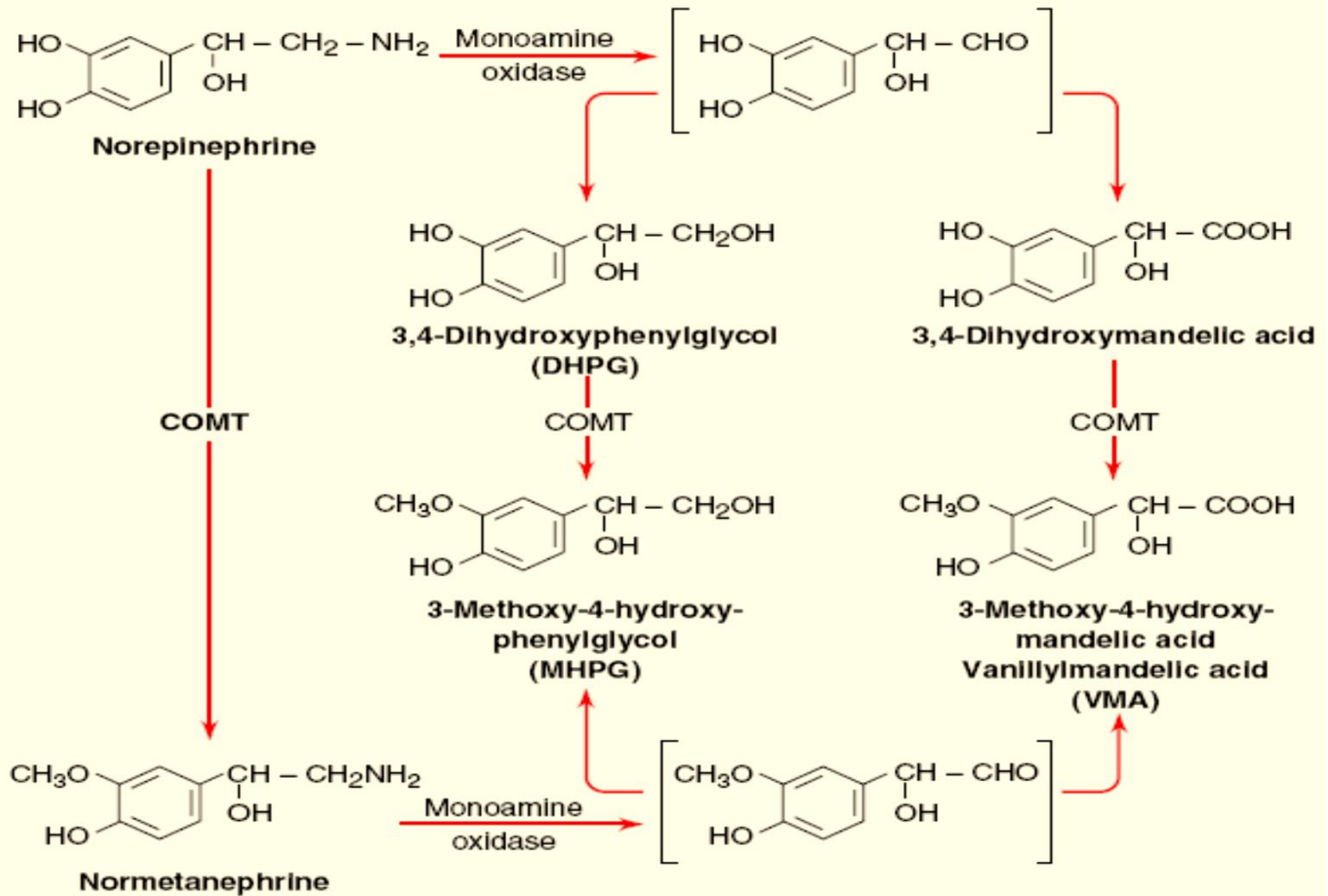
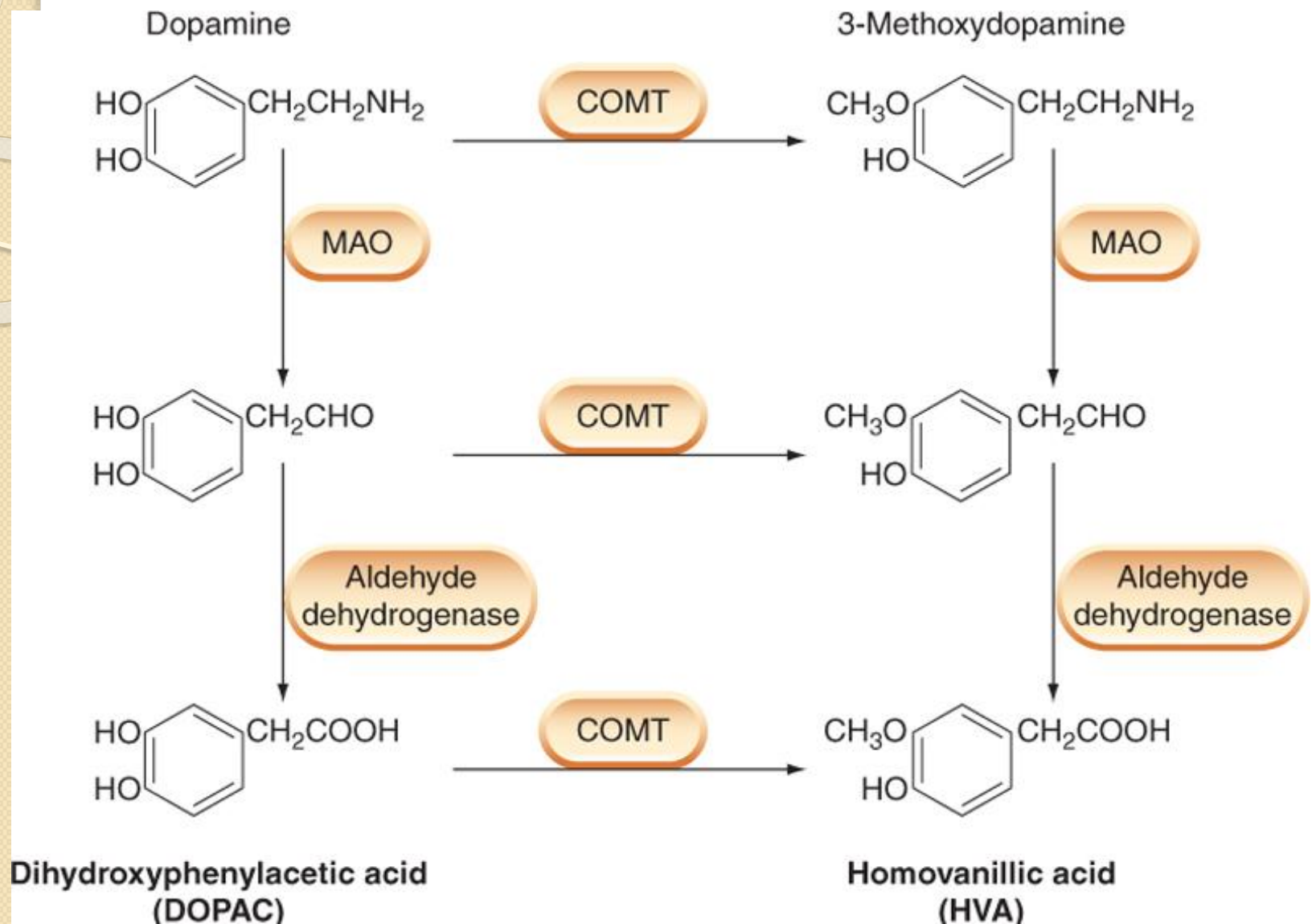
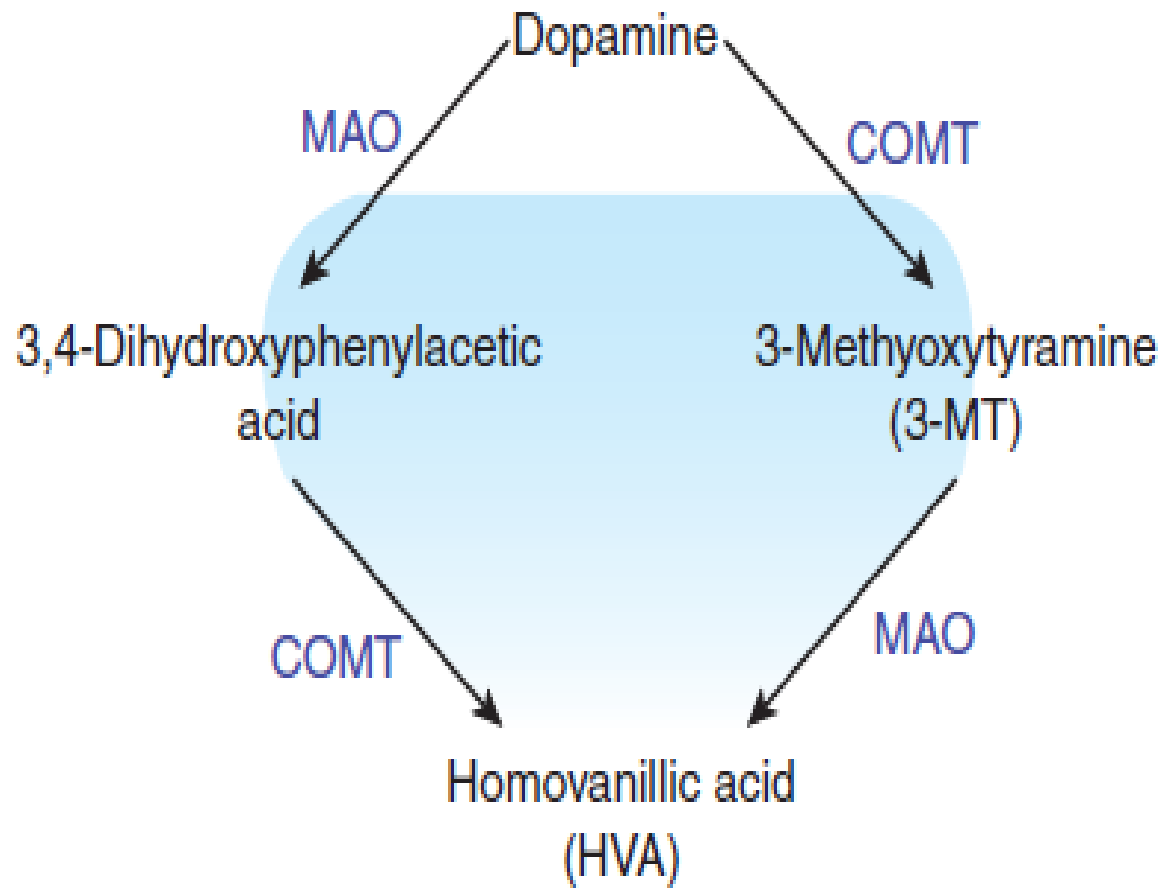


Fig. Pathways of norepinephrine degradation. COMT, catechol-O-methyltransferase.



Dopamine Metabolism



dopamine to homovanillic acid (HVA),

Catecholamine Receptors

- ➡ The brain contains many different catecholamine receptors.
- ➡ The effects of dopamine are mediated through interaction with five different receptors; usually referred to as D1-like (D1, D5) and D2-like (D2, D3, D4).
- ➡ The effects of norepinephrine and epinephrine are mediated through nine distinct receptors grouped into three families (α_1 , α_2 , β), each containing three subtypes encoded by separate genes.

D1-like receptor
family

D1

D5

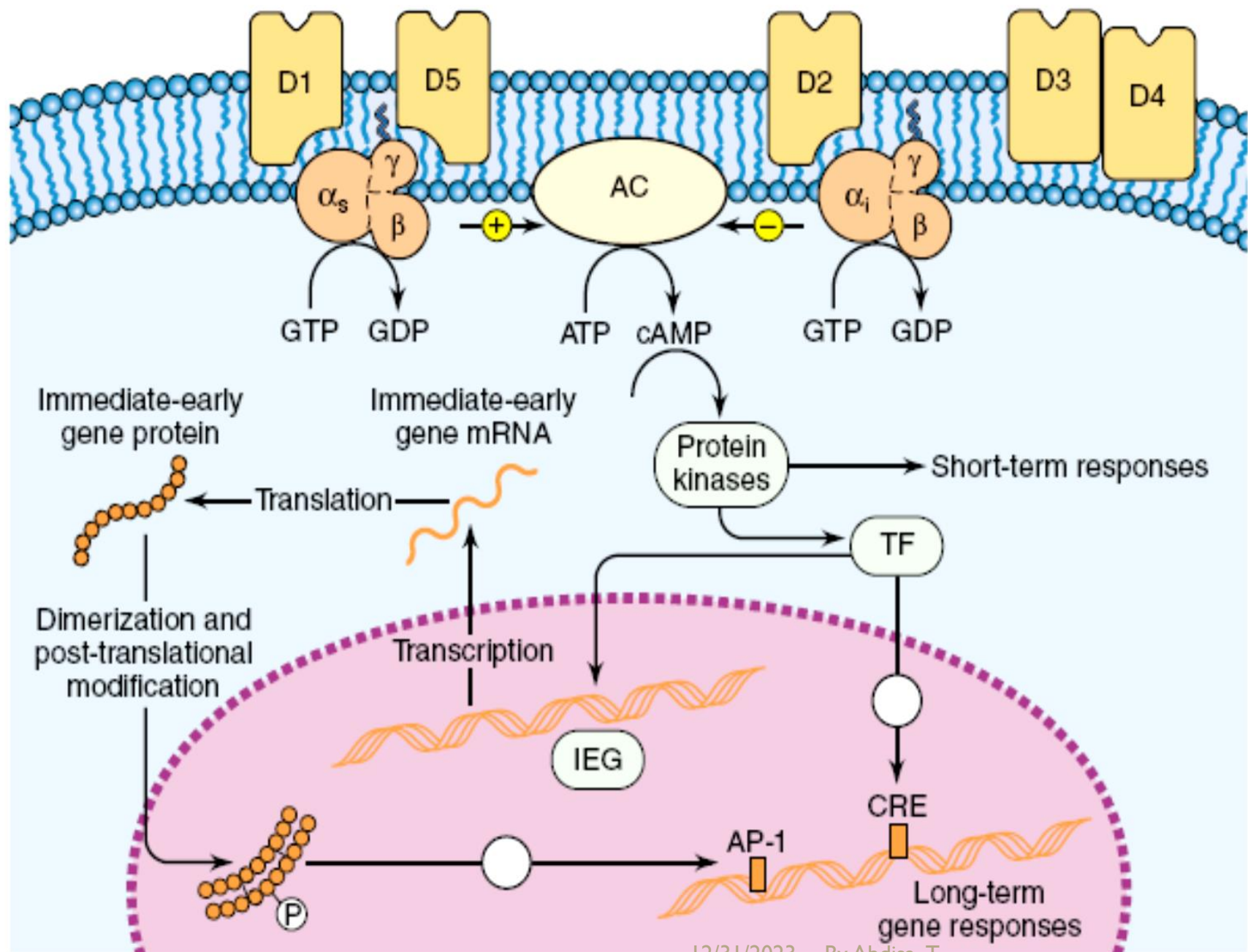
D2-like receptor
family

D2

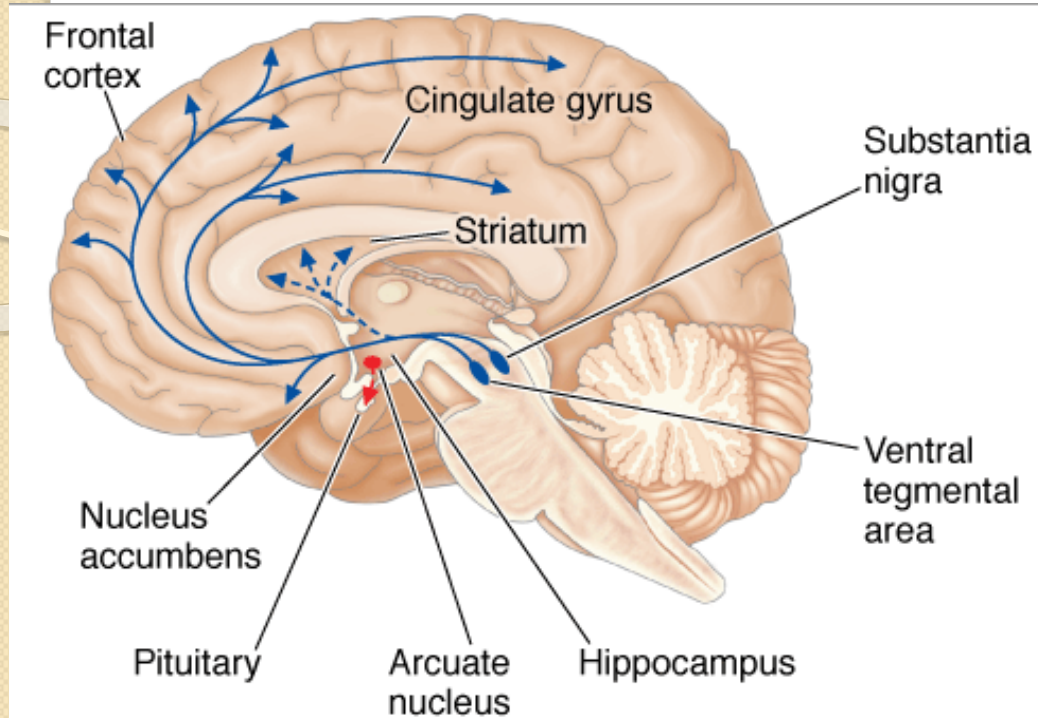
D3

D4

	D1-Like		D2-Like		
	D1	D5	D2	D3	D4
Amino acids	446	477	415/444	400	387
Effector pathways	↑cAMP	↑cAMP	↓cAMP ↑K ⁺ channel ↓Ca ²⁺ channel	↓cAMP ↑K ⁺ channel ↓Ca ²⁺ channel	↓cAMP ↑K ⁺ channel ↓Ca ²⁺ channel
Distributions	Olfactory tubercle, cerebral cortex	Hippocampus, hypothalamus, cerebral cortex	Nucleus accumbens, midbrain	Olfactory tubercle, hypothalamus,	Frontal cortex, medulla, midbrain, nucleus accumbens

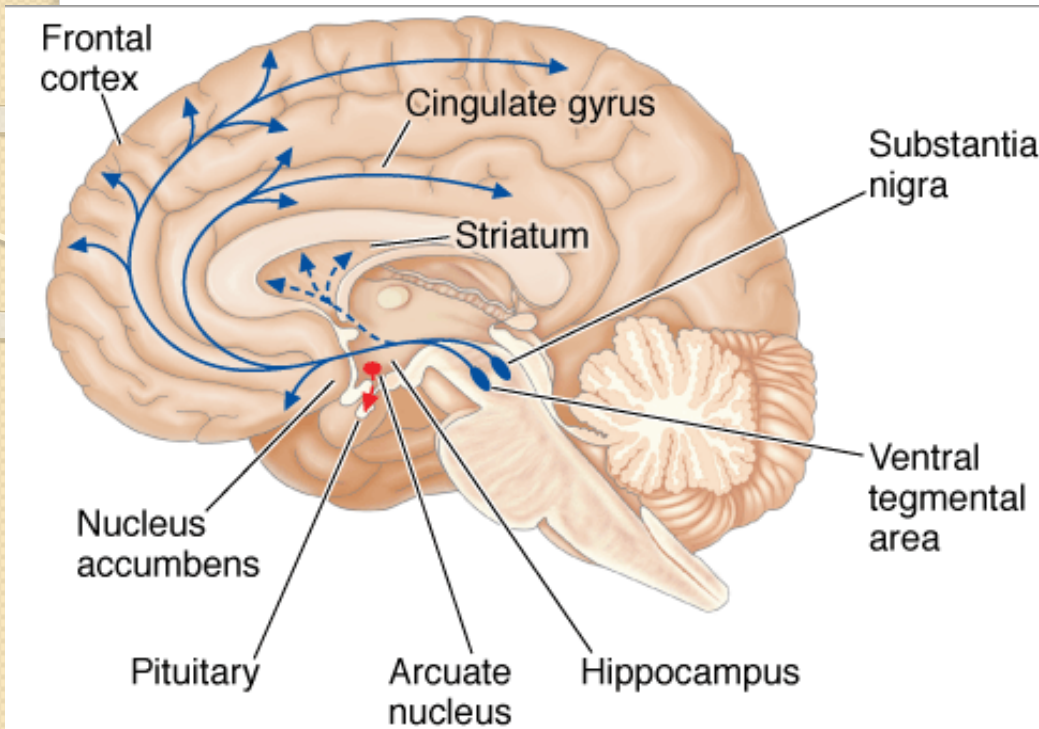


Dopamine Pathways & Function



1. **Mesostriatal (or nigrostriatal) pathway:** 75% of the dopamine in brain. Neurons in the substantia nigra pars compacta (SNc) project to the dorsal striatum (*upward dashed blue arrows*); **motor control**-this is the pathway that degenerates in Parkinson disease.
2. **Tuberohypophyseal system:** is a group of short neurons running from the ventral hypothalamus to the median eminence and pituitary gland (*red arrows*).
Regulate secretions of pituitary gland

Dopamine Pathways & Function



3. Mesolimbic pathway: The third pathway projects from the ventral tegmentum to the mesolimbic forebrain, especially the nucleus accumbens and the amygdaloid nucleus.

4. Mesocortical pathway: whose cell bodies also lie in the VTA and which project via the medial forebrain bundle to the frontal cortex (*solid blue arrows*).

These pathways - **cognitive, reward and emotional behavior**

Mesocortical dopamine deficiency - ADHD

Mesolimbic dopamine deficiency –

Schizophrenia

Adrenergic Receptors

- ☞ Since NE & EP are important messengers in both the peripheral sympathetic nervous system & the brain, adrenergic receptors are widely distributed in peripheral tissues as well as existing in high concentrations in the brain.
- ☞ Adrenergic receptor subclassified into 3 major families ($\alpha 1$, $\alpha 2$, β) based on their pharmacological properties, amino acid sequences and signaling mechanisms.
- ☞ Like the dopamine receptors, are members of the GCPR

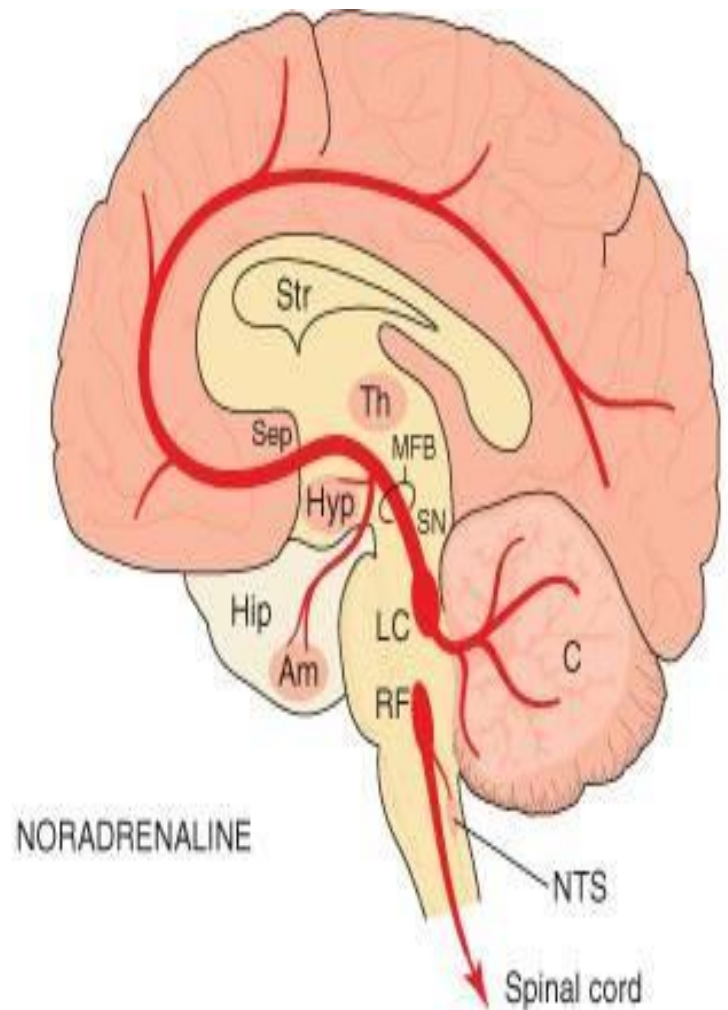
Adrenoceptors

<u>Sub-type</u>	<u>Location of receptors</u>	<u>Type of G-protein coupled to receptor</u>	<u>Basic pathway</u>
$\alpha 1$	Smooth Muscle	Gq	Increase in PLC Increase in IP3 Increase in intracellular Ca^{2+} => m. contraction
$\alpha 2$	Presynaptic nerves	Gi	Decrease in activation of Adenylate cyclase Decrease in cAMP
$\beta 1$	Heart	Gs	Increase in activation of Adenylate cyclase, Increase in cAMP Increase in intracellular signaling pathways
$\beta 2$	Smooth muscle		
$\beta 3$	Fat tissue		

The actions of noradrenaline

- are mainly inhibitory (β -receptors), but some are excitatory (α - or β -receptors)
- **Noradrenergic transmission**
 - controls wakefulness and alertness
 - regulates blood pressure
 - regulates response to stress
 - takes a part in the control of mood (functional deficiency contributing to depression)

- The principal centers for noradrenergic neurons are the **locus coeruleus** and the **caudal raphe nuclei**. The ascending nerves of the locus coeruleus project to the **frontal cortex, thalamus, hypothalamus** and **limbic system** (influence emotional behaviour).
- Noradrenaline is also transmitted from the **locus coeruleus to the cerebellum**.
- Nerves projecting from the **caudal raphe nuclei** ascend to the **amygdala** and descend to the **midbrain**.



Clinical Comment

- In albinism, either the copper dependent tyrosine hydroxylase of melanocytes (which is distinct from the tyrosine hydroxylase found in the adrenal medulla) or other enzymes that convert tyrosine to melanins may be defective.
- Individuals with albinism suffer from a lack of pigment in the skin, hair, and eyes, and they are sensitive to sunlight.





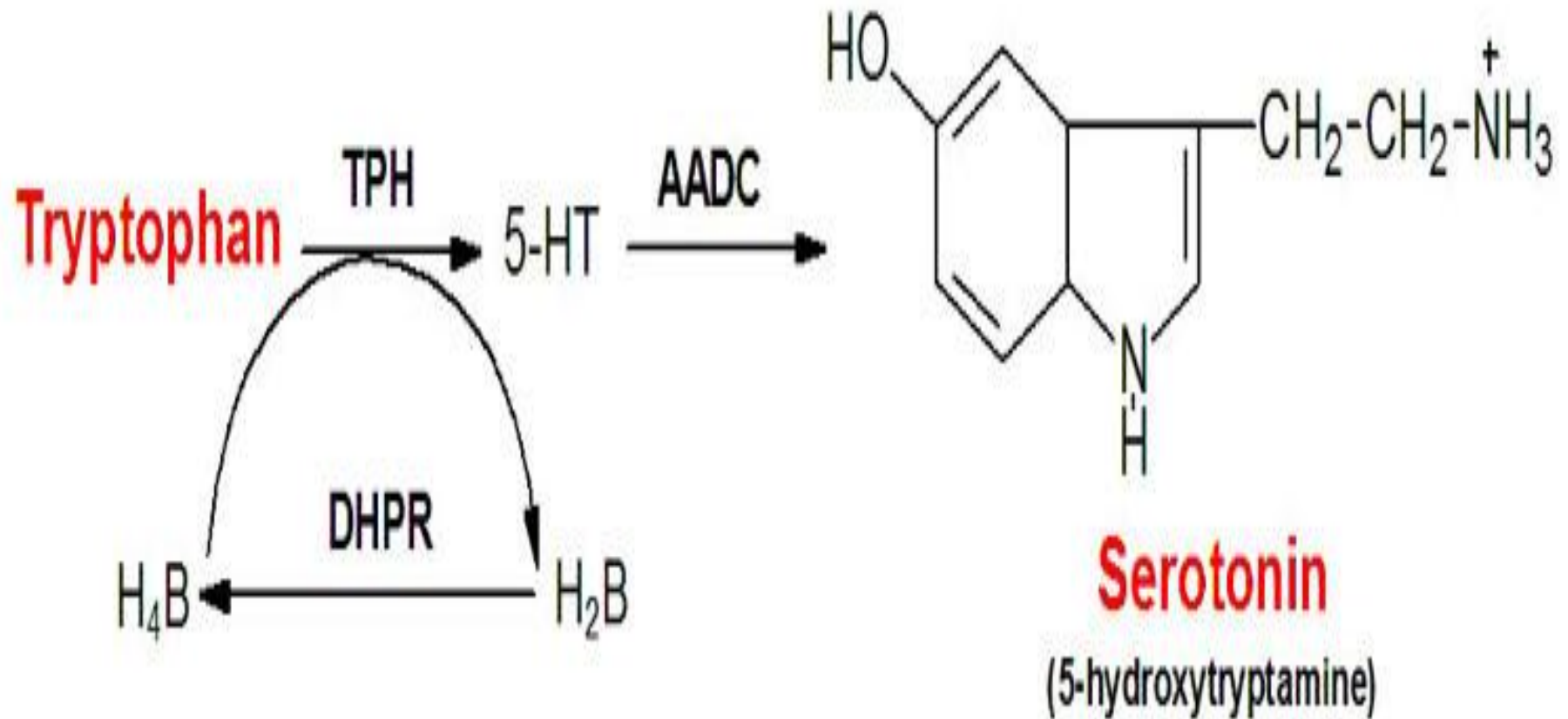
Treatment

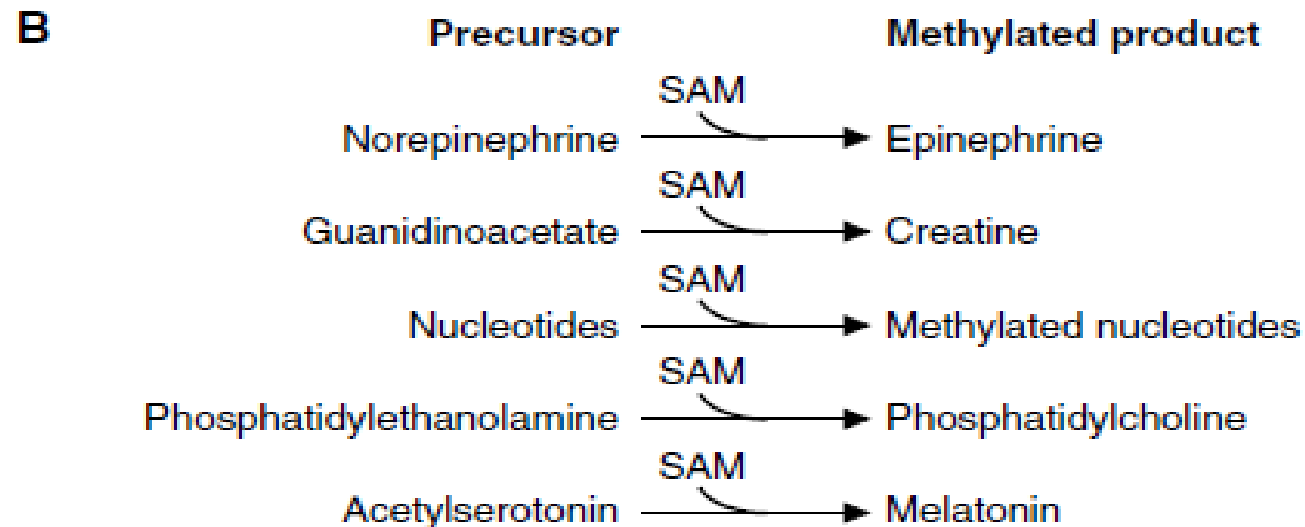
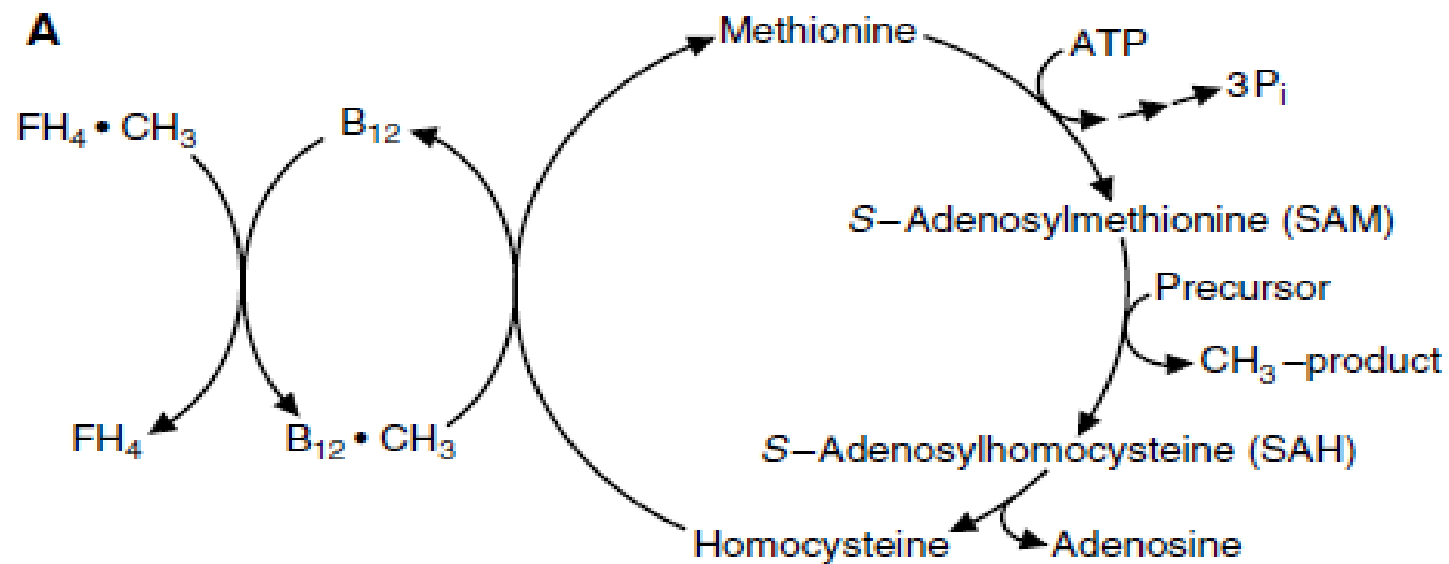
- There is no cure for albinism
- Treatments only ease the symptoms
- Visual Rehabilitation
- Wear dark glasses to shield light
- Always wear sunscreen
- Special UV proof clothing/swimsuits
- Does not effect life span
- There will never be a cure (genetics)

Metabolism of Serotonin

- The pathway for the synthesis of **serotonin** from **tryptophan** is very similar to the pathway for the synthesis of norepinephrine from tyrosine .
 1. The first enzyme of the pathway, **tryptophan hydroxylase**, uses an enzymic mechanism similar to that of tyrosine and **phenylalanine hydroxylase** and requires **BH₄** to hydroxylate the ring structure of tryptophan.
 2. The second step of the pathway is a **decarboxylation reaction catalyzed by the same enzyme that decarboxylates DOPA**. Serotonin, like the catecholamine neurotransmitters, can be *inactivated* by **MAO**.

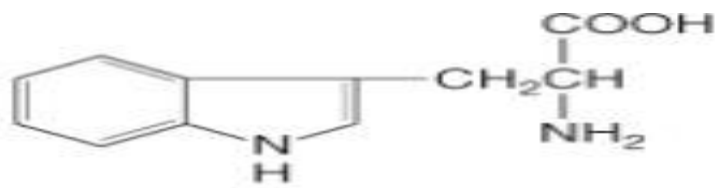
- The neurotransmitter melatonin is also synthesized from tryptophan.
- Melatonin is produced in the ***pineal gland in response to the light–dark cycle***, its level in the blood rising in a dark environment.





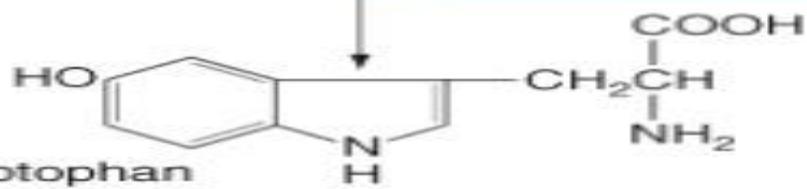
- **Fig. Relationship between FH₄, B₁₂, and SAM.**
A. Overall scheme. B. Some specific reactions requiring SAM

Tryptophan



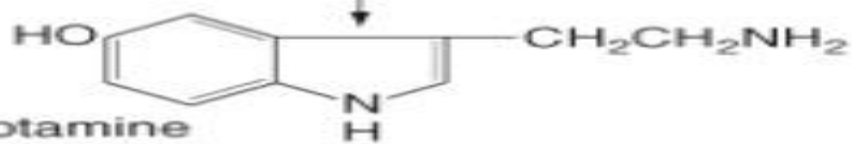
Tryptophan hydroxylase

5-Hydroxytryptophan

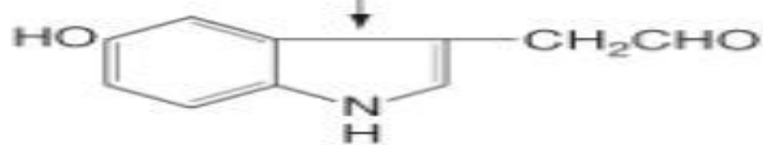


L-Aromatic acid decarboxylase
(= dopa decarboxylase)

5-Hydroxytryptamine
(serotonin)

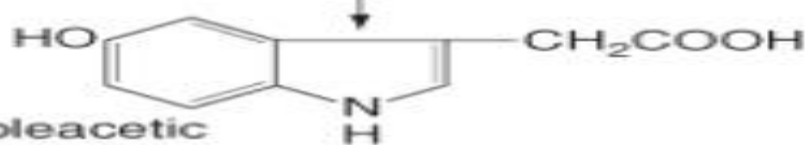


Monoamine oxidase

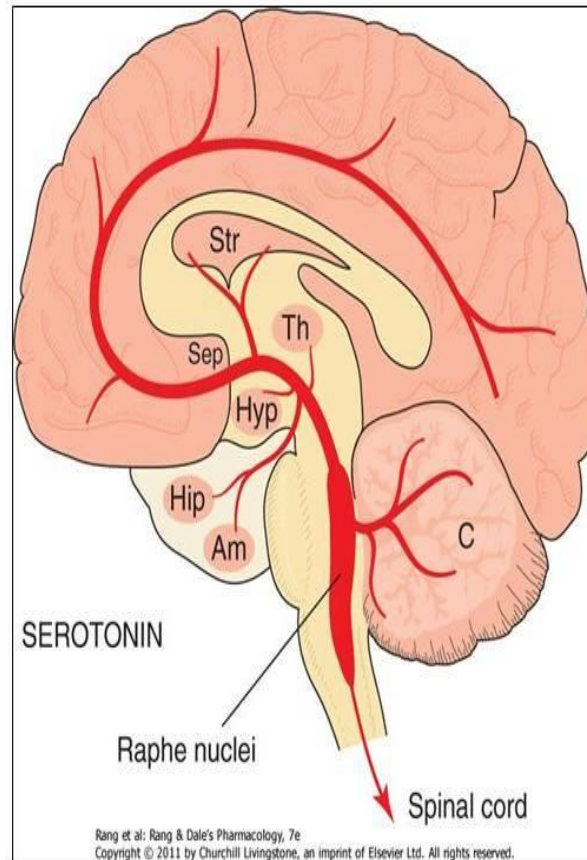


Aldehyde dehydrogenase

5-Hydroxyindoleacetic
acid (5-HIAA)



MAJOR SEROTONERGIC PATHWAYS/SYSTEMS IN CNS:



- the most of the serotonin in the brain is in the brainstem, specifically in the raphe nuclei - these neurons control **muscle activity**
- considerable amounts are present in areas of the hypothalamus, the limbic system - **regulate memory and mood**
- in frontal cortex - regulate **cognition and memory** and the **pituitary gland**

Serotonin pathways

Receptor Family	Associated G-Proteins	Result of Receptor Activation
5HT ₁	G _i /G _o	inhibits cAMP production, inhibitory neurotransmission
5HT ₂	G _q /G ₁₁	increased production of DAG and IP ₃ , excitatory neurotransmission
5HT ₃	ligand-gated Na ⁺ and K ⁺ channels	depolarizes axonal membrane, excitatory neurotransmission

Metabolism Acetylcholine

I. SYNTHESIS

- The synthesis of acetylcholine from **acetyl CoA and choline** is catalyzed by the enzyme **choline acetyltransferase (ChAT)**.
- This synthetic step occurs in the presynaptic terminal.
- The compound is stored in vesicles and later released through calcium-mediated exocytosis. **Choline** is taken up by the presynaptic terminal from the blood via a low-affinity transport system (**high Km**) and from the synaptic cleft via a high-affinity transport mechanism (**low Km**).

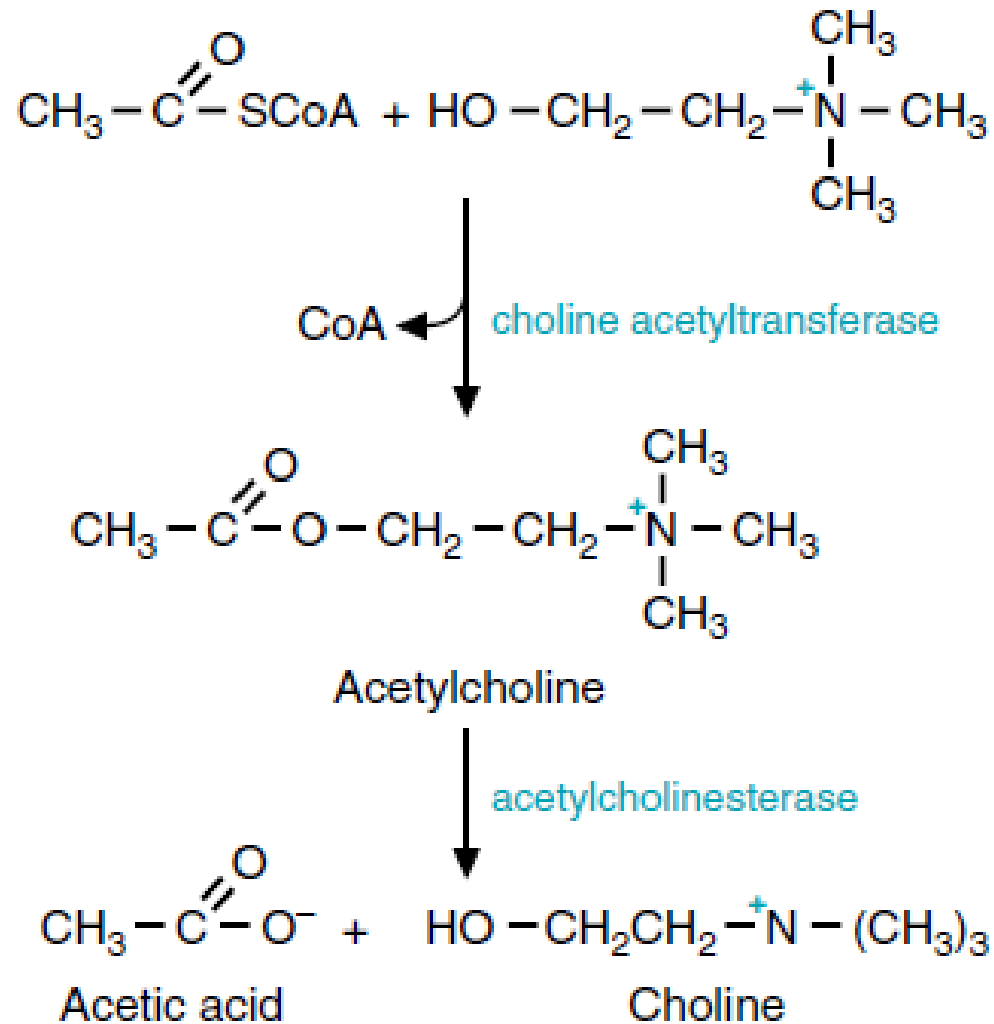


Fig. Acetylcholine synthesis and degradation.

Clinical Comment

- It is believed that the vitamin B12 requirement for choline synthesis contributes to the neurologic symptoms of vitamin B12 deficiency.
- The methyl groups for choline synthesis are donated by SAM (adenosylmethionine), which is converted to (SAH) S-adenosylhomocysteine in the reaction.
- Recall that formation of SAM through recycling of homocysteine requires both **tetrahydrofolate** and **vitamin B12**.

Clinical Comment

- The supply of **choline** in the brain can become rate-limiting for acetylcholine synthesis, and supplementation of the diet with **lecithin** (phosphatidylcholine) has been used to increase brain acetylcholine in patients suffering from tardive dyskinesia (often persistent involuntary movements of the facial muscles and tongue).

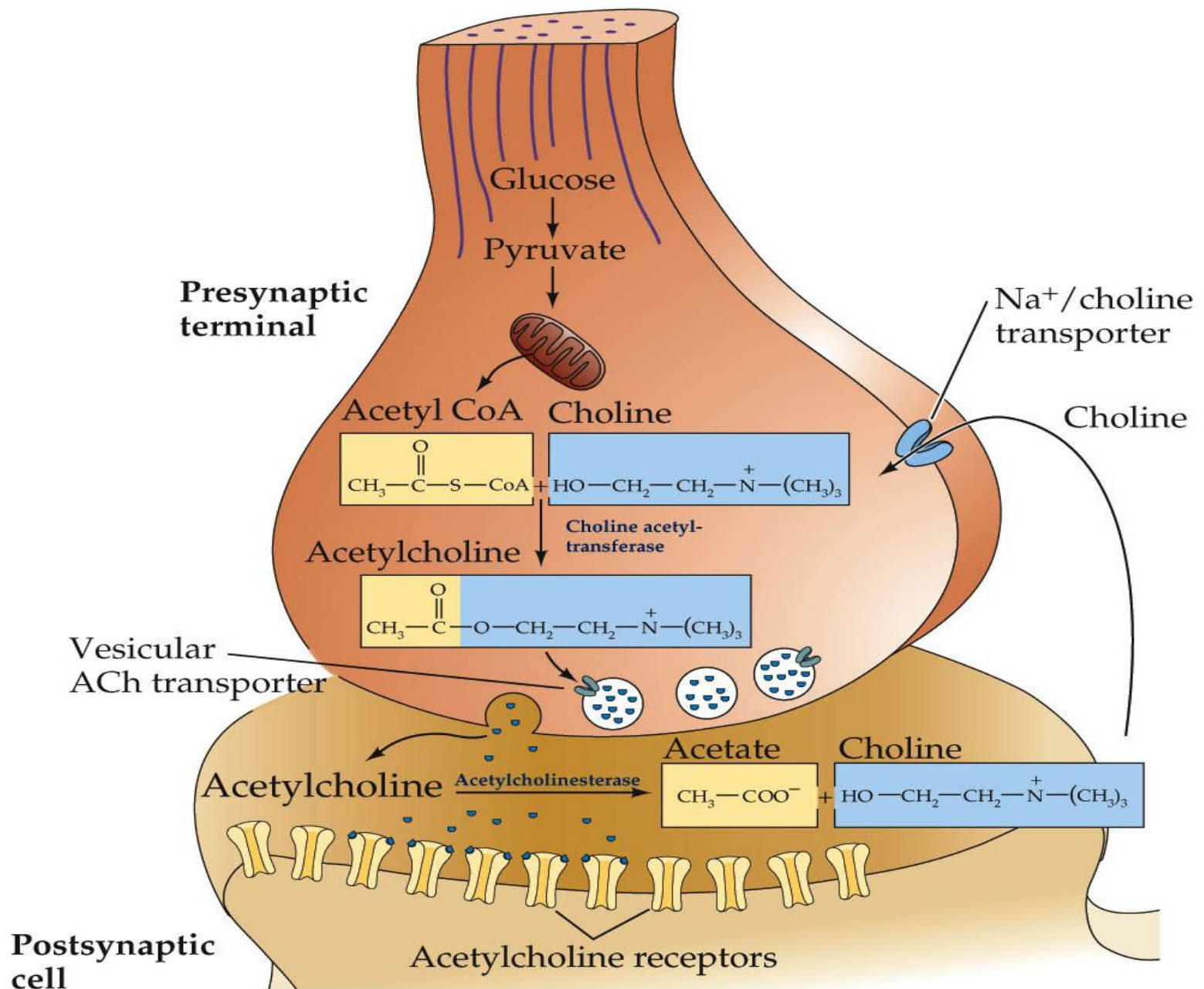
Comment

- The neonate has a very high demand for choline, for both brain lipid synthesis (phosphatidylcholine and sphingomyelin) and acetylcholine biosynthesis.
- High levels of **phosphatidylcholine** in the maternal milk and a high activity of a high-affinity transport system through the **blood-brain barrier** for choline in the neonate help to maintain brain choline concentrations.
- The fetus also has a **high demand** for choline, and there is a high affinity transport system for choline across the placenta.

Clinical Comment

- The choline is derived from maternal stores, maternal diet, and synthesis primarily in the maternal liver.
- *Because choline synthesis is dependent on **folate** and **vitamin B12**, the high fetal demand may contribute to the increased maternal requirement for both vitamins during pregnancy.*

- The acetyl group used for acetylcholine synthesis is derived **principally from glucose oxidation** to pyruvate and decarboxylation of pyruvate to form **acetyl CoA** via the pyruvate dehydrogenase reaction.
- This is because **neuronal tissues have only a limited capacity to oxidize fatty acids to acetyl CoA** so that glucose oxidation is the major source of acetyl groups.
- **Pyruvate dehydrogenase** is found only in mitochondria. The acetyl group is probably transported to the cytoplasm as part of **citrate**, which is then cleaved in the cytosol to form **acetyl CoA** and **oxaloacetate**.



Biochemical Comment

- An inherited **pyruvate dehydrogenase deficiency**, a **thiamine deficiency**, or **hypoxia** deprives the brain of a source of acetyl CoA for acetylcholine synthesis, as well as a source of acetyl CoA for ATP generation from the TCA cycle.

2. INACTIVATION OF ACETYLCHOLINE

- Acetylcholine is inactivated by **acetylcholinesterase**, which is a **serine esterase** that forms a covalent bond with the acetyl group.
- The enzyme is inhibited by a wide range of compounds (pharmacologic **drugs** and **neurotoxins**) that form a covalent bond with this reactive serine group. Neurotoxins such as **Sarin** (the gas used in Japanese subways by a terrorist group)
- Acetylcholine is the major neurotransmitter at the neuromuscular junctions; inability to inactivate this molecule leads to constant activation of the nerve–muscle synapses, a condition that leads to varying degrees of **paralysis**.

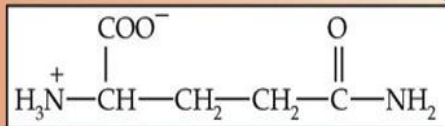
Glutamate and GABA

I. SYNTHESIS OF GLUTAMATE

- Glutamate functions as an **excitatory neurotransmitter** within the **central nervous system**, leading to the depolarization of neurons.
- Within nerve terminals, glutamate is generally synthesized **de novo from glucose rather than taken up from the blood because its plasma concentration is low and it does not readily cross the blood-brain barrier.**

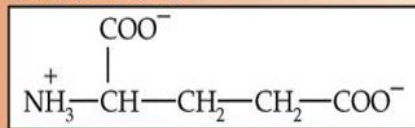
Presynaptic terminal

Glutamine



Glutaminase

Glutamate



VGLUT

Glutamate

Postsynaptic cell

Glutamate receptors

Glial cell

Glutamine

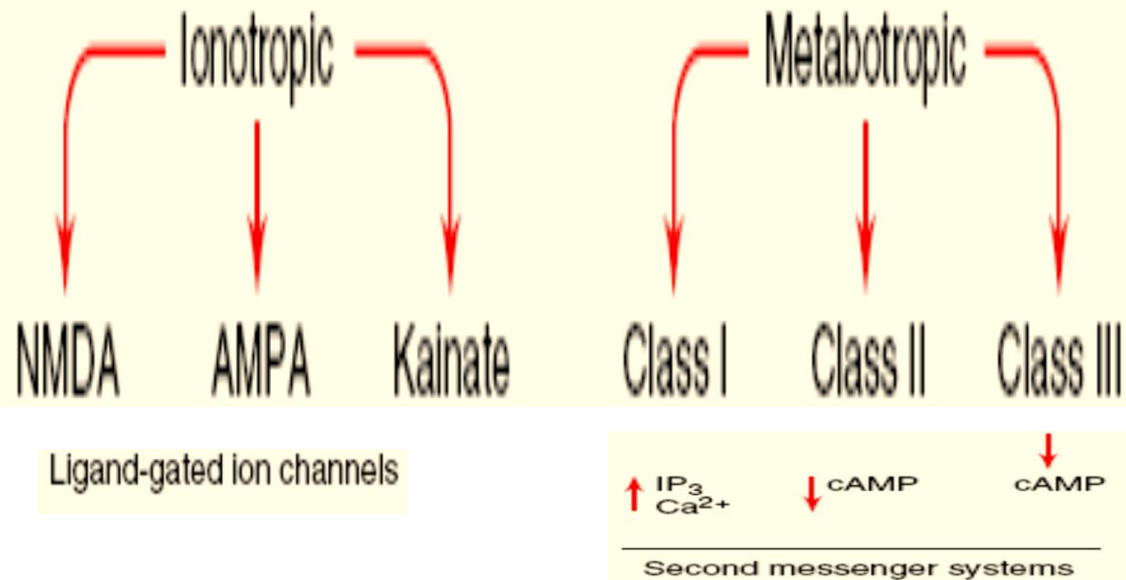
Glutamine synthetase

Glutamate

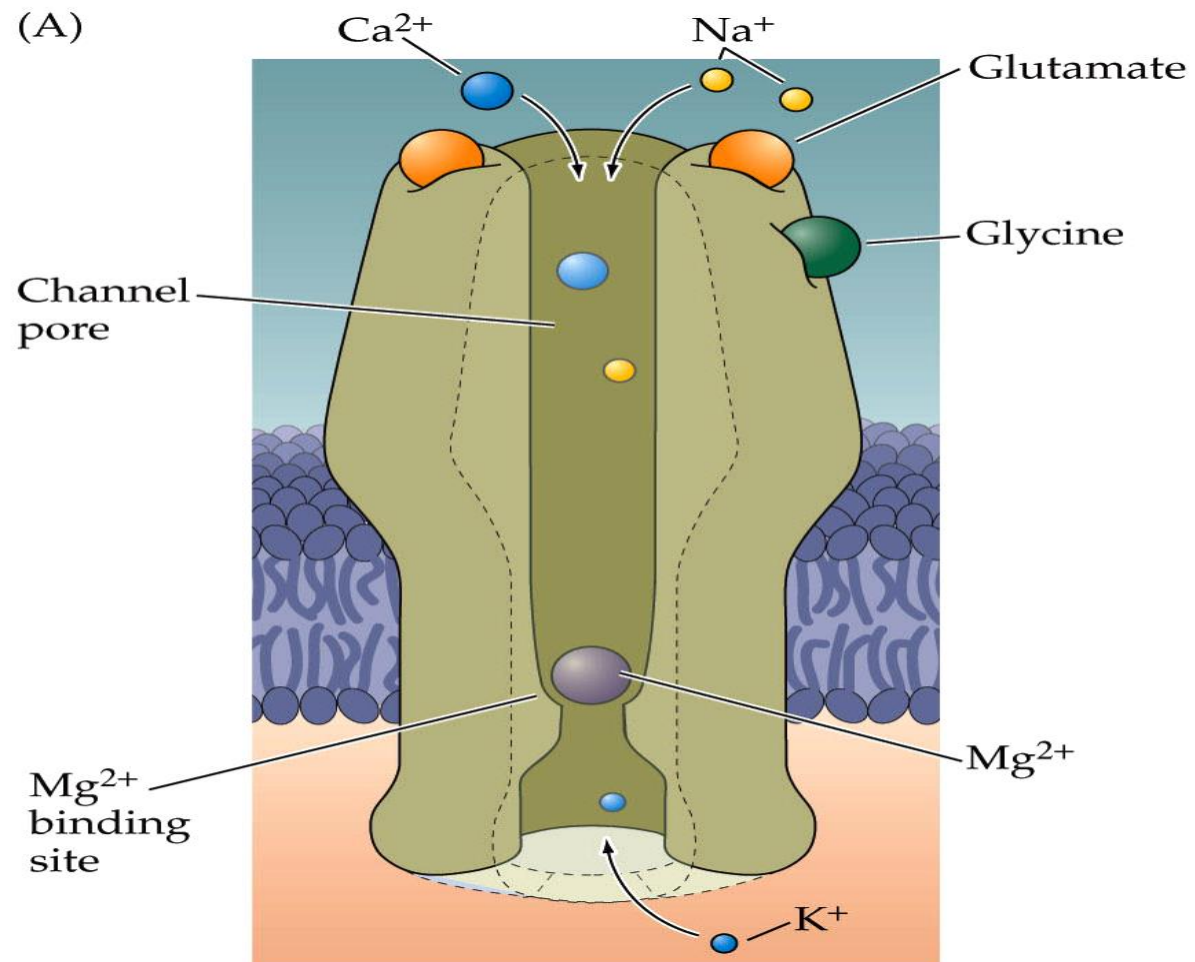
EATT

(error - should be EAAT)

Glutamate Receptor Subtypes

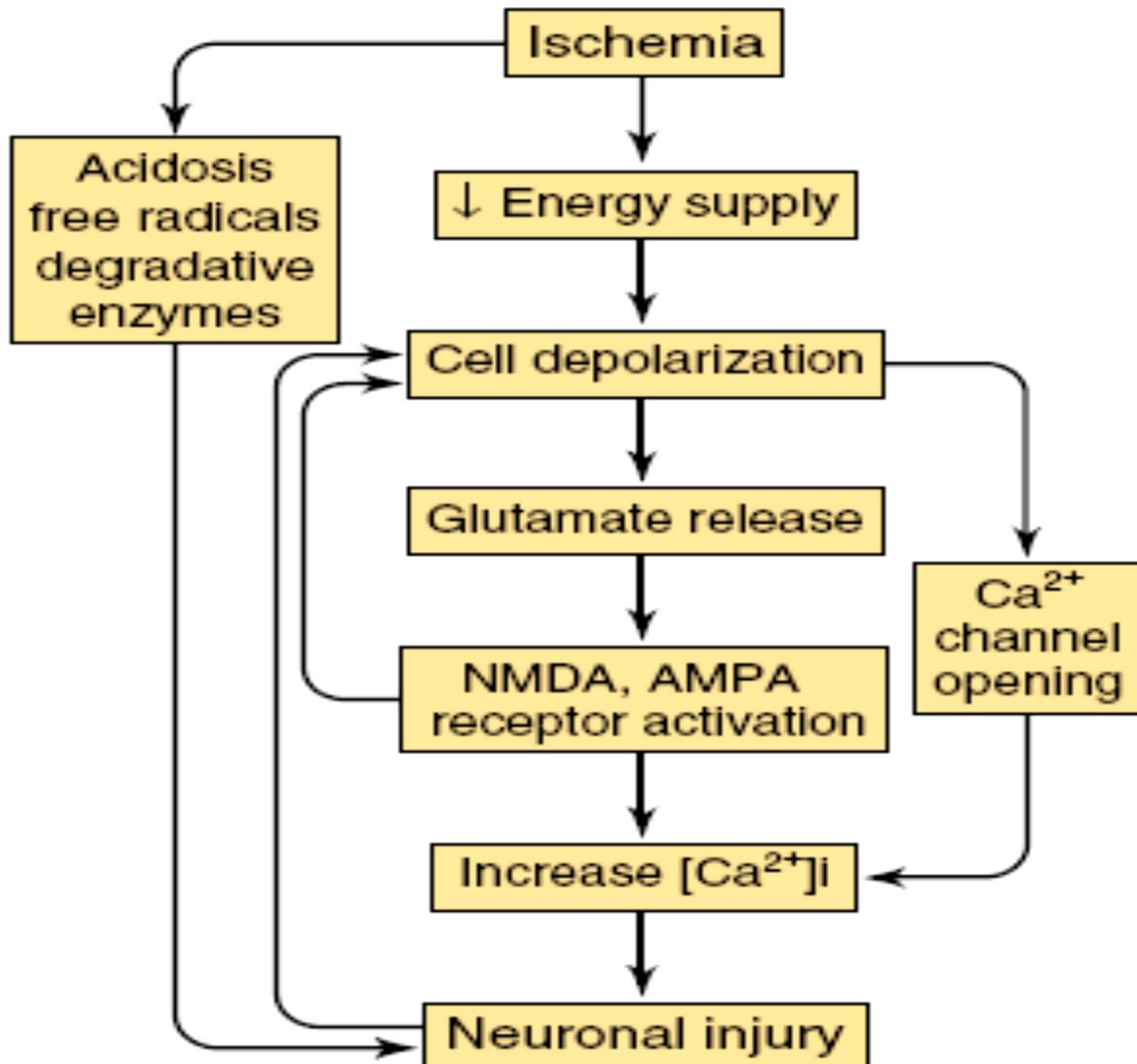


N-methyl-D-aspartate (NMDA) receptors



Neurotoxic effects of Glutamate

- Excessive glutamate receptor activation may mediate certain neurological disorders.
- Glutamate and its analogs can be neurotoxins and cause excitotoxicity.
- Glutamate and structurally related ligands, such as N-methyl-D-aspartate (NMDA) and kainate, in addition to their powerful excitatory effects at glutamate receptors, are potent neurotoxins.
- The toxic effect of glutamate and its analogs is related to their excitatory properties at glutamate receptors, and this type of toxicity is therefore called excitotoxicity.





1. Which of the following is the primary excitatory neurotransmitter in the central nervous system?

- a) GABA
- b) Glycine
- c) Glutamate
- d) Dopamine

2. The NMDA receptor is a type of glutamatergic receptor involved in:

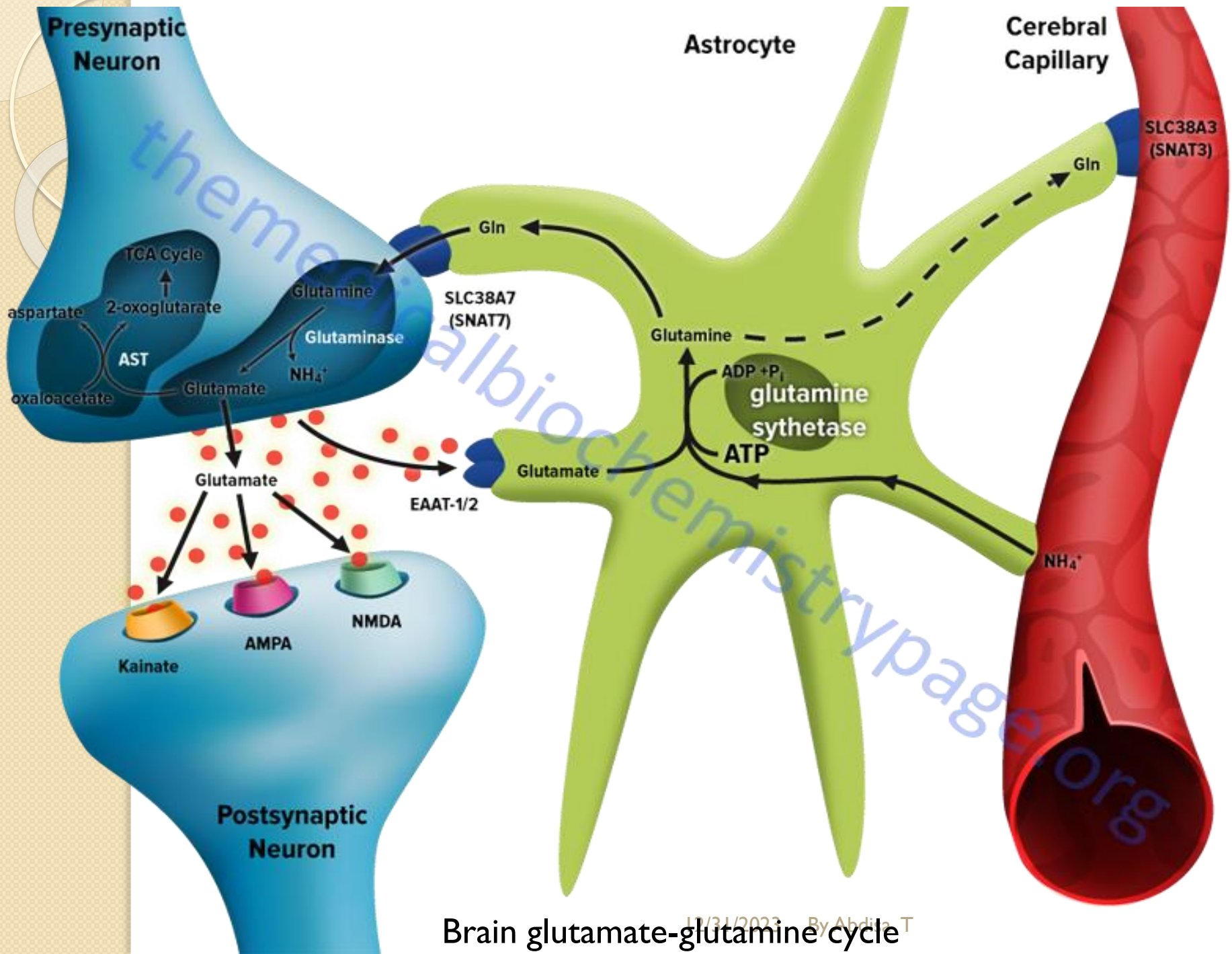
- a) Fast synaptic transmission
- b) Learning and memory
- c) Inhibitory signaling
- d) Muscle contraction

3. Excitotoxicity, a pathological process involving excessive glutamate signaling, is associated with:

- a) Parkinson's disease
- b) Alzheimer's disease
- c) Schizophrenia
- d) Myasthenia gravis

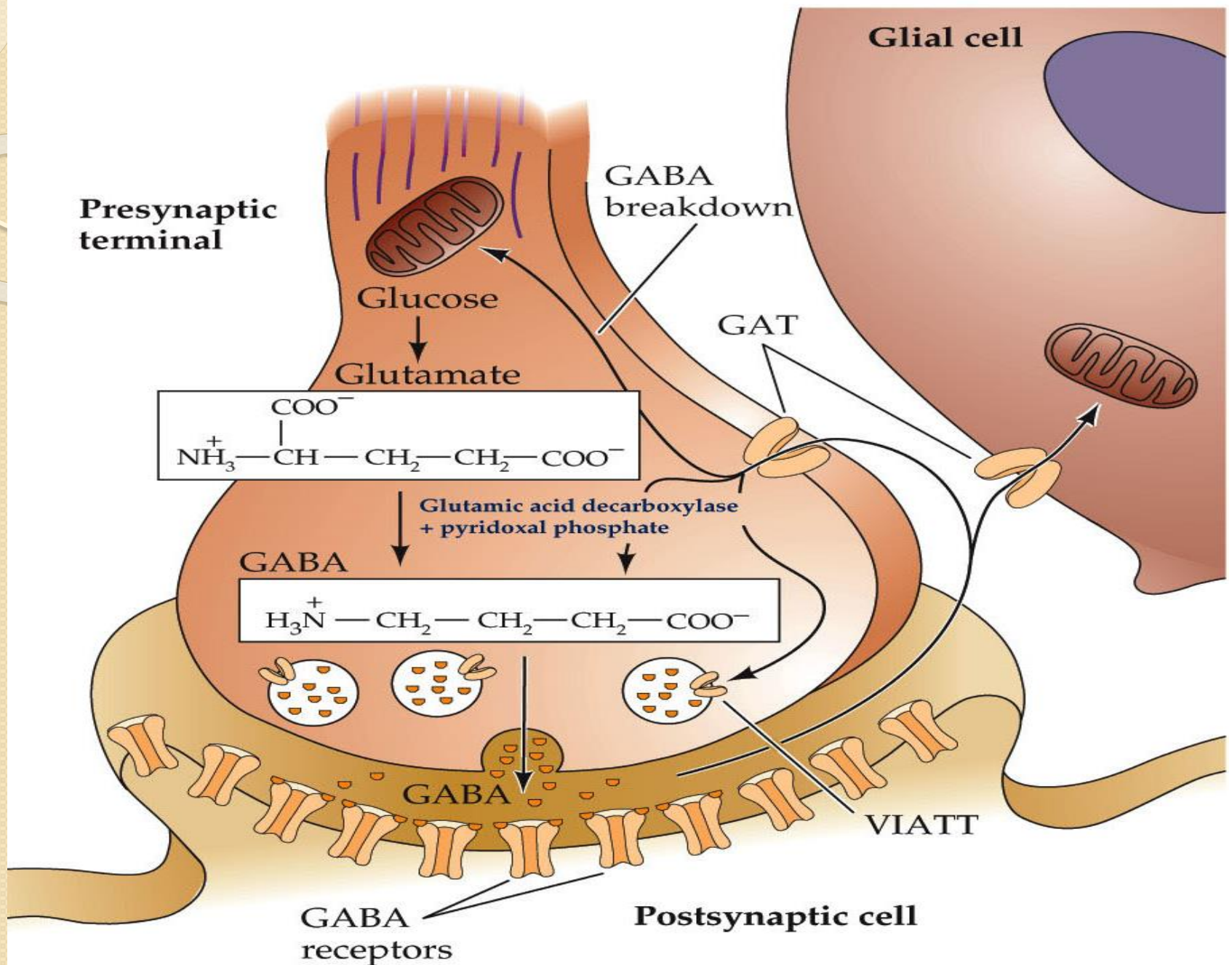
4. Adrenaline, also known as epinephrine, primarily acts on which of the following types of adrenergic receptors?

- a) Alpha-adrenergic receptors
- b) Beta-adrenergic receptors
- c) Gamma-adrenergic receptors
- d) Delta-adrenergic receptors



Brain glutamate-glutamine cycle

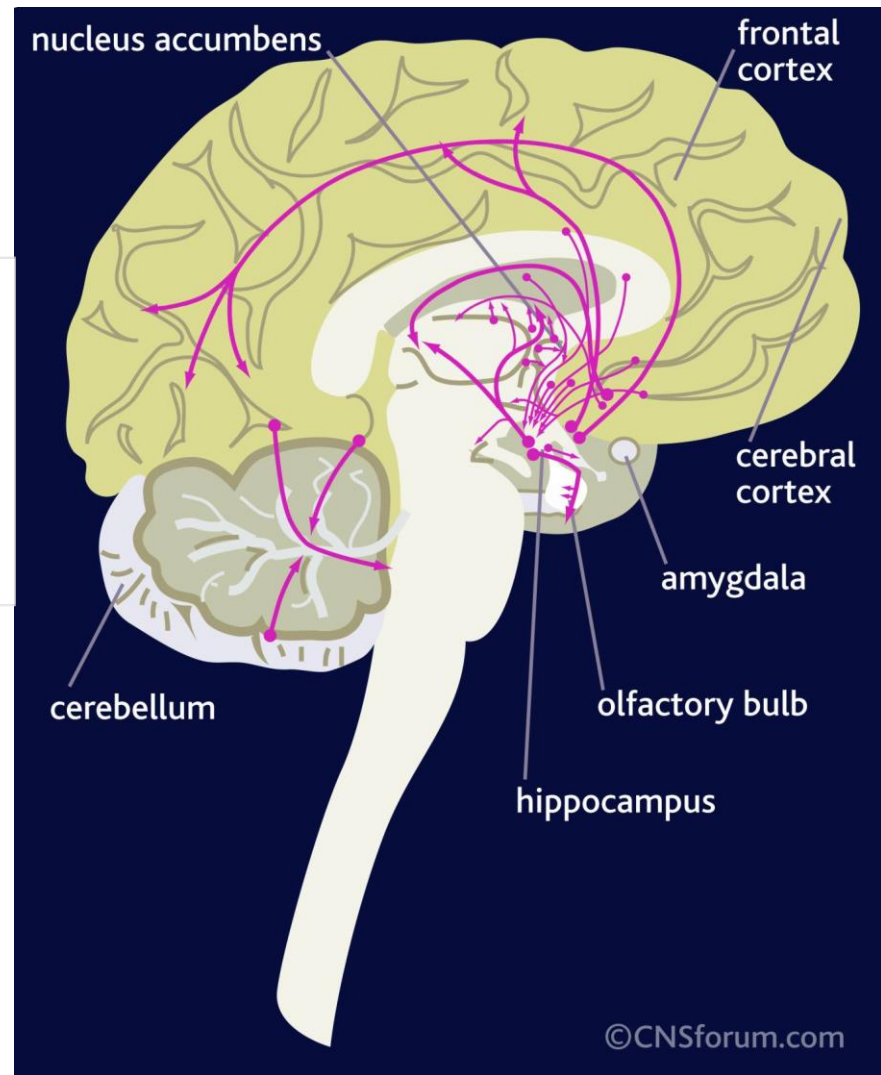
11/31/2023 By Abdi T



γ-Amino butyric acid (GABA)

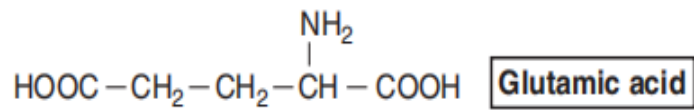
GABA pathways

GABAergic inhibition is seen at all levels of the CNS, including the hypothalamus, hippocampus, cerebral cortex and cerebellar cortex.

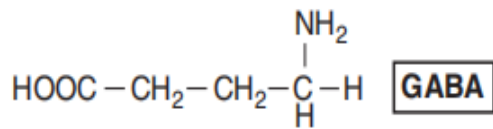


GABA receptors:

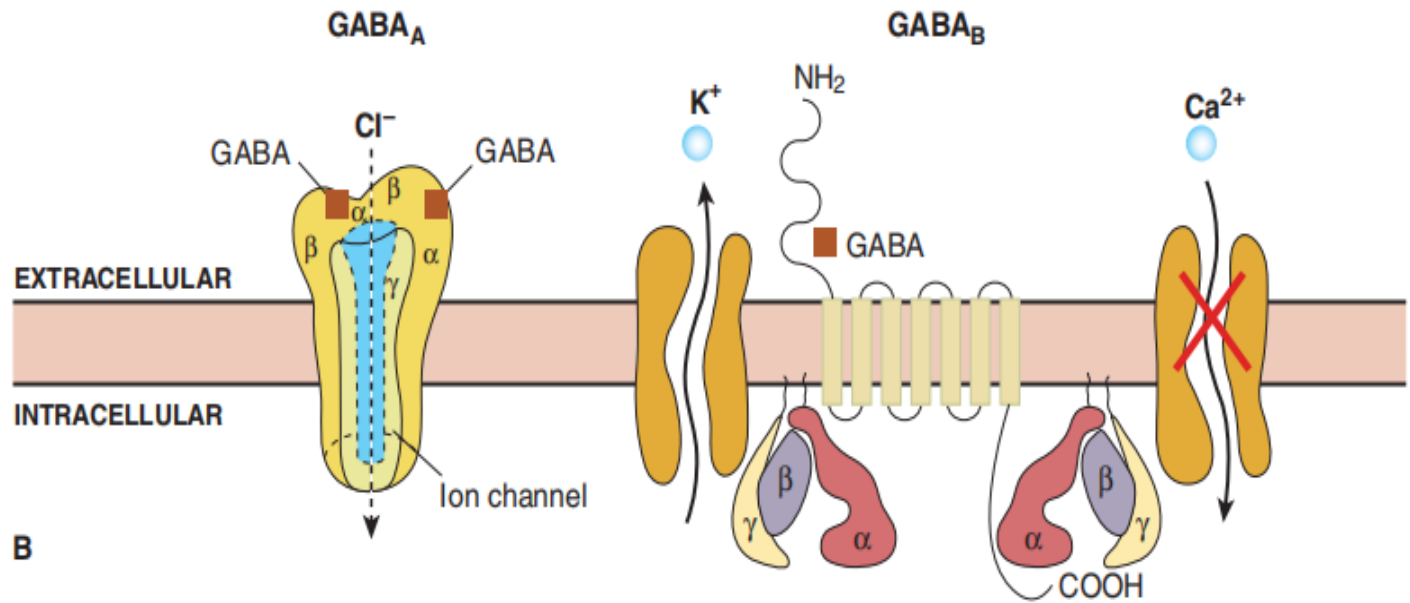
- ➡ Fast GABA transmission mediated mainly by GABA_A receptors, which are ligand-activated chloride channels.
- ➡ Some fast GABA transmission mediated by so-called GABA_C receptors, which are a closely-related sub-family of GABA_A receptors
- ➡ GABA also utilizes a metabotropic receptor called the GABA_B receptor, described in Neuromodulation section.
- ➡ GABA_A receptors belong to the 'ligand-gated ion channel superfamily', which also includes nicotinic acetylcholine receptors, glycine receptors, and the 5-HT₃ serotonin receptor.



*Glutamic acid
decarboxylase*



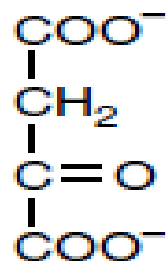
A



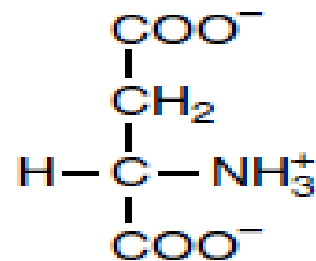
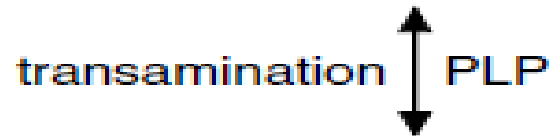
Other Amino Acid Neurotransmitters

I. ASPARTATE

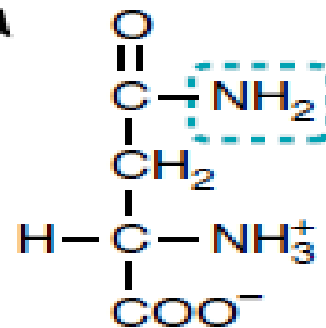
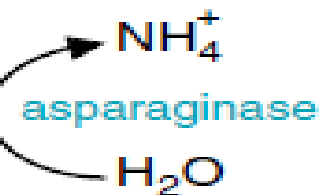
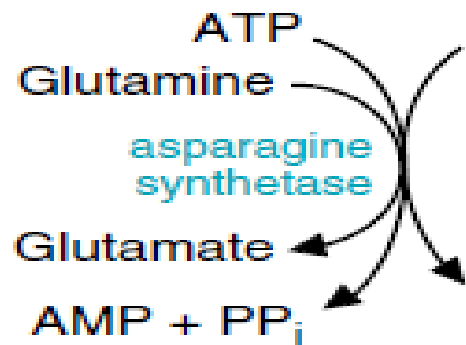
- Aspartate, like glutamate, is an **excitatory neurotransmitter**.
- It is synthesized from the TCA cycle intermediate oxaloacetate via transamination reactions.
- Aspartate synthesis uses oxaloacetate that must be replaced through **anaplerotic reactions**.
- Aspartate cannot pass through the **BBB**.



Oxaloacetate



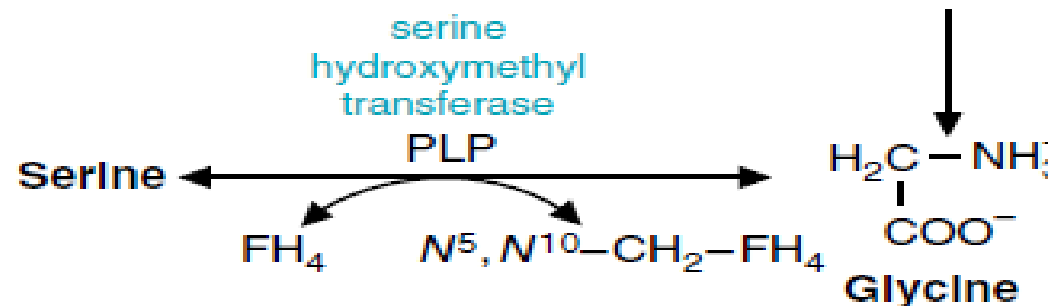
Aspartate



Asparagine

2. GLYCINE

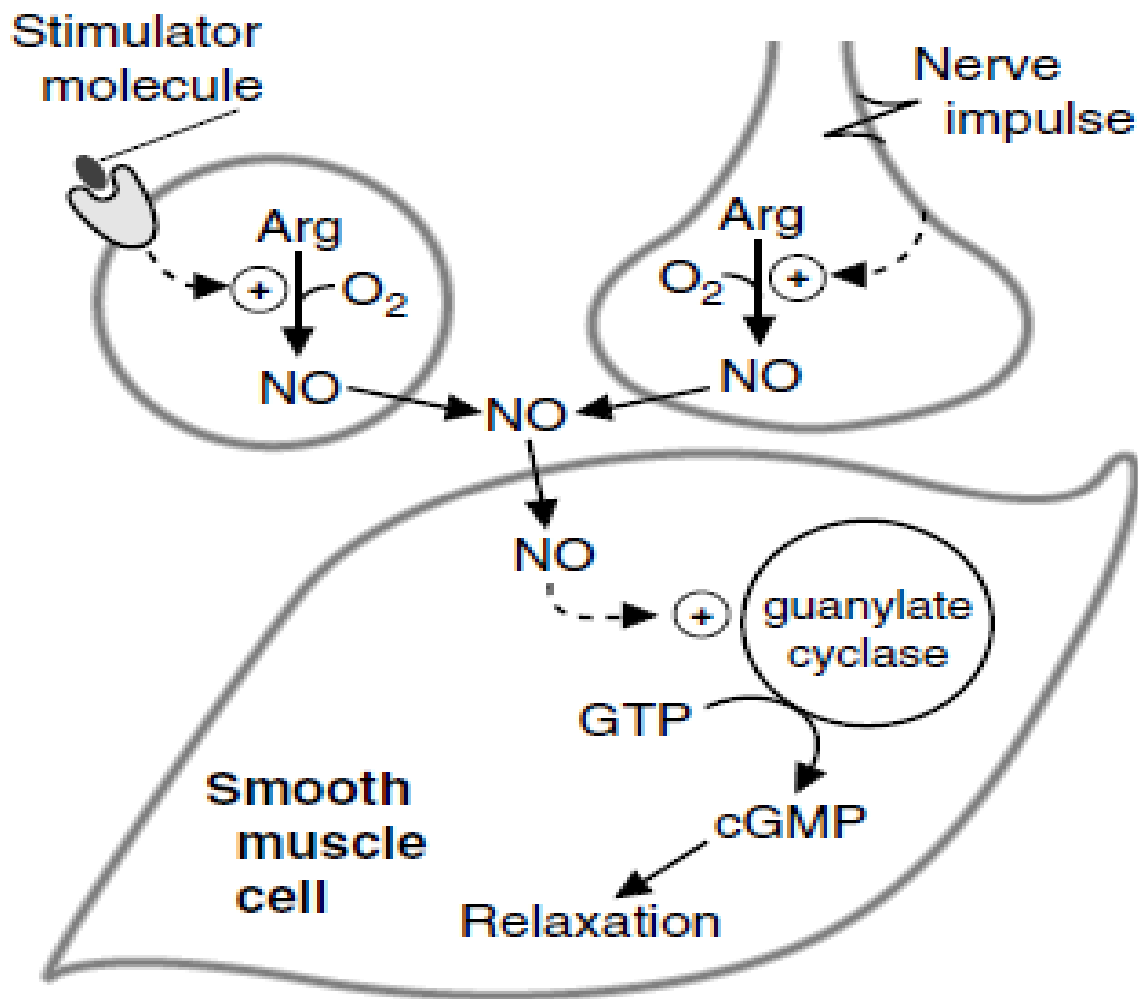
- **Glycine** is the *major inhibitory neurotransmitter* in the spinal cord.
- Most of the glycine in neurons is synthesized *de novo* within the nerve terminal from serine by the enzyme **serine hydroxymethyltransferase**, which requires *folic acid*.
- **Serine**, in turn, is synthesized from the intermediate 3-phosphoglycerate in the glycolytic pathway. The action of glycine is probably terminated via uptake by a high-affinity transporter.



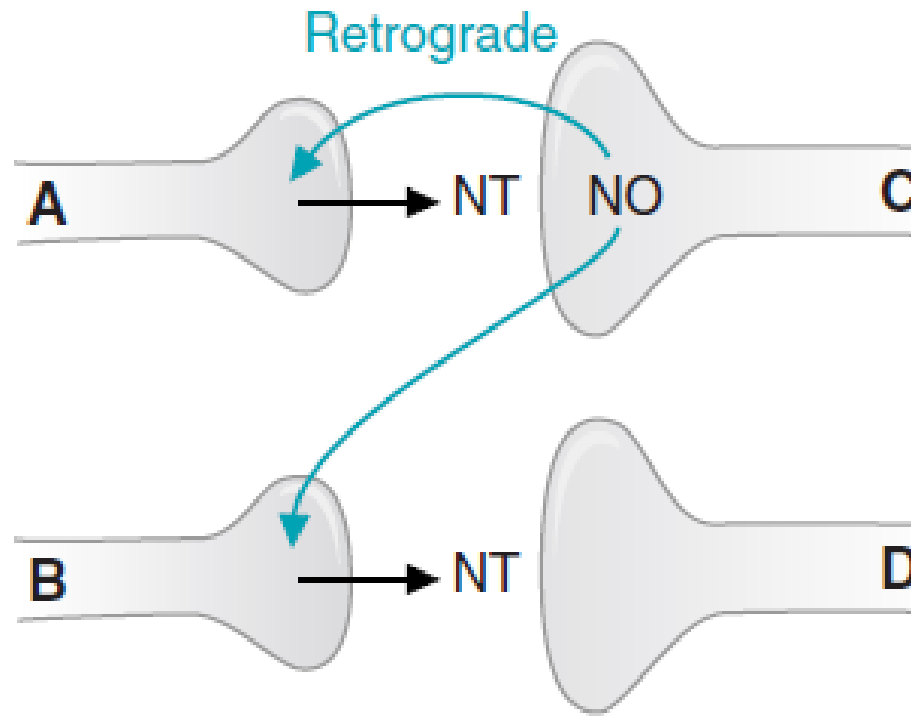
3. CONVERSION OF ARGININE TO NITRIC OXIDE

- **Nitric oxide** (NO) is a biologic messenger in a variety of physiologic responses, including **vasodilation**, **NT**, and the ability of the immune system to kill **tumor cells** and **parasites**.
- NO is synthesized from arginine in a reaction catalyzed by **NO synthase**.
- NO synthase exists as **tissue-specific** forms of two families of enzymes.
- *The form present in **macrophages** is responsible for the overproduction of **NO**, leading to its **cytotoxic** actions on **parasites** and **tumor cells**.*

- The enzyme present in nervous tissue, vascular endothelium, platelets, and other tissues is responsible for the physiologic responses to NO such as **vasodilation** and **neural transmission**.
- In target cells, NO activates a soluble **guanylate cyclase**, which results in increased cellular levels of **cGMP** (3, 5 - cyclic GMP)
- In smooth muscle cells, cGMP, like cAMP, activates one or more protein kinases, which are responsible for the **relaxation of smooth muscle** and the subsequent **dilation of vessels**.
- NO stimulates **penile erection** by acting as a neurotransmitter, stimulating **smooth muscle relaxation** that permits the **corpus cavernosum** to fill with blood.



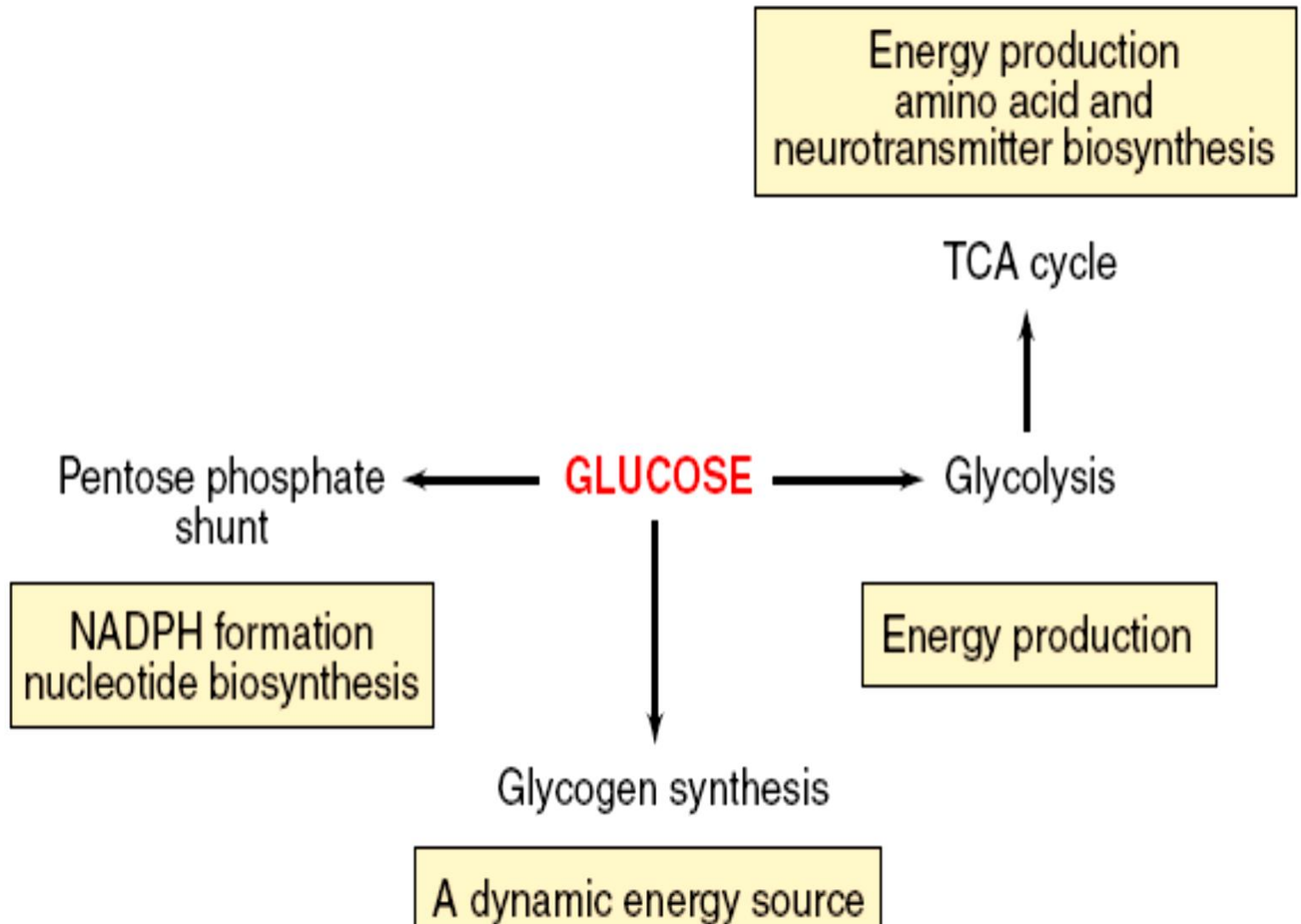
- **Fig. Action of nitric oxide (NO) in vasodilation.** The synthesis of NO occurs in response to a stimulator binding to a receptor on some cells or to a nerve impulse in neurons. NO enters smooth muscle cells, stimulating guanylate cyclase to produce cGMP, which causes smooth muscle cell relaxation. When the smooth muscle cells relax, blood vessels dilate.



- **NO as a retrograde messenger.**
- Cell A releases a neurotransmitter, which stimulates cell C to produce NO. NO, being a gas, can diffuse back and regulate cell A's production and release of NTs. NO can also diffuse to cell B, and to stimulate cell B to produce a different NT to elicit a response from cell D.

Fuels and energy metabolism in the brain

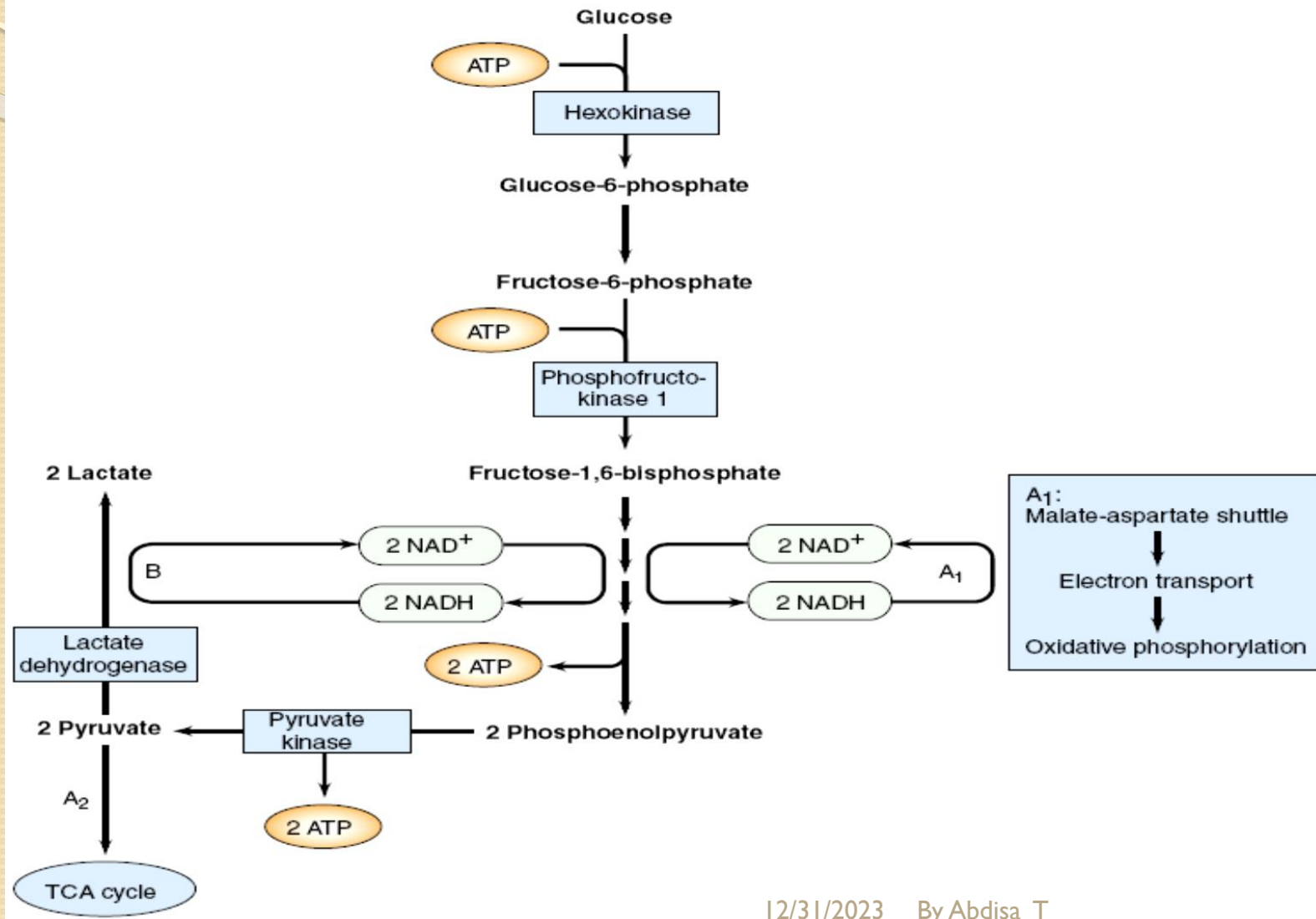
- Despite comprising only 2% of body weight, the brain takes 20% of the resting cardiac output to supply glucose and oxygen.
- The blood supply is required to generate sufficient ATP for various energy-requiring processes in the brain.
- These include not only maintenance of ion gradients for generation of electrical impulses, on which neural communication is based, but also to *synthesize the many different neurotransmitter molecules and to package them into vesicles, to move them within the cell, release them into the synaptic cleft and then remove them from the synapse.*



Glucose Metabolism in the Brain

GLYCOLYSIS:

- Hexokinase is one key point in the regulation of carbohydrate metabolic rate in brain.
- The second major regulatory enzyme is phosphofructokinase, discussed later.



Phosphocreatine has a role in maintaining ATP levels in brain

- ➡ The brain ATP metabolism is mainly controlled by ATPase and creatine kinase (CK) reactions that are coupled together and constitute a complex chemical exchange system involving ATP, phosphocreatine (PCr), and intracellular inorganic phosphate (Pi)

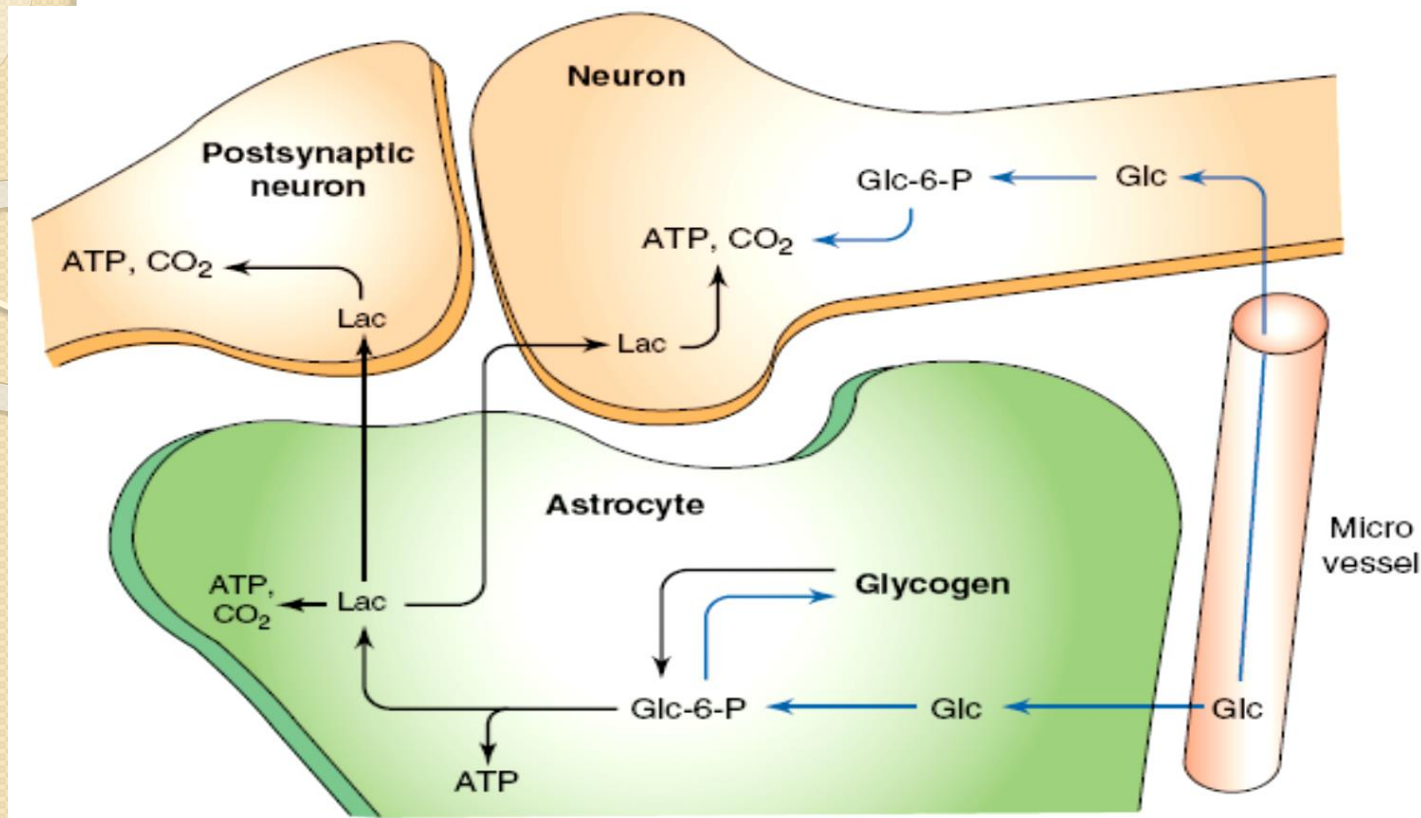
i.e., a $\text{PCr} \rightleftharpoons \text{ATP} \rightleftharpoons \text{Pi}$ -----chemical exchange system.

- ➡ The CPK system also may function in regulating mitochondrial activity (energy buffer or as transport of ATP equivalents).
- ➡ In neurons with a very heterogeneous mitochondrial distribution, the creatine/phosphocreatine shuttle may play a critical role in energy transport.

Substrates other than glucose provide energy for brain cells

(e.g. glutamate, glutamine, lactate, fatty acids, ketone bodies)

- ➡ All of the machinery for using monocarboxylic acids for energy is present in adult brain.
- ➡ Normally, there are no significant cerebral arteriovenous differences for the ketone bodies β -hydroxybutyrate and acetoacetate, which are formed in the course of the catabolism of fatty acids by liver.

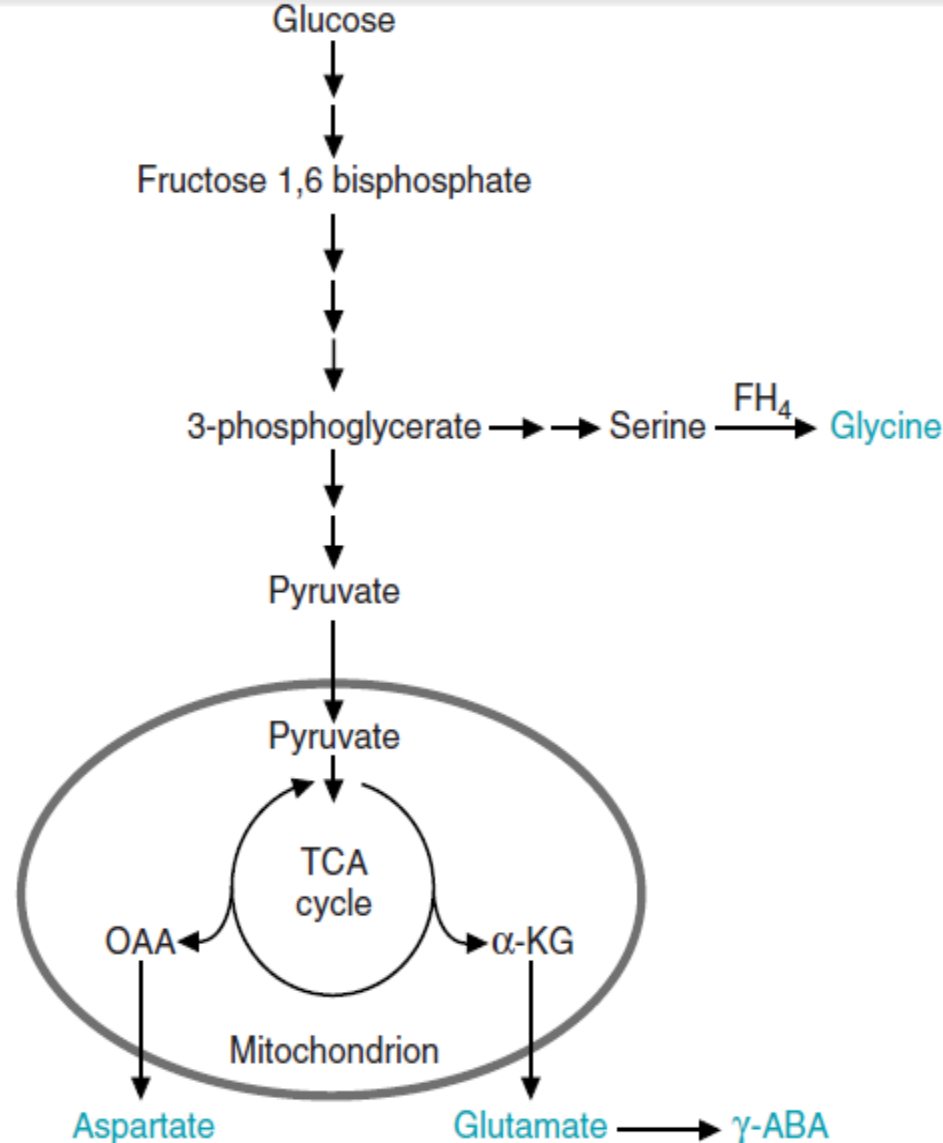


When glucose phosphorylation is limited by the low brain glucose concentration, astrocytic glycogenolysis can provide the necessary glucosyl units to maintain ATP synthesis in the glial compartment. Glycogen can provide fuel to neurons, presumably in the form of lactate, during hypoglycemia and thus reduce the energy deficit in the neuronal compartment.

III. METABOLIC ENCEPHALOPATHIES AND NEUROPATHIES

A. Hypoglycemic Encephalopathy

- During the progression of hypoglycemic encephalopathy, as blood glucose falls below 2.5 mM (45 mg/dL), the brain attempts to use internal substrates such as glutamate and TCA cycle intermediates as fuels.
- Because the pool size of these substrates is quite small, they are quickly depleted.
- If blood glucose levels continue to fall below 1 mM (18 mg/dL), ATP levels become depleted.
- As the blood glucose drops from 2.5 to 2.0 mM (45 to 36 mg/dL), the symptoms appear to arise from decreased synthesis of neurotransmitters in particular regions of the brain rather than a global energy deficit.



- Fig. Glucose metabolism leading to the synthesis of the neurotransmitters glycine, aspartate, glutamate, and GABA.** As blood glucose levels drop and brain glucose levels diminish, synthesis of these neurotransmitters may be compromised.

B. Hypoxic Encephalopathy

- Experimental studies with human volunteers show that cerebral energy metabolism remains normal when **mild** to **moderate** hypoxia (partial pressure of oxygen, or $\text{PaO}_2 = 25\text{--}40$ mm Hg) results in **severe cognitive dysfunction**.
- The diminished cognitive function is believed to result from **impaired neurotransmitter synthesis**.

- In mild hypoxia, cerebral blood flow **increases** to maintain oxygen delivery to the brain. In addition, anaerobic glycolysis is accelerated, resulting in maintenance of **ATP levels**.
- This occurs, however, at the expense of an increased lactate production and a **fall of pH**. Acute hypoxia ($\text{PaO}_2 \rightarrow 20 \text{ mm Hg}$) generally results in a coma.

- Hypoxia can result from insufficient oxygen reaching the blood (e.g., at high altitudes), severe anemia (e.g., iron deficiency), or a direct insult to the oxygen-utilizing capacity of the brain (e.g., cyanide poisoning).
- *All forms of hypoxia result in diminished neurotransmitter synthesis.*
- Inhibition of pyruvate dehydrogenase diminishes acetylcholine synthesis, which is acutely sensitive to hypoxia.

- Glutamate and GABA synthesis, which depend on a functioning TCA cycle, are decreased as a result of **elevated NADH** levels, which inhibit **TCA cycle enzymes**.
- NADH levels are increased when oxygen is unavailable to accept electrons from the electron transport chain and NADH cannot be converted back into NAD.
- Even the synthesis of catecholamine neurotransmitters may be decreased because the hydroxylase reactions require O_2 .

C. The Relationship Between Glutamate Synthesis and the Anaplerotic Pathways of Pyruvate Carboxylase and Methylmalonyl CoA Mutase

- Synthesis of glutamate removes α -ketoglutarate from the TCA cycle, thereby decreasing the regeneration of oxaloacetate in the TCA cycle.
- Because oxaloacetate is necessary for the oxidation of acetyl CoA, oxaloacetate must be replaced by anaplerotic reactions.

- 
- There are two major types of anaplerotic reactions:

(1) pyruvate carboxylase and

(2) the degradative pathway of the branched-chain amino acids, valine and isoleucine, which contribute succinyl CoA to the TCA cycle. This pathway uses B12 (but not folate) in the reaction catalyzed by methylmalonyl CoA mutase.

IV. LIPID SYNTHESIS IN THE BRAIN AND PERIPHERAL NERVOUS SYSTEM

- A number of features of lipid synthesis and degradation in the nervous system distinguish it from most other tissues.
- The first is that the portion of the neuronal cell membrane involved in ***synaptic transmission has a unique role and a unique composition.***
- At the presynaptic terminal, the lipid composition is rapidly changing as storage vesicles containing the neurotransmitter fuse with the cell membrane and release their contents.

- 
- Portions of the membrane are also lost as endocytotic vesicles.
 - On the postsynaptic terminal, the membrane contains the receptors for the neurotransmitters as well as a high concentration of membrane signaling components, such as phosphatidylinositol.
 - A second important feature of brain lipid metabolism is that the blood-brain barrier restricts the entry of nonessential fatty acids such as palmitate, which are released from adipose tissue or present in the diet.

- 
- Conversely, essential fatty acids are taken up by the brain.
 - Because of these considerations, the brain is constantly synthesizing those lipids (cholesterol, fatty acids, glycosphingolipids, and phospholipids), which it needs for various neurologic functions.
 - Neuronal signaling also requires that non-neuronal glial cells synthesize myelin, a multilayered membrane that surrounds the axons of many neurons.

- 
- Myelin is lipid rich and has a different lipid composition than the neuronal membranes.
 - The white matter in the brain contains significantly more myelin than the gray matter; it is the presence of myelin sheaths that is responsible for the characteristic color differences that exist between the two types of brain tissue.

A. Brain Lipid Synthesis and Oxidation

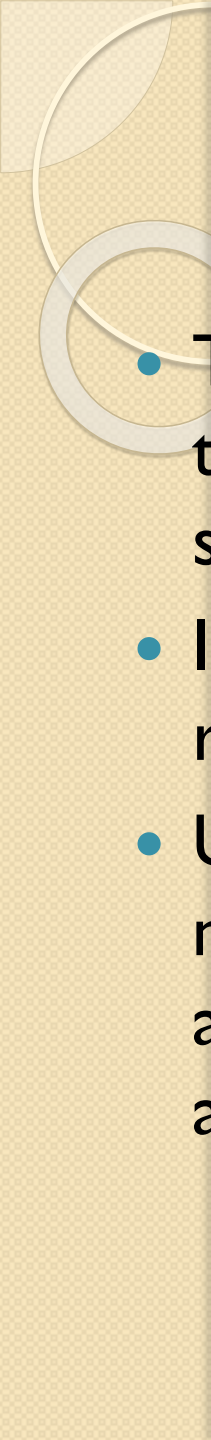
- Because the blood-brain barrier significantly inhibits the entry of certain fatty acids and lipids into the CNS, virtually all lipids found there must be synthesized within the CNS.
- The exceptions are the essential fatty acids (linoleic and linolenic acid), which do enter the brain, where they are elongated or further desaturated.
- cholesterol, Glycerol, sphingolipids, and phospholipids are all synthesized using the pathway of others.
- Of particular note is that ***very-long-chain fatty acids are synthesized in the brain, where they play a major role in myelin formation.***

- Oxidation and turnover of brain lipids using Beta oxidation.
- Peroxisomal fatty acid oxidation is important in the brain because the brain contains very-long-chain fatty acids and phytanic acid (from the diet), both of which are oxidized in the peroxisomes by α -oxidation.
- Thus, disorders that affect peroxisome biogenesis (such as Refsum's disease) severely affect brain cells because
of the inability to metabolize both branched-chain and very-long-chain fatty acids.

- 
- If there is a disorder in which the degradation of glycosphingolipids or mucopolysaccharides is reduced, lysosomes in brain cells will become engorged with partially digested glycolipids, leading to varying degrees of neurologic dysfunction.

B. Myelin Synthesis

- A rapid rate of nerve conduction in the peripheral and central motor nerves depends on the ***formation of myelin***, a multilayered lipid and protein structure that is formed from the plasma membrane of glial cells.
- In the peripheral nervous system, the Schwann cell is responsible for myelinating one portion of an axon of one nerve cell.

- 
- The Schwann cell does this by wrapping itself around the axon multiple times such that a multilayered sheath of membrane surrounds the axon.
 - In the central nervous system, the oligodendrocyte is responsible for myelination.
 - Unlike the Schwann cell, oligodendrocytes can myelinate portions of numerous axons (up to 40), and do so by extending a thin process that wraps around the axon multiple times.

- 
- Thus, CNS axons are only surrounded by the membranes of oligodendrocytes, whereas axons in the PNS are surrounded by the entire Schwann cell.
 - To maintain the myelin structure, the oligodendrocyte synthesizes 4 times its own weight in lipids per day.

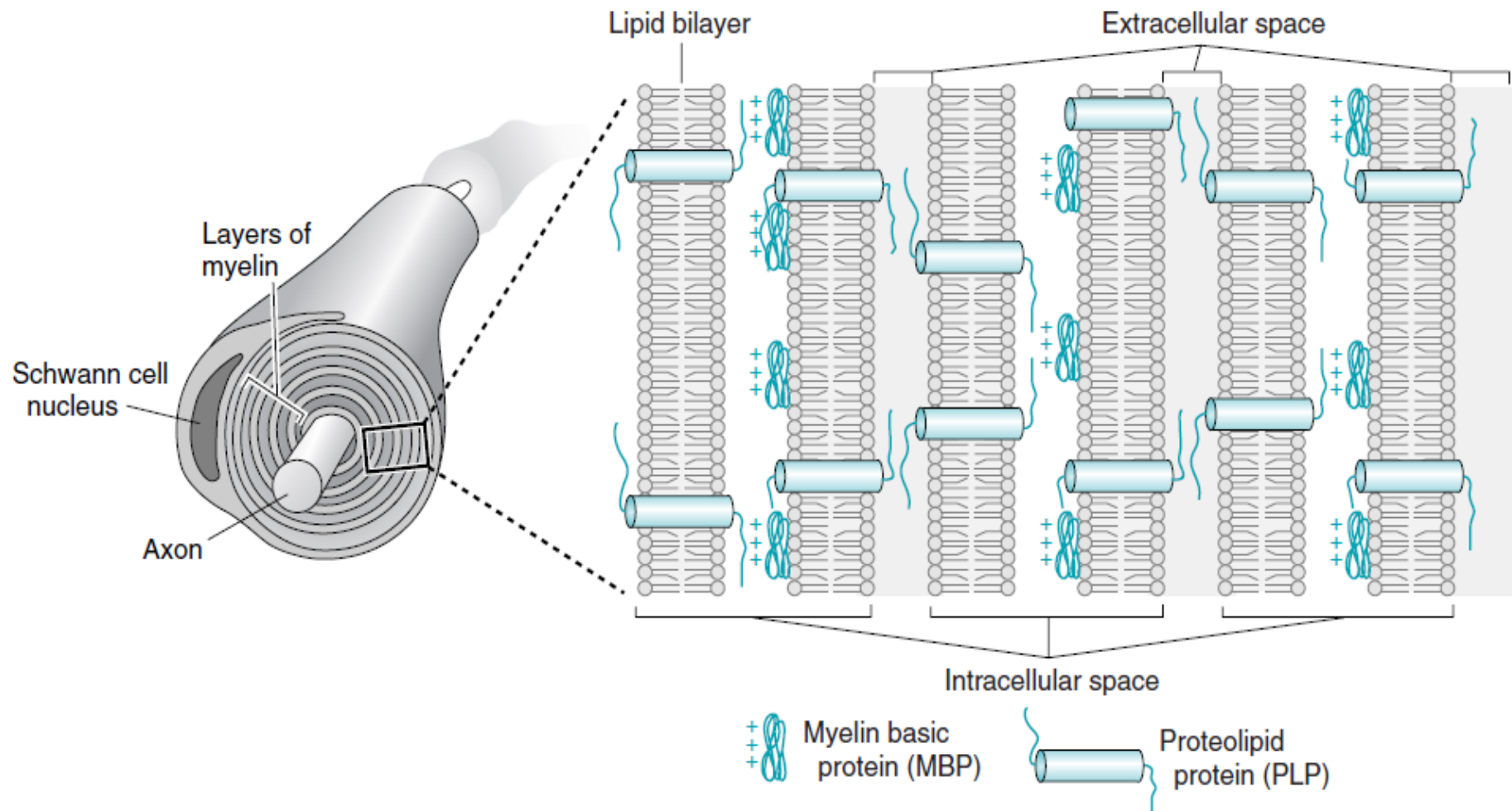


Fig. A composite diagram indicating a Schwann cell that has wrapped around a portion of an axon, forming the myelin sheath. The expansion represents a portion of the myelin sheath. PNS myelin is shown, although it is similar to CNS myelin except that Po would take the place of PLP (proteolipid protein). Recall that there are multiple layers of membrane surrounding the axon; PLP protrudes into the extracellular space and aids in compaction of the membranes through hydrophobic interactions. MBPs help to stabilize the structure from within the membrane.

I. MYELIN LIPIDS

- As the plasma membrane of the glial cell is converted into myelin, the lipid composition of the brain changes .
- The lipid-to-protein ratio is greatly increased, as is the content of sphingolipids.
- The myelin is a tightly packed structure, and there are significant ***hydrophobic interactions*** between the lipids and proteins to allow this to occur.

Table. Protein and Lipid Composition of CNS Myelin and Human Brain

Substance ^a	Myelin	White Matter	Gray Matter
Protein	30.0	39.0	55.3
Lipid	70.0	54.9	32.7
Cholesterol	27.7	27.5	22.0
Cerebroside	22.7	19.8	5.4
Sulfatide	3.8	5.4	1.7
Total galactolipid	27.5	26.4	7.3
Ethanolamine phosphatides	15.6	14.9	22.7
Phosphatidylcholine	11.2	12.8	26.7
Sphingomyelin	7.9	7.7	6.9
Phosphatidylserine	4.8	7.9	8.7
Phosphatidylinositol	0.6	0.9	2.7
Plasmalogens	12.3	11.2	8.8
Total phospholipids	43.1	45.9	69.5

- 
- Cerebrosides constitute approximately 16% of total myelin lipid and are almost completely absent from other cell-type membrane lipids.
 - The predominant cerebroside, galactosylcerebroside, has a single sugar attached to the hydroxyl group of the sphingosine.
 - In contrast, sphingomyelin, which one might guess is the predominant lipid of myelin, is present in roughly the same low concentration in all membranes.

2. MYELIN STRUCTURAL PROTEINS

- The layers of myelin are held together by protein/lipid and protein/protein interactions, and any disruption can lead to demyelination of the membrane
- Although numerous proteins are found in both the CNS and PNS
- The major proteins in the CNS and PNS are different.
- In the CNS, two proteins constitute between 60 and 80% of the total proteins ***proteolipid protein (PLP) and myelin basic proteins (MBP)***.

- The PLP is a very hydrophobic protein that forms large aggregates in aqueous solution and is relatively resistant to proteolysis. Its molecular weight is 30,000 Daltons.
- Its role is thought to be one of promoting the ***formation and stabilization of the multilayered myelin structure.***
- The MBPs are a family of proteins. Unlike PLP, MBPs are easily extracted from the membrane and are soluble in aqueous solution.
- The major MBP has no tertiary structure and has a molecular weight of 15,000 Daltons. MBP is located on the cytoplasmic face of myelin membranes

- 
- In the PNS, the major myelin protein is Po, a glycoprotein that accounts for greater than 50% of the PNS myelin protein content.
 - The molecular weight of Po is 30,000, the same as PLP. Po is thought to play a similar structural role in maintaining myelin structure, as does PLP in the CNS.
 - Myelin basic proteins are also found in the PNS, with some similarities and differences to the MBPs found in the CNS.
 - The major PNS- specific MBP has been designated P2.

Ethanol

Neuronal dysfunction

- An effect of chronic ethanol consumption is neurological degeneration resulting in dementia, loss of memory and peripheral neuropathies.
- The dementia is called the Wernicke–Korsakoff syndrome type (or Wernicke's encephalopathy).
- One explanation for this is a deficiency of thiamine (vitamin B1), which is necessary in the reactions catalysed by pyruvate dehydrogenase and oxoglutarate dehydrogenase and hence the oxidation of glucose and ATP generation are decreased.

Ethanol

- The deficiency may be the result of a poor diet, since ethanol oxidation can provide a considerable proportion of the daily energy requirement, leading to malnutrition in alcoholics.

Acute neurological effects are probably caused by

- alcohol 'dissolving' in the plasma membrane and increasing fluidity of membranes , which affects ion channels and receptors, particularly neurotransmitter receptors.
- Three proposed biochemical explanations for effects of ethanol are:
 - (i) Enhancement of the action of γ -aminobutyric acid which is an inhibitory neurotransmitter.
 - (ii) Inhibition of Ca^{2+} ion entry into neurones by inhibition (blocking) of voltage-gated Ca^{2+} ion channels, particularly those involved in control of exocytosis of the neurotransmitters from presynaptic terminals .

Mental illnesses: biochemical causes

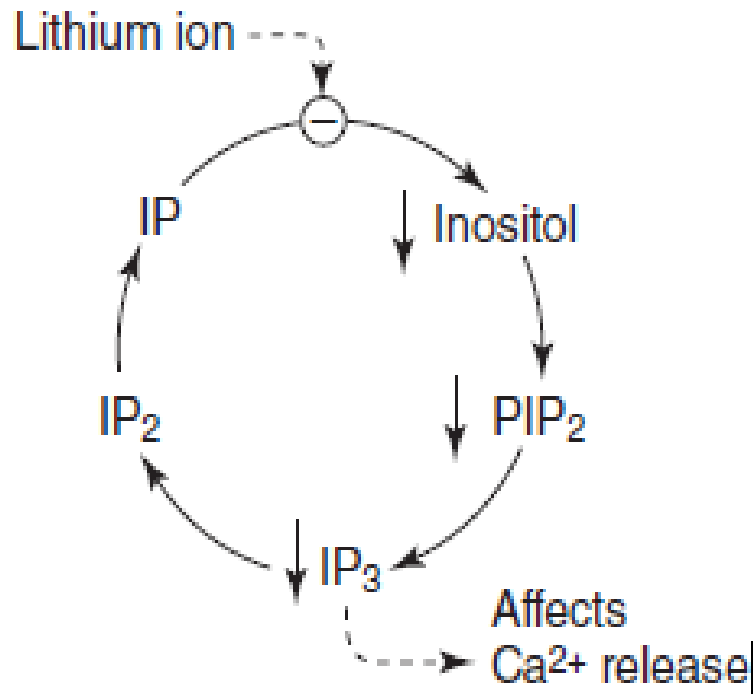
- The biochemical causes include the following:
 - disturbances in neurotransmitter systems;
 - impairment of electrical activity;
 - energy stress;
 - disturbance of membrane structure.

Bipolar disorder (mania)

- ***This disorder is characterised*** by marked swings in mood from depression to elation.
- It is considered to be caused by ***too high a concentration of a catecholamine,***

- The effectiveness of the neurotransmitter could be reduced by ***inhibiting exocytosis or by blocking the binding of the neurotransmitter to its receptor.***
- This has been achieved by use of the antipsychotic drug which blocks dopamine receptors.
- However, the most frequently used drug is a ***lithium salt***. Its mode of action is interesting since it interferes with regulation of Ca^{2+} ion concentration in the cytosol of the presynaptic terminal and hence in exocytosis.

- 
- This is achieved indirectly by inhibiting a phosphatase that degrades inositol phosphates so that the inositol concentration is markedly decreased and phosphatidylinositol bisphosphate cannot be synthesised.
 - This lowers the concentration of the bisphosphate in the plasma membrane so that insufficient inositol trisphosphate is formed to stimulate Ca^{2+} ion release from the reticulum



- **Figure** *Phosphatidylinositol bisphosphate cycle and treatment of bipolar disease.* The metal ion lithium inhibits inositol monophosphate phosphatases and, therefore, inhibits the flux from IP₃ to inositol, so that the concentration of the latter decreases. This can restrict formation of phosphatidylinositol the bisphosphate (PIP₂) so that the amount in the membrane decreases

Secondary defects in neurotransmitter systems

- Neurodegenerative disorders are due to chronic damage to neurones from which they fail to recover and eventually die.
- This results in failure of the neurotransmitter system

in which these neurones are involved. Since the damage progressively accumulates throughout life, neurodegenerative disorders increase with age.

Alzheimer's disease

- Signs and symptoms of the disease are as follows: progressive impairment of cognitive function; behavioural change, such as aggression and depression, and loss of short-term memory.

There are two main hypotheses to explain the damage

to neurones in Alzheimer's disease:

- I. The presence of neurofibrillary tangles within the neurones (both in the cell body and the axons).

- 
- Tangles are filaments of proteins twisted around each other.
 - The major protein in the tangle is known as tau, which is phosphorylated to an abnormal extent and contains attached ubiquitin molecules.
 - This suggests that the neuron is ‘attempting’ to degrade the protein by ‘tagging’ it with ubiquitin

- 
- The protein tau is normally a component of microtubules and hence it is involved in axonal transport.
 - This suggests that degradation of tubules is one of the first changes that give rise to neuronal damage but there is no explanation as to why the tubules are being degraded.

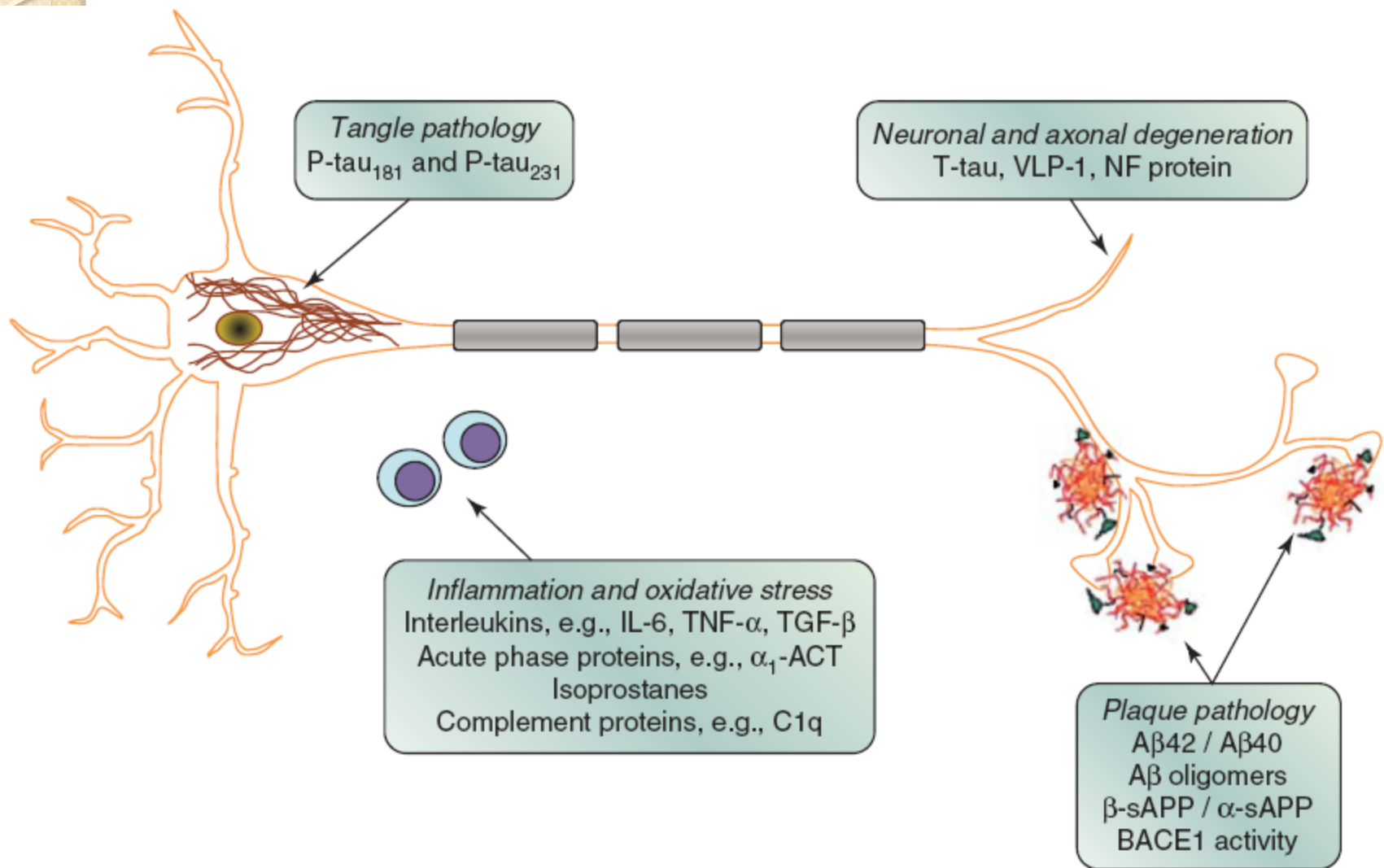


Fig. Neuron with intracellular neurofibrillary tangles and three neuritic plaques, together with two lymphocytes. Candidate cerebrospinal fluid biomarkers for different pathogenic processes.

2. A protein known as the amyloid precursor protein (APP) spans the plasma membrane of the neurone. It possesses an extracellular domain but its function is unknown.

- The extracellular domain is partially hydrolyzed by proteolytic enzymes, known as **secretases**.
- One of the products is the amyloid peptide, of which there are two forms.
- The larger form, contains 42 amino acids and readily polymerises to form plaques in the extracellular space, damaging the neurons.

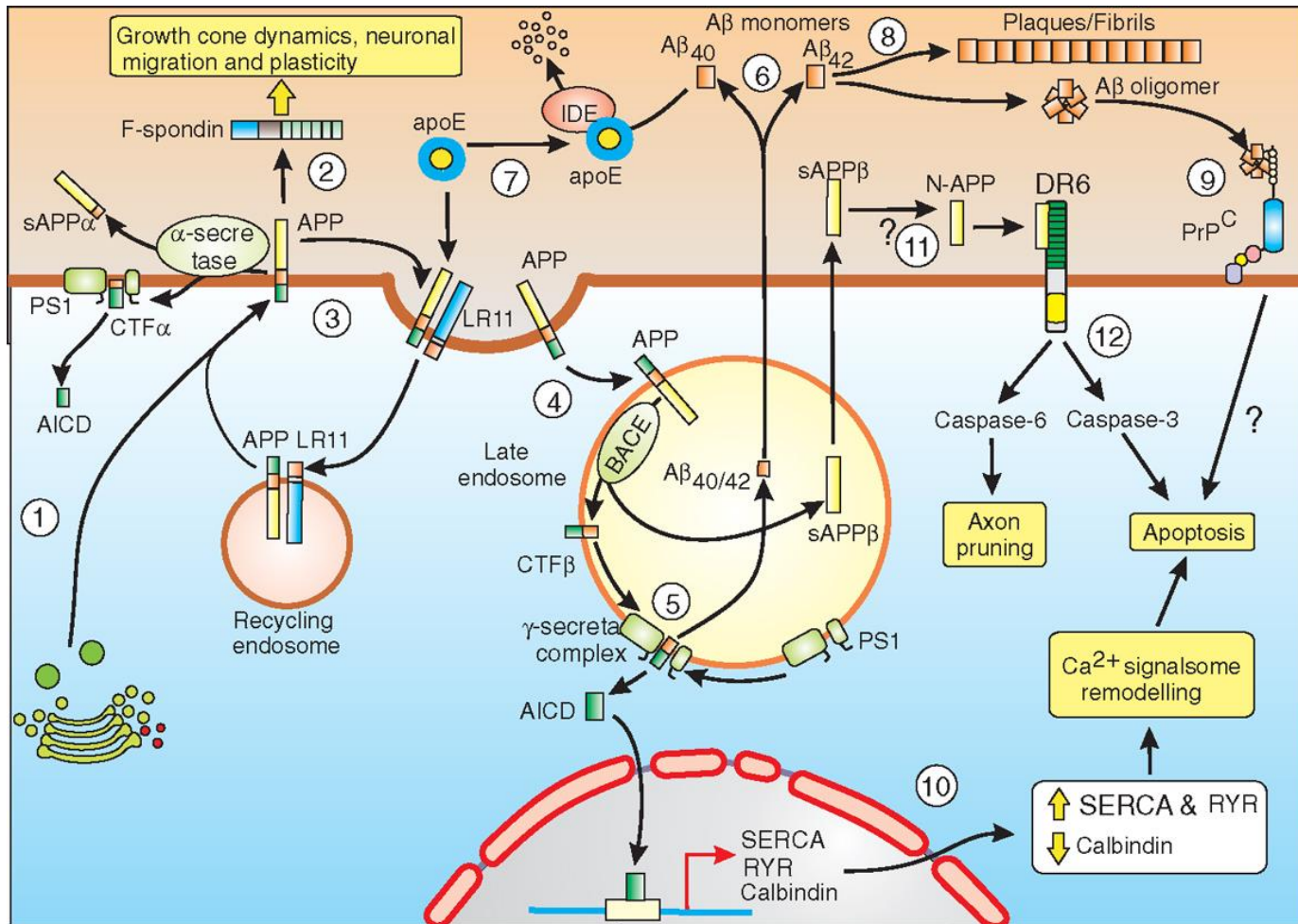
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- Some sufferers possess a mutated form of the APP protein which more readily produces the larger peptide upon proteolysis, so that more toxic plaques are produced.
 - It is the progressive accumulation of these plaques that is considered to be one cause of Alzheimer's disease.

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- several factors are associated with Alzheimer's disease.
 - An elevated plasma level of homocysteine is associated with the disease and it is considered as a risk factor for development of the disease.
 - The damage may be caused by a degradation product of homocysteine a thiolactone which causes oxidative damage to cells.

- 
- ❖ high levels of plasma cholesterol and increased incidence of Alzheimer's disease.
 - A possible explanation is that if cholesterol enters the plasma membranes of neuron it might reduce membrane fluidity, which could enhance cleavage of APP and hence increased formation of the larger peptide and hence the plaques.
 - Cholesterol can reduce fluidity by filling the gaps in the membrane between the polyunsaturated fatty acids

- 
- A high plasma level of the lipoprotein, apoE, is also associated with Alzheimer's disease. ApoE is necessary for transport of cholesterol into cells, there, cholesterol can form cholesteral ester which, in excess can damage cells.

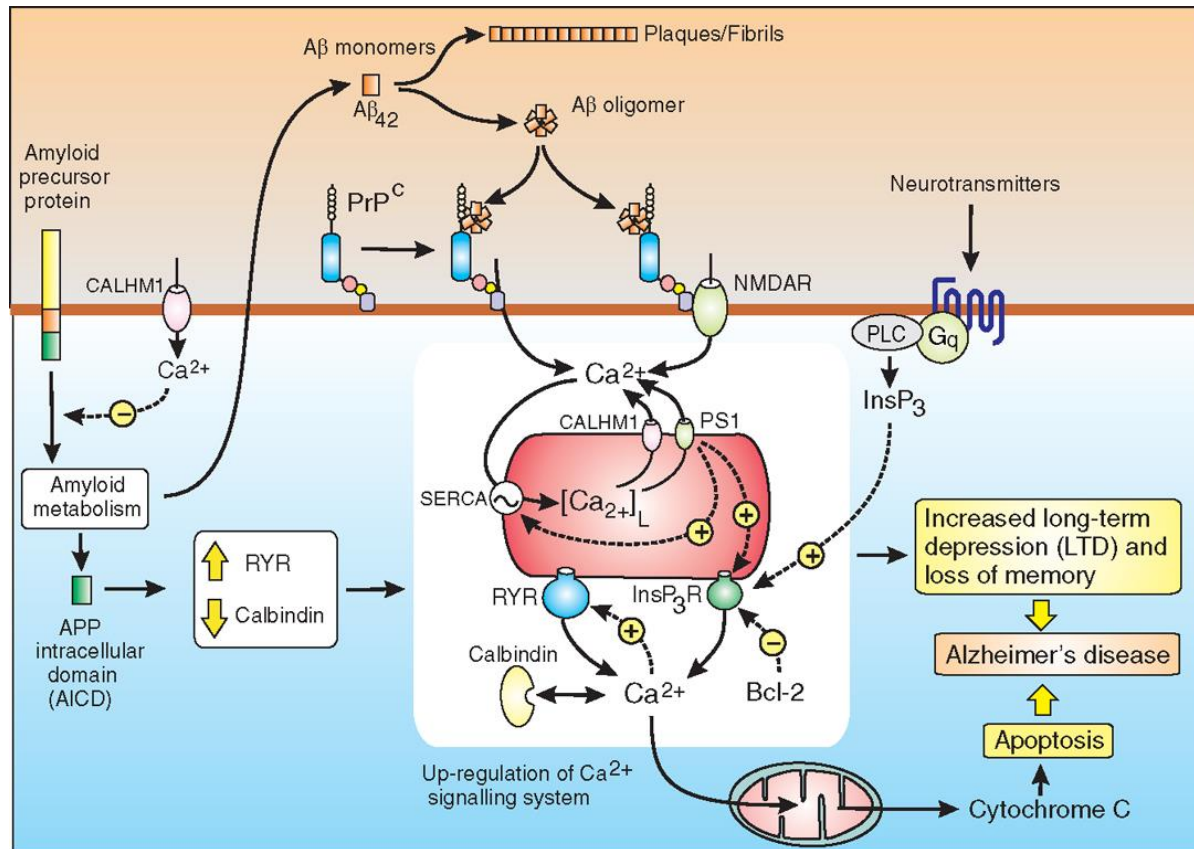
Module 12: Figure amyloid cascade hypothesis



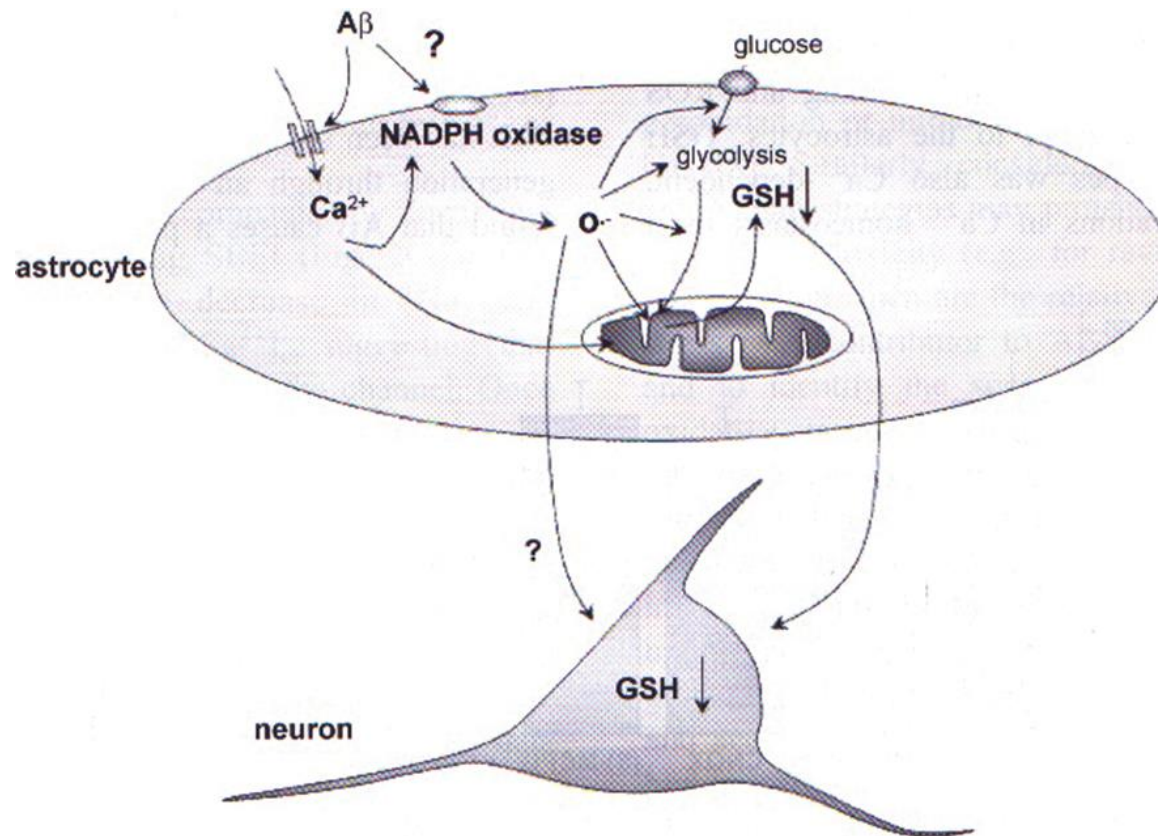
Amyloids and Ca^{2+} signalling

- The calcium hypothesis of Alzheimer's disease suggests that alterations in amyloid processing results in the formation of $\text{A}\beta$ oligomers that bring about an overall increase in Ca^{2+} signalling that then induces neuronal apoptosis and the onset of dementia. The way in which the amyloidogenic pathway functions to remodel the Ca^{2+} signalling system is described in [Module 12: Figure amyloid cascade hypothesis](#)

Module 12: Figure amyloids and Ca²⁺ signalling



Module 12: Figure astrocyte-induced neuronal death



Cell Signalling Biology - Michael J. Berridge - www.cellsignallingbiology.org - 2012

DOMINANTLY INHERITED FORMS OF AD

Missense mutations in the APP or Presenilin 1 or 2 genes



Increased A β 42 production throughout life



Accumulation and oligomerization of A β 42 in limbic and association cortices



Subtle effects of A β 42 oligomers on synaptic efficacy



Gradual deposition of A β 42 oligomers as diffuse plaques



Microglial and astrocytic activation and attendant inflammatory responses



Altered neuronal ionic homeostasis; oxidative injury



Altered kinase/phosphatase activities lead to tangles



Widespread neuronal/synaptic dysfunction and selective neuronal loss, with attendant neurotransmitter deficits



DEMENTIA

NONDOMINANT FORMS OF AD (Including "Sporadic" AD)

Failure of A β clearance mechanisms (e.g., inheritance of ApoE4, faulty A β degradation, etc.)



Gradually rising A β levels with age



Accumulation and oligomerization of A β 42 in limbic and association cortices



Subtle effects of A β 42 oligomers on synaptic efficacy



Gradual deposition of A β 42 oligomers as diffuse plaques



Microglial and astrocytic activation and attendant inflammatory responses



Altered neuronal ionic homeostasis; oxidative injury



Altered kinase/phosphatase activities lead to tangles



Widespread neuronal/synaptic dysfunction and selective neuronal loss, with attendant neurotransmitter deficits



DEMENTIA

Parkinson's disease (PD)

- Is a slow progressive and common neurodegenerative disease that presents with tremor, rigidity and postural instability central nervous system.
- Pathologically characterized by selective degeneration of the dopaminergic neurons of the substantia nigra in association with formation of Lewy body inclusions and Lewy dystrophic neuritis
 - ➔ Muscle rigidity, tremor, slowing of physical movement (Bradykinesia) and, in extreme cases, loss of physical movement (Akinesia).

Blood Biomarkers

- Parkinson's disease is thought to be due, at least in part, to increased oxidative stress.
- patients show a selective reduction in mitochondrial complex I in their substantia nigra and that platelets exhibit a similar deficiency.
- oxidative stress and mitochondrial respiratory failure are implicated in loss of dopaminergic neurons in PD

Oxidative stress

- It is considered that the brain and nervous tissue are more susceptible to oxidative damage than other tissues or organs.
- Chronic oxidative stress may lead to degeneration of neurones and hence the degenerative diseases.
- Susceptibility may be due to the exclusive dependence on mitochondrial ATP generation, with the opportunity of formation of oxygen free radicals from the continuous flux of electrons along the electron transfer chain and the presence of the high concentration of oxygen in the neurones (hence they are totally dependent on aerobic ATP generation.)

Thank you



